

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

SCHEM, MLB, D1

8/8/12

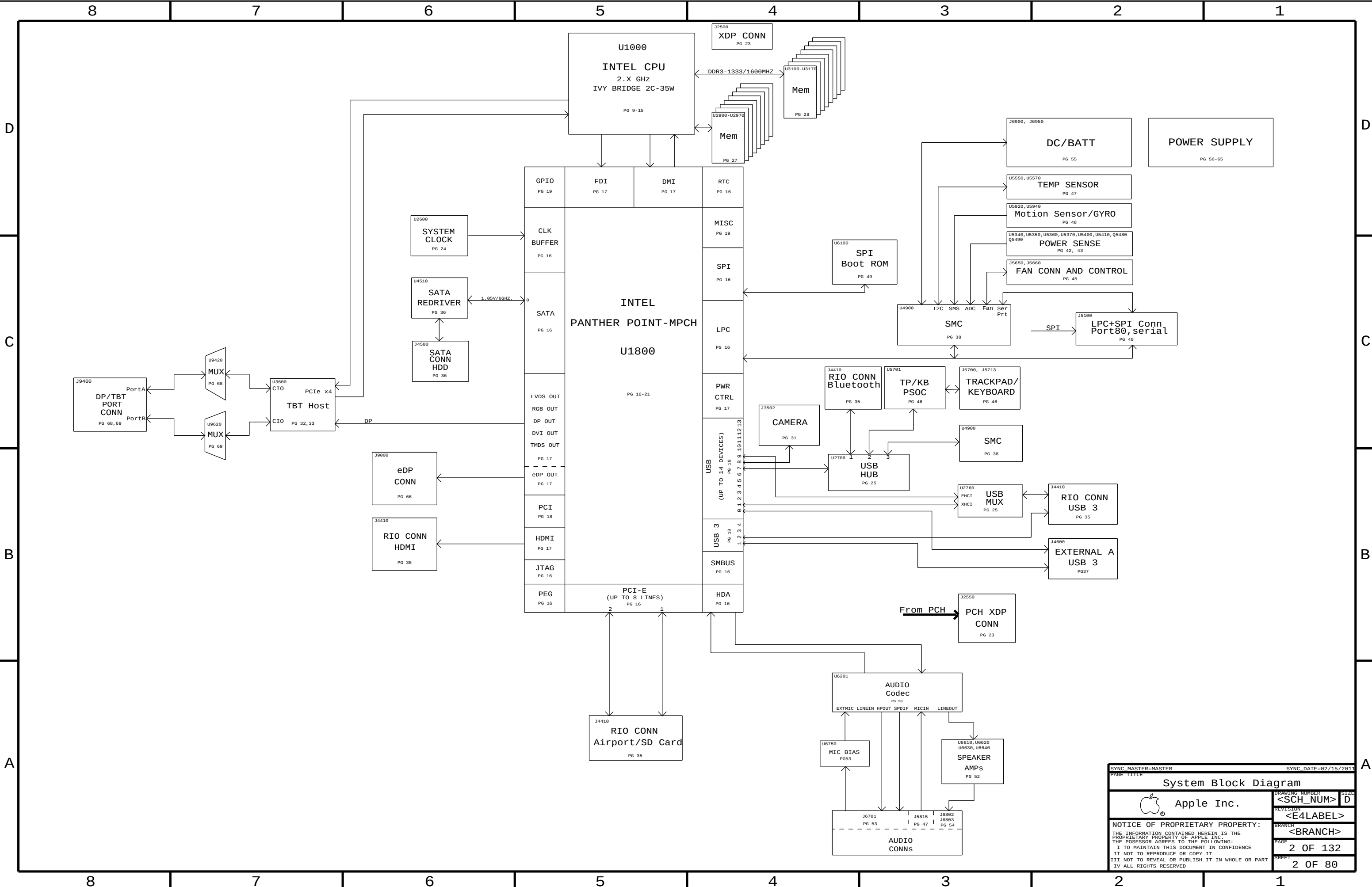
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3	Power Block Diagram	K17_REF	06/30/2009
4	Revision History	MASTER	MASTER
5	BOM Configuration	MASTER	MASTER
6	BOM Configuration	MASTER	MASTER
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8	Power Aliases	MASTER	MASTER
9	Signal Aliases	D1_MLB_TEST	01/27/2012
10	CPU DMI/PEG/FDI/RSVD	J30_MLB	07/14/2011
11	CPU CLOCK/MISC/JTAG	J30_MLB	07/14/2011
12	CPU DDR3 INTERFACES	J30_MLB	07/14/2011
13	CPU POWER	J30_MLB	07/14/2011
14	CPU GROUNDS	J30_MLB	07/14/2011
15	CPU DECOUPLING-I	MASTER	MASTER
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17	PCH SATA/PCIE/CLK/LPC/SPI	J13_MLB	09/15/2011
18	PCH DMI/FDI/PM/Graphics	J13_MLB	09/15/2011
19	PCH PCI/USB/TP/RSVD	J13_MLB	09/15/2011
20	PCH GPIO/MISC/NCTF	J13_MLB	09/15/2011
21	PCH POWER	J13_MLB	09/15/2011
22	PCH GROUNDS	J13_MLB	09/15/2011
23	PCH DECOUPLING	J13_MLB	09/15/2011
24	CPU & PCH XDP	J30_MLB	07/14/2011
25	Chipset Support	MASTER	MASTER
26	USB HUB & MUX	J5_AMD	08/17/2011
27	CPU Memory S3 Support	J5_MLB	07/29/2011
28	DDR3 SDRAM Bank A (Rank 0)	J5_MLB	07/14/2011
29	DDR3 SDRAM Bank B (Rank 0)	J5_MLB	07/14/2011
30	DDR3 Termination	MASTER	MASTER
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44	Power Sensor: High Side	D1_SENSORS	02/20/2012
45	Thermal Sensors	D1_SENSORS	02/20/2012

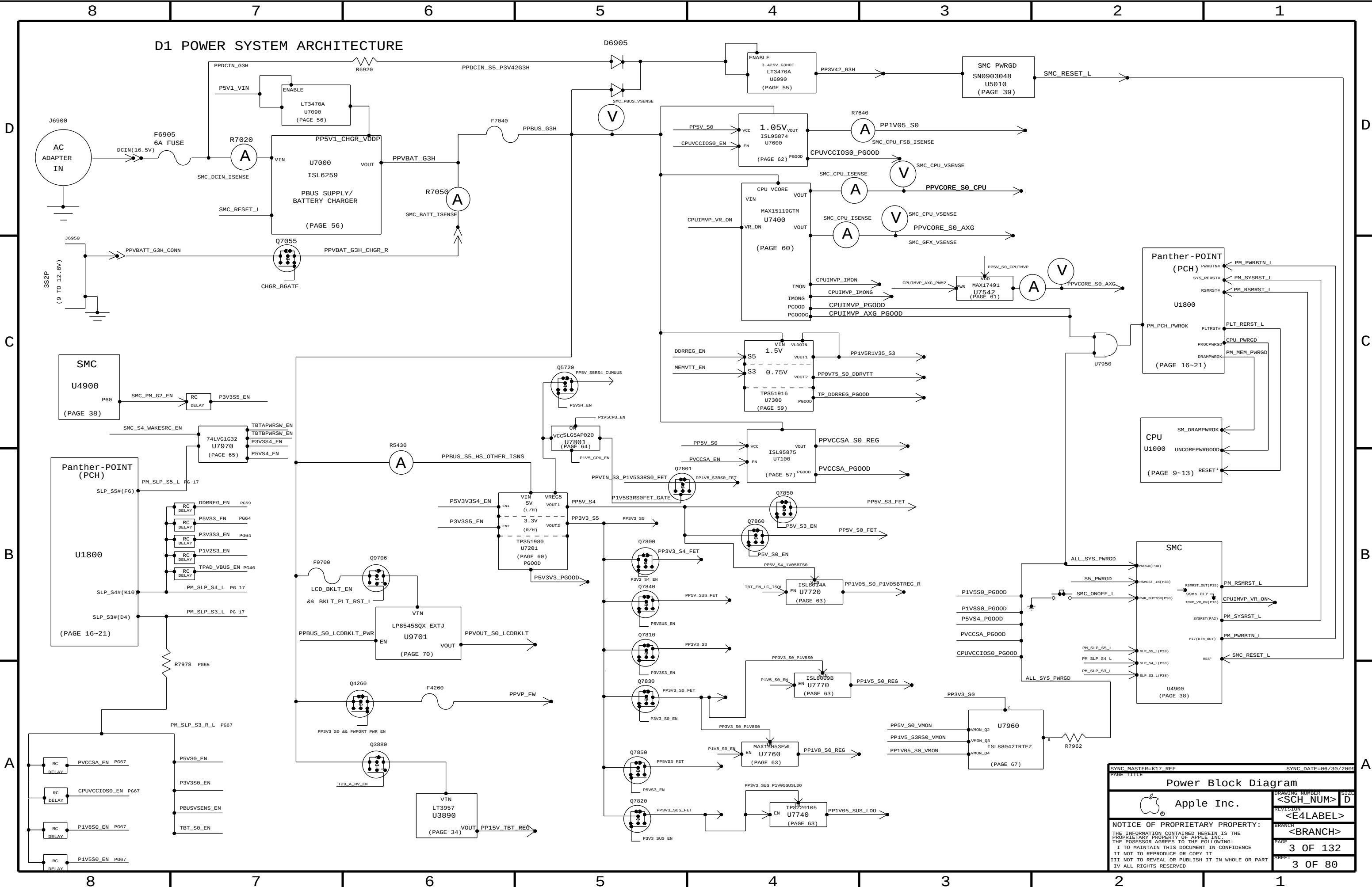
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48	58	KEYBOARD/TRACKPAD (2 OF 2)	D2_MLB_KEPLER	12/08/2011
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50	61	SPI ROM	J13_MLB	01/20/2012
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71	97	LCD Backlight Driver (LP8545)	J5_MLB_KEPLER	09/21/2011
72	100	CPU Constraints	J5_MLB	09/13/2011
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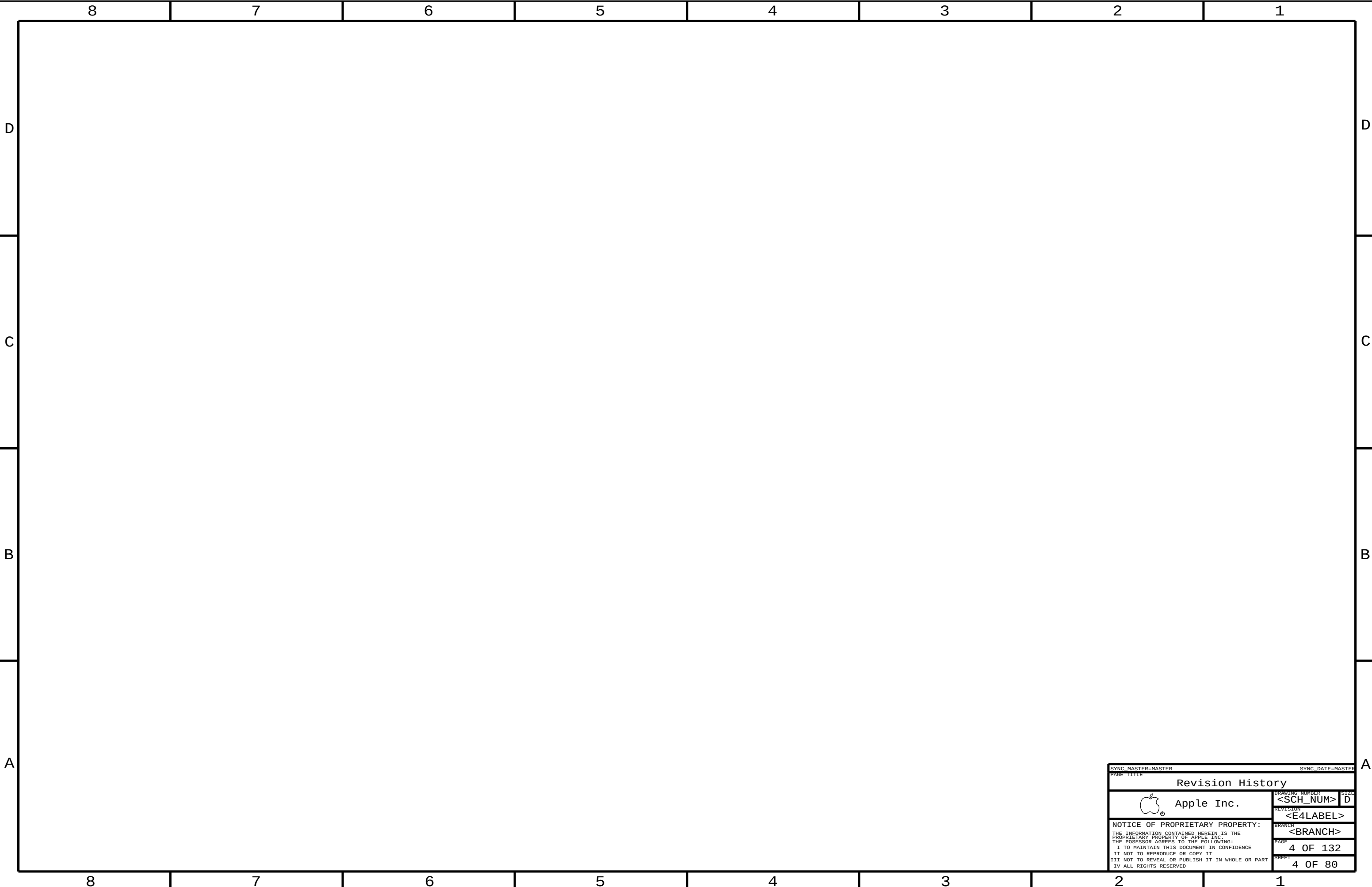
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9216	1	SCHEM,MLB,D1	SCH	CRITICAL	
820-3462	1	PCBF,MLB(NEW),D1	PCB	CRITICAL	

DRAWING
TITLE=MLB
ABBREV=ABBREV
LAST_MODIFIED=Thu Aug 9 12:34:09 2012

DRAWING TITLE		SCHEM, MLB, D1	
 Apple Inc.	DRAWING NUMBER	<SCH_NUM>	SIZE
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	BRANCH	<BRANCH>	
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Revision History

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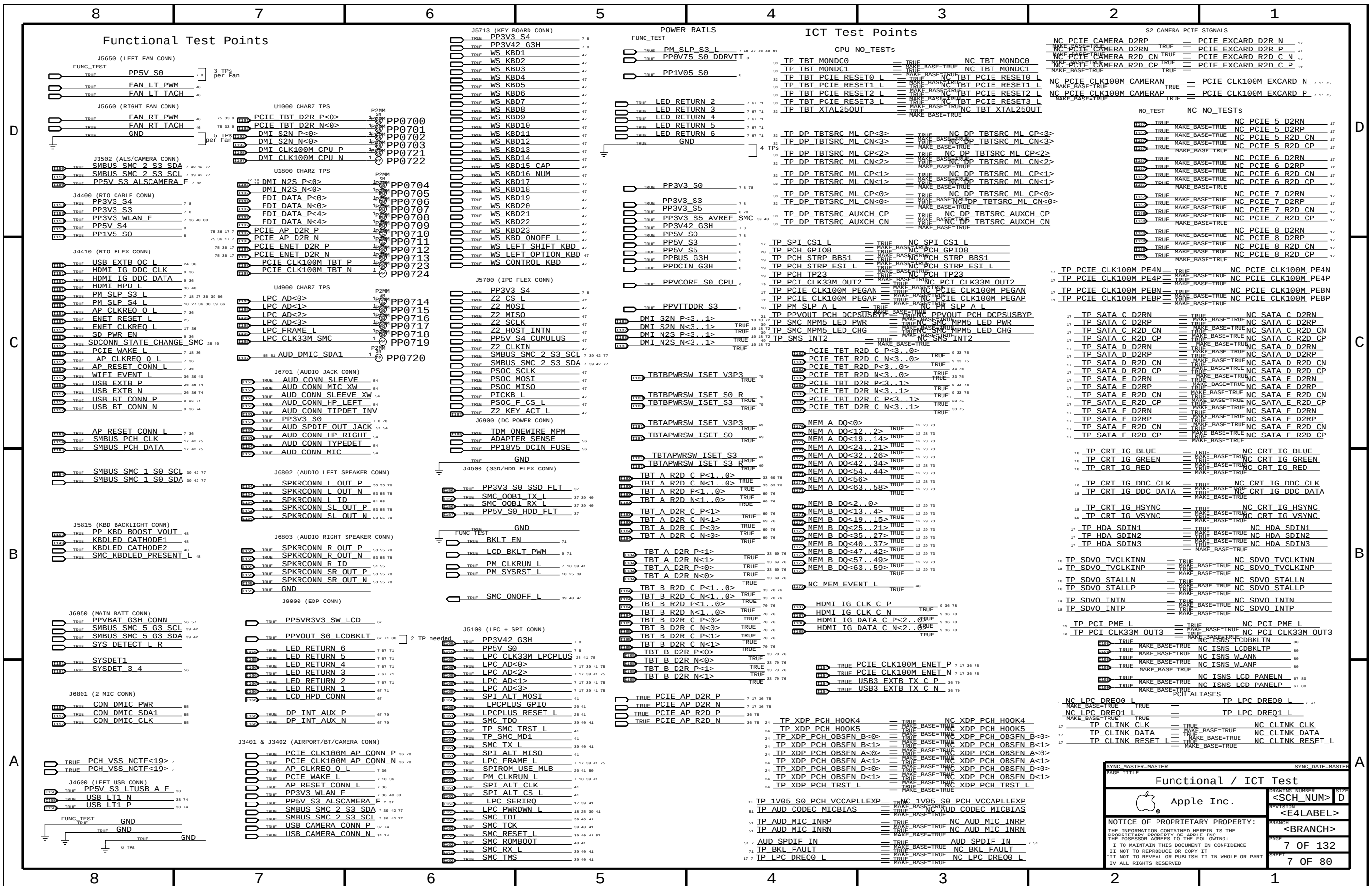
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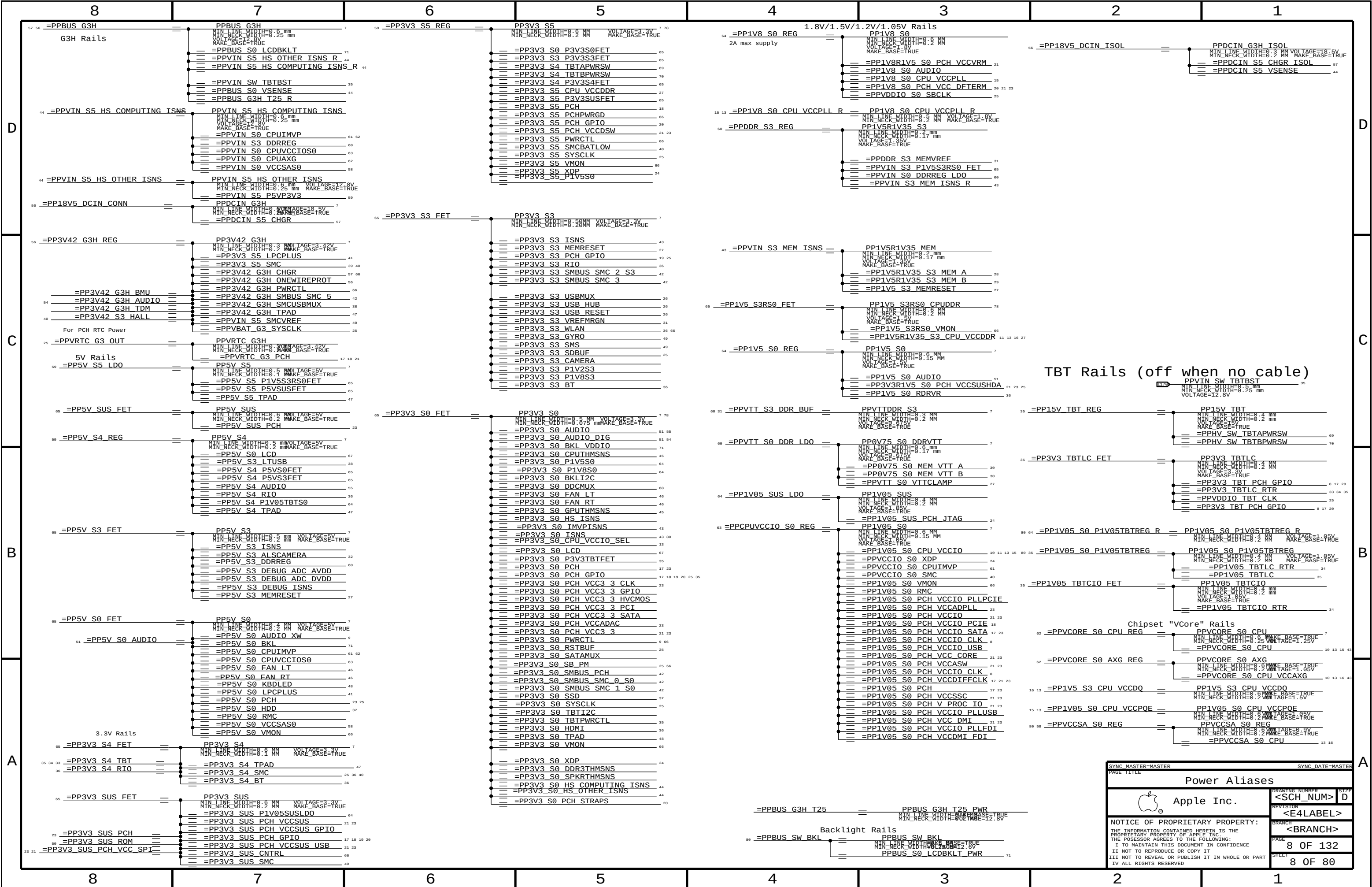
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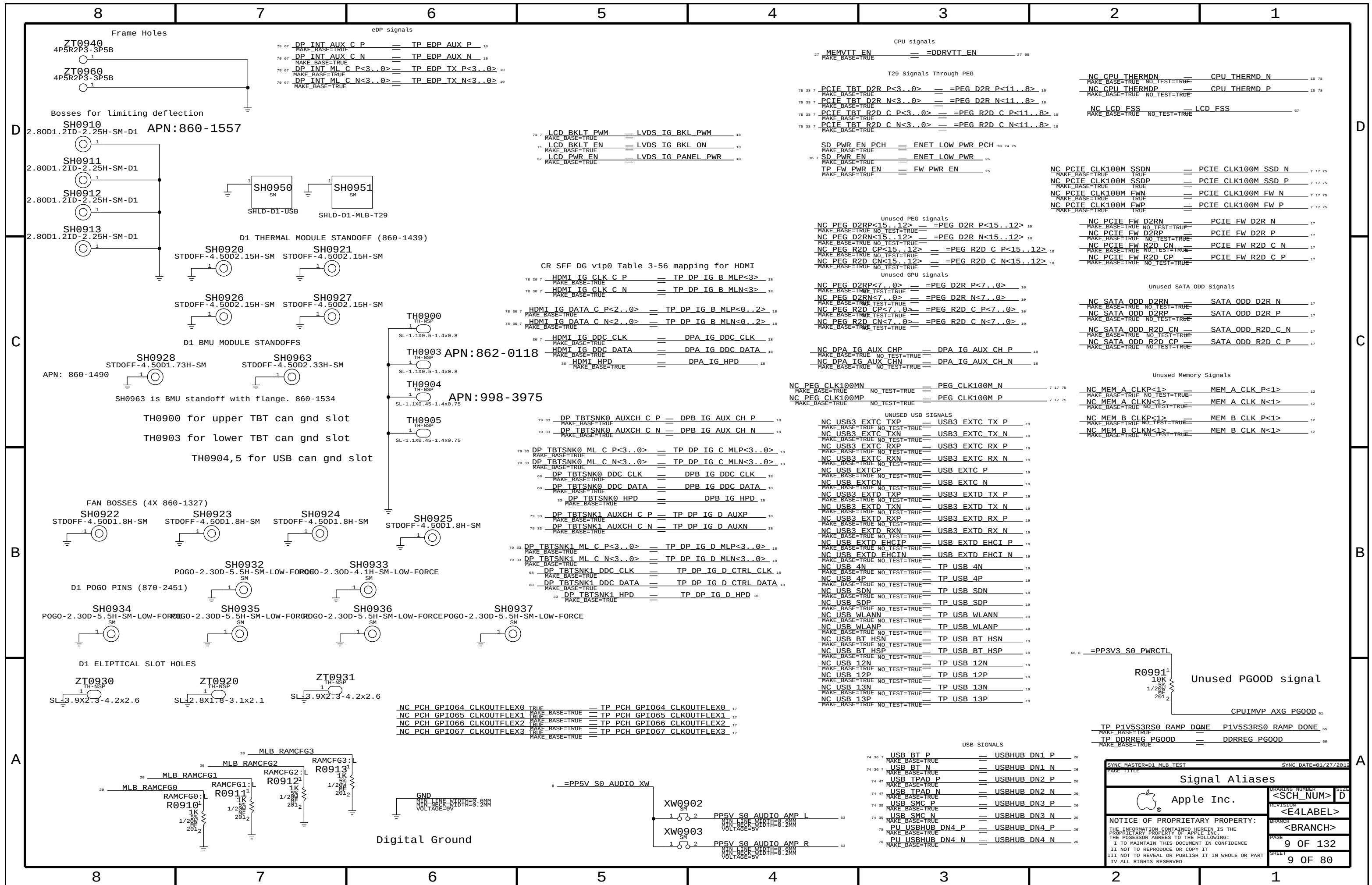
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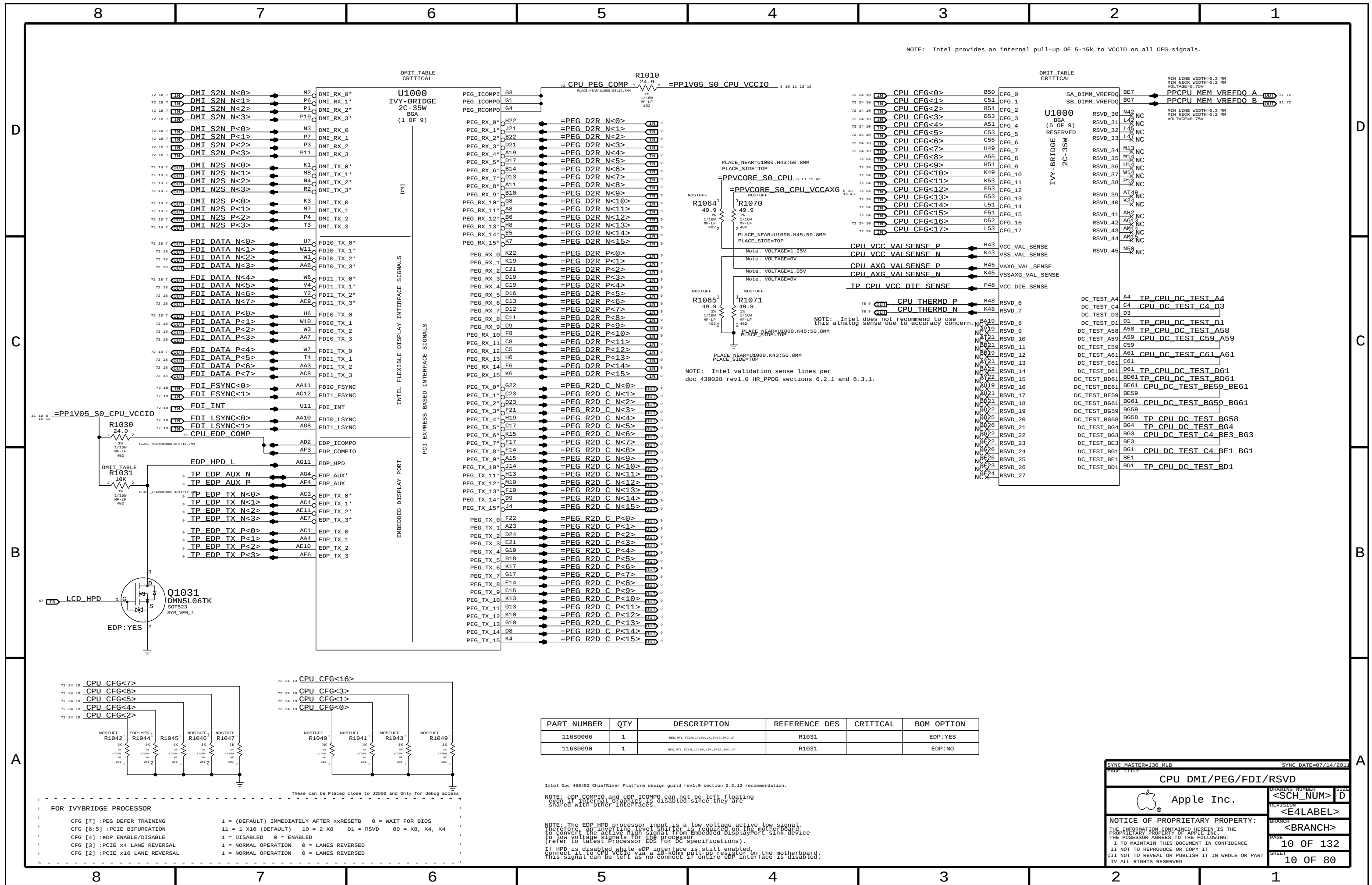
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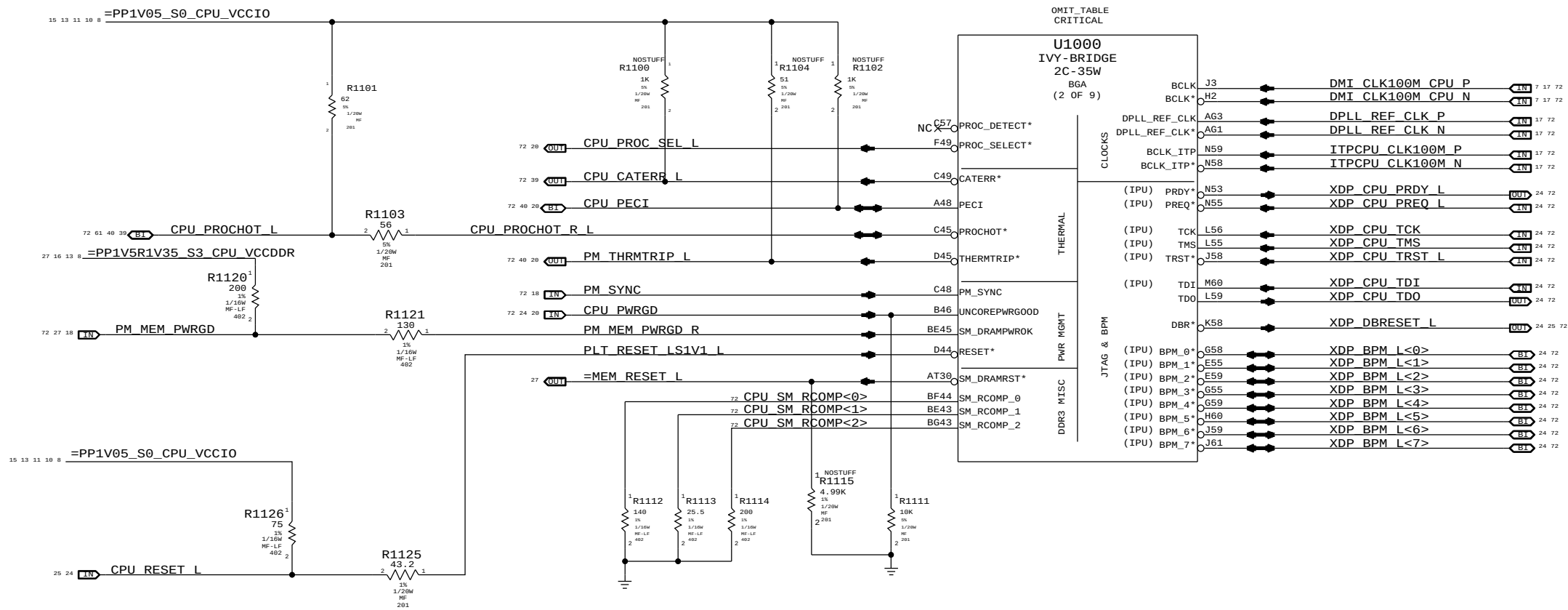
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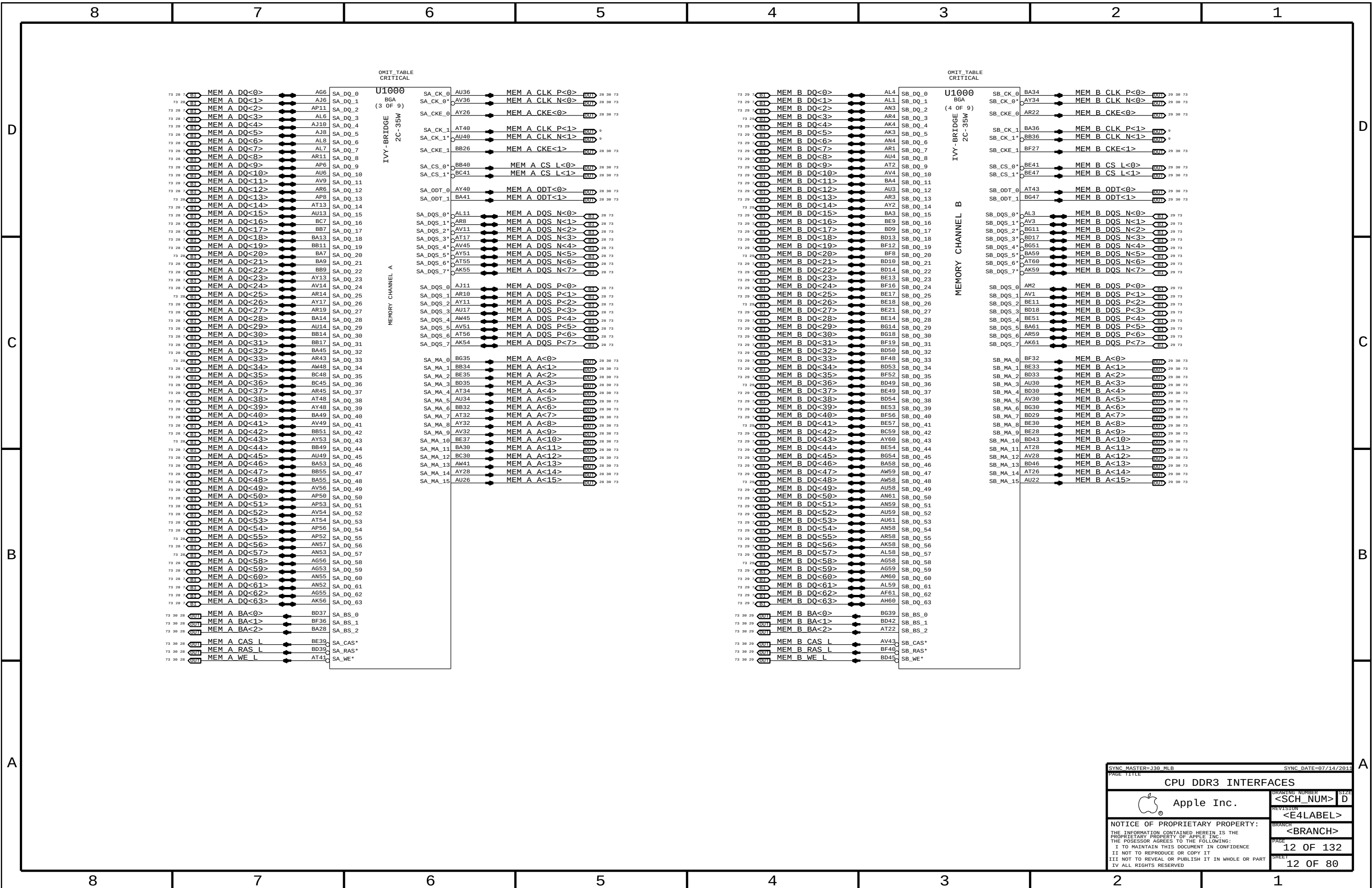


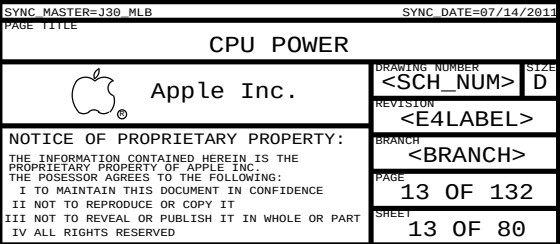


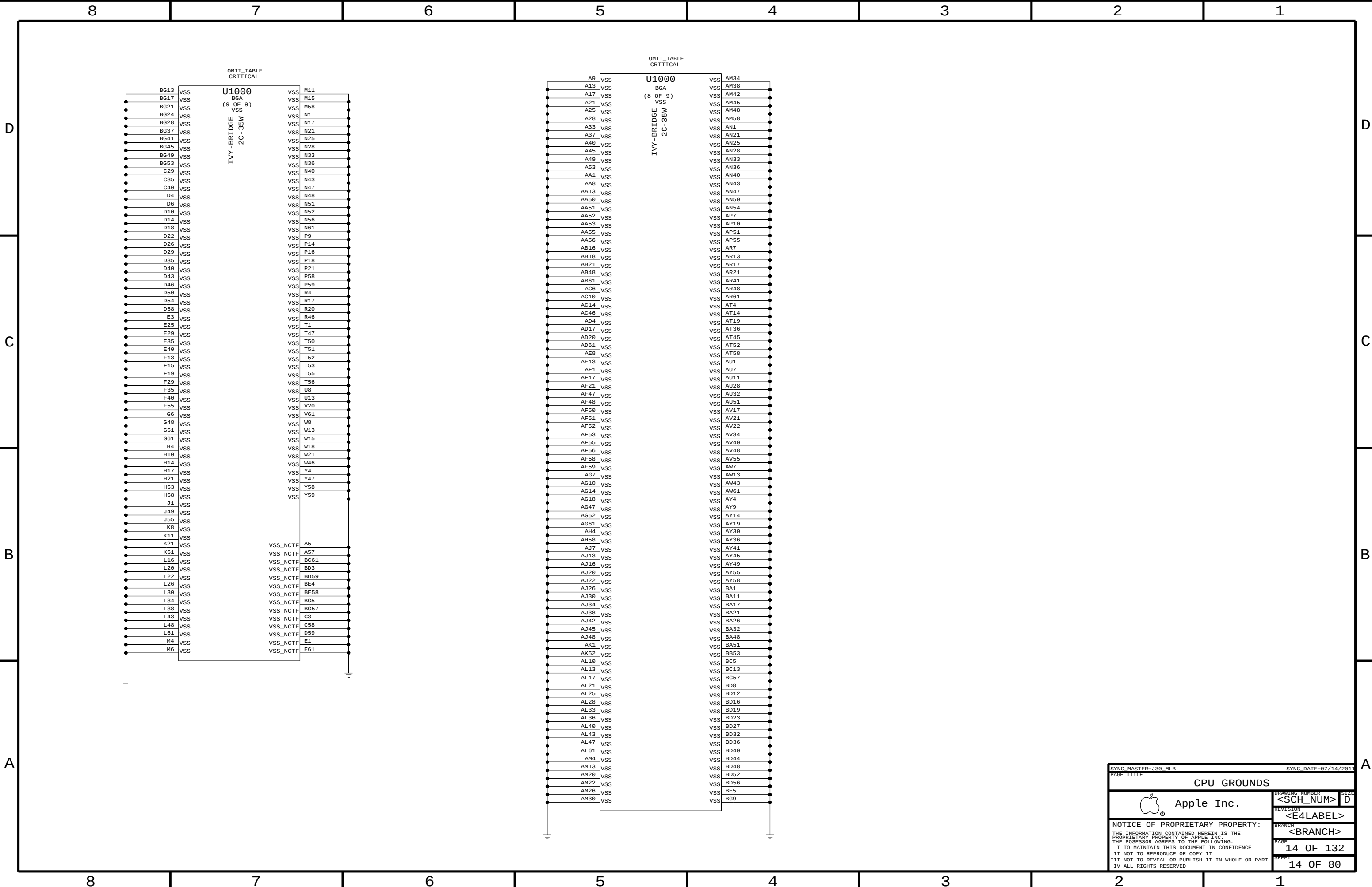




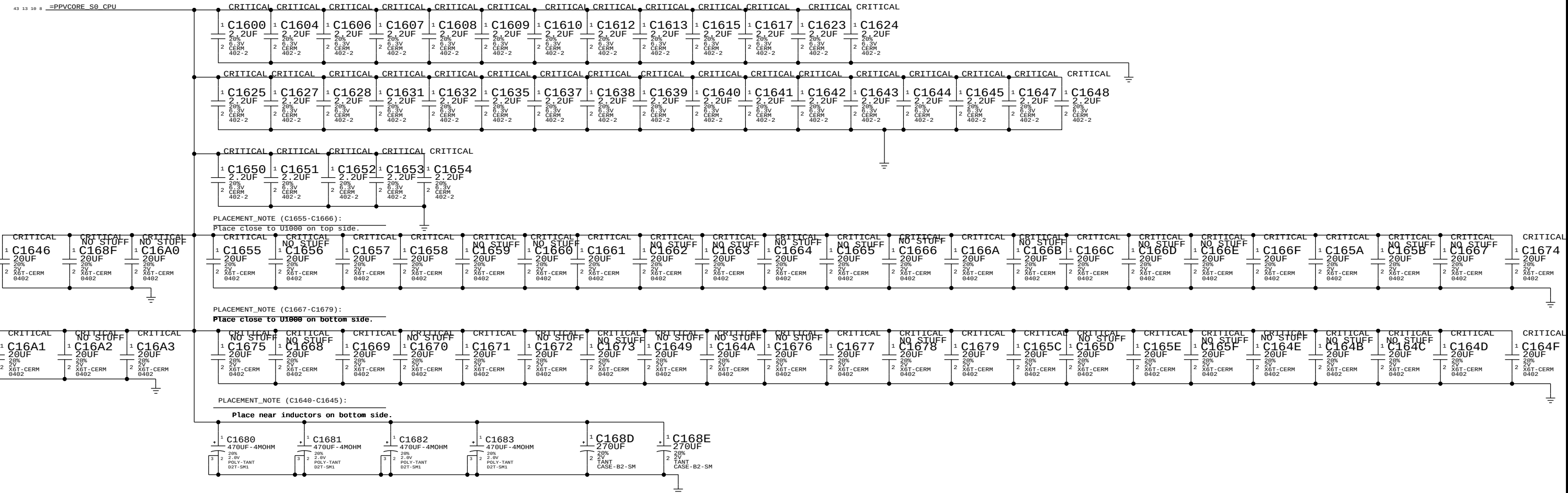






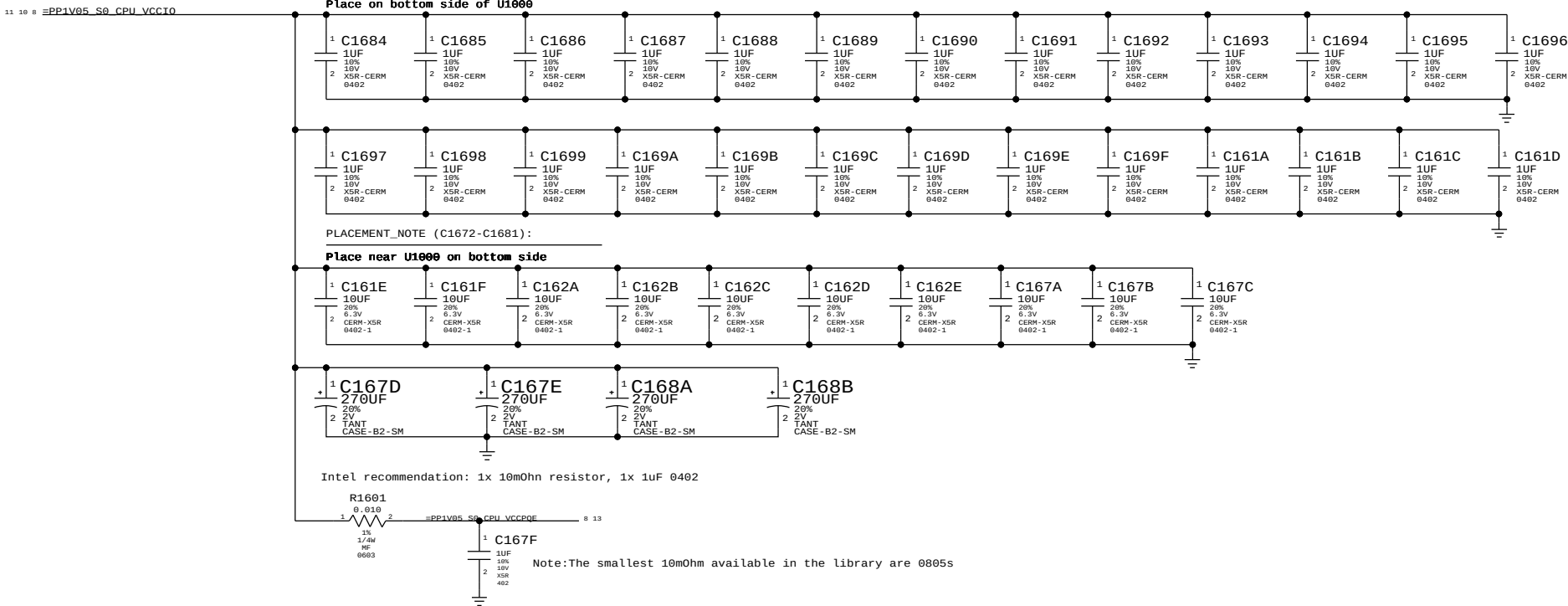


Intel recommendation (Table 7-2): Option 2: 35x 2.2uF, 12x 22uF, 4x 470uF, or Option 3: 35x 2.2uF, 6x 22uF, 6x 330 uF



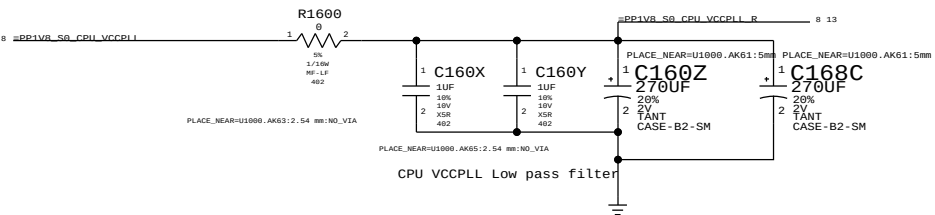
Intel recommendation (Table 7-7): 26x 1uF, 10x 10uF, 2x 330uF

LACEMENT_NOTE (C1684-C167F):

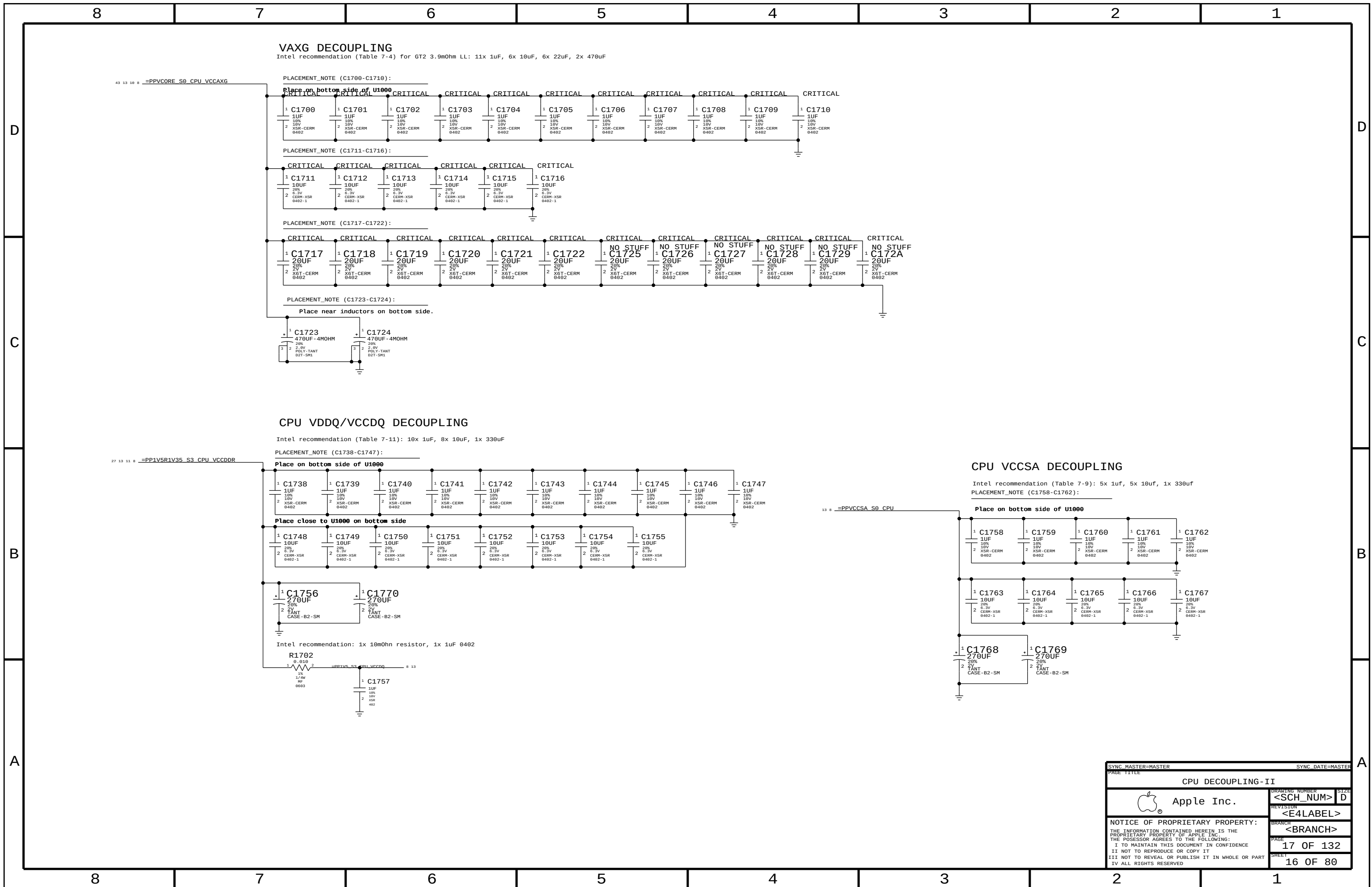


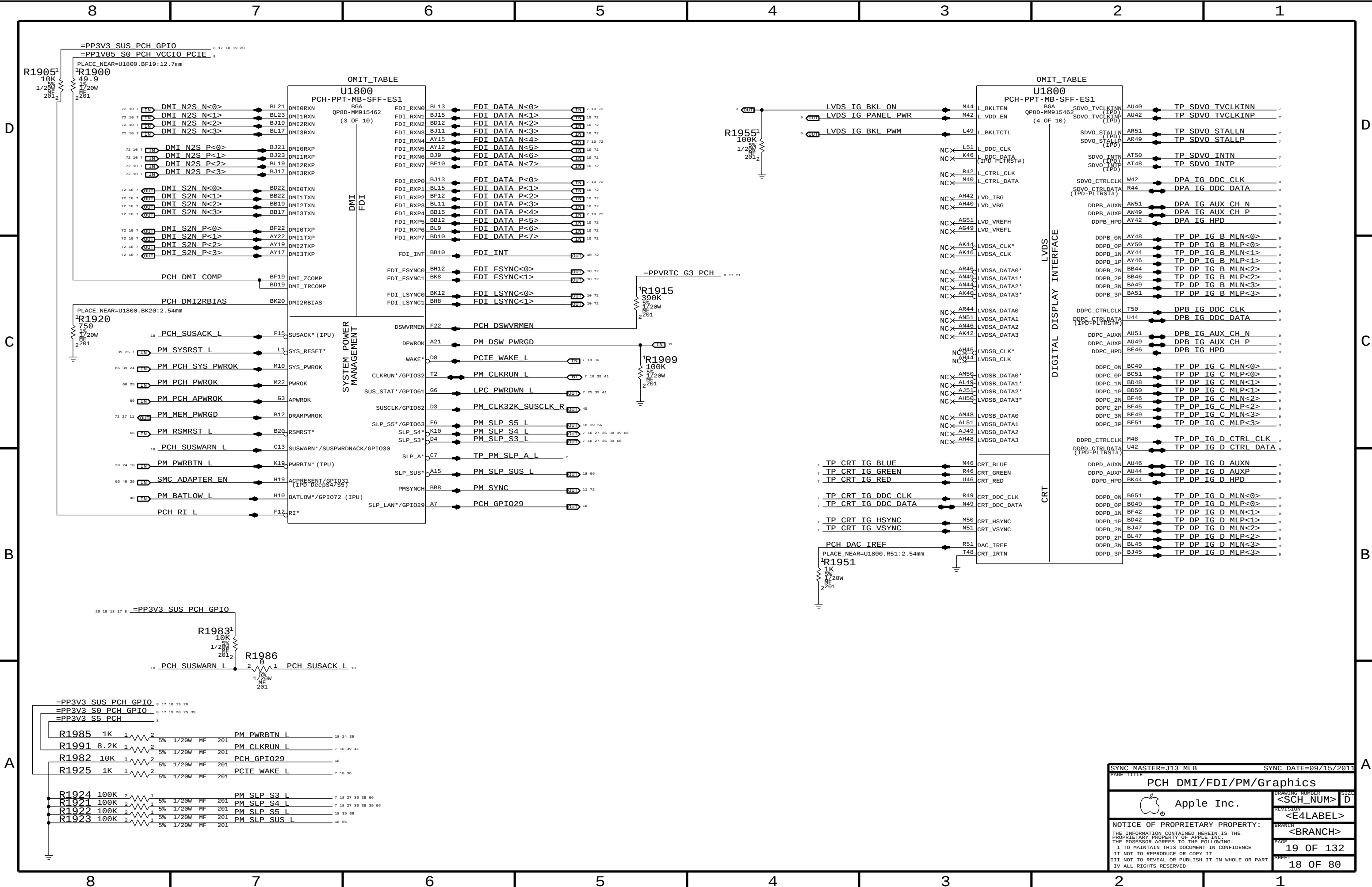
intel recommendation (table 7-5): 2x 1uF, 1x 330uF

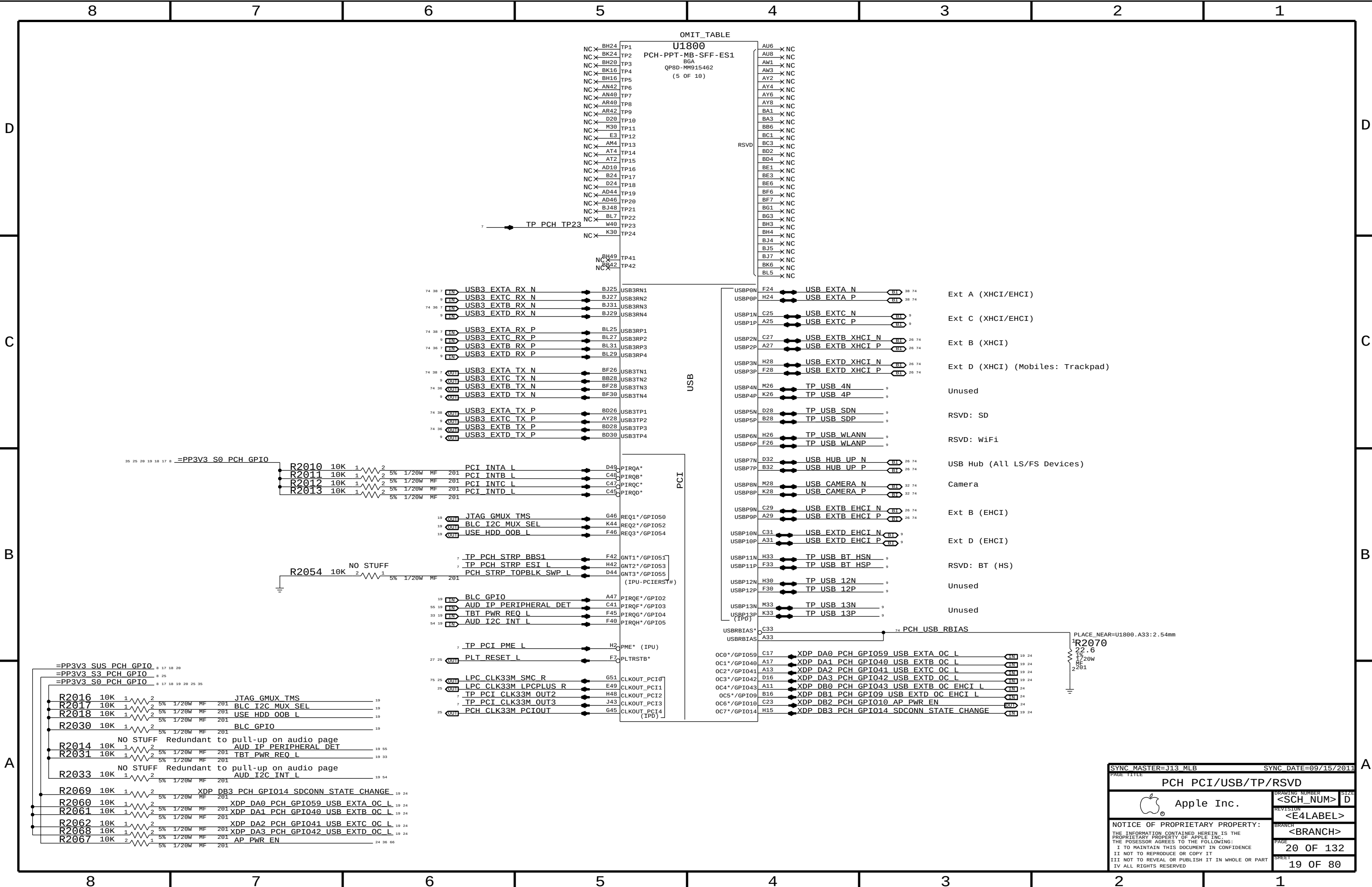
PLACEMENT_NOTE (C1646-C1671):

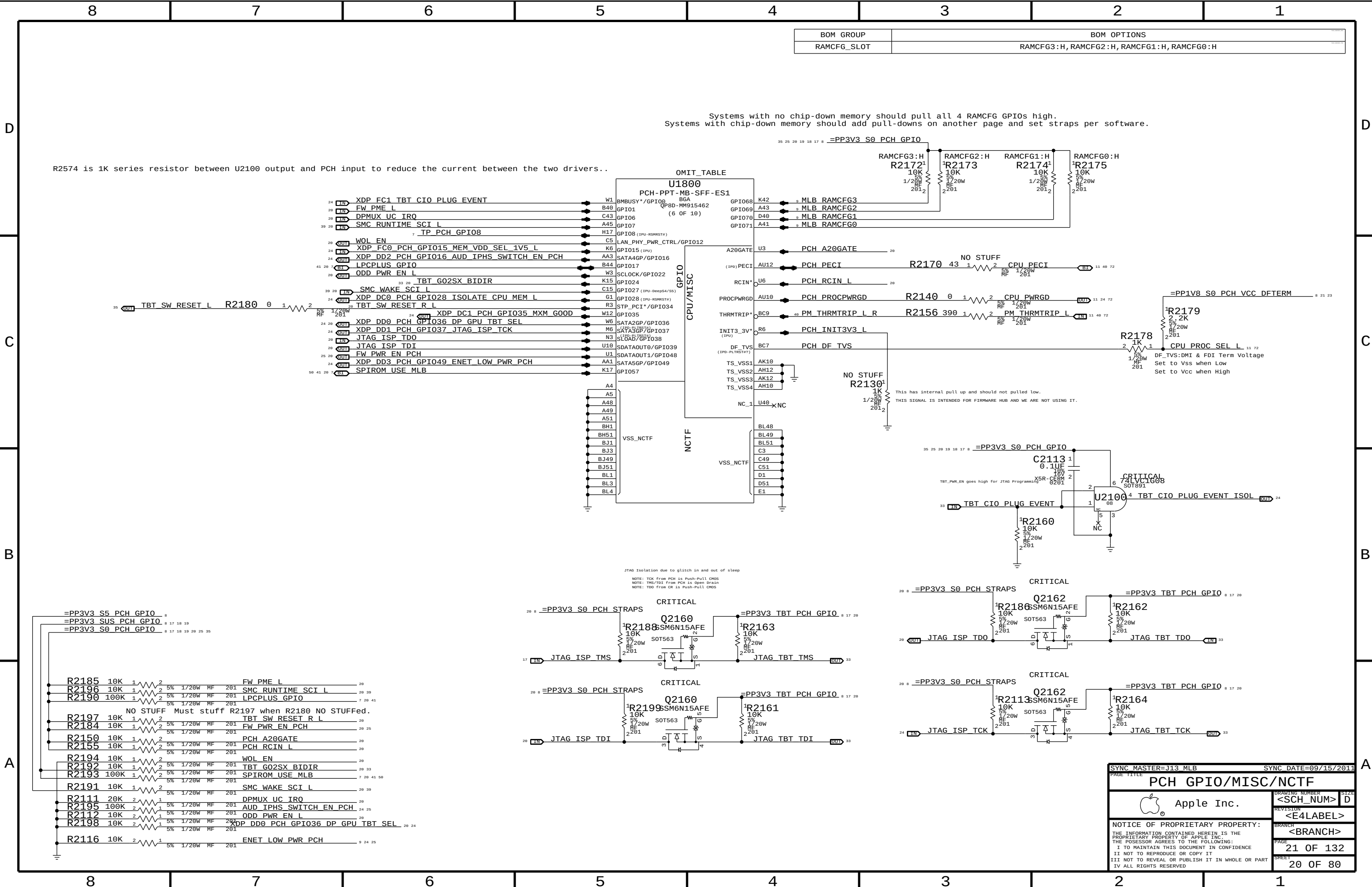


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CPU DECOUPLING-I			
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BOM GROUP	BOM OPTIONS
RAMCFG_SL0T	RAMCFG3:H, RAMCFG2:H, RAMCFG1:H, RAMCFG0:H

Systems with no chip-down memory should pull all 4 RAMCFG GPIOs high.
Systems with chip-down memory should add pull-downs on another page and set straps per software.

R2574 is 1K series resistor between U2100 output and PCH input to reduce the current between the two drivers..

OMIT_TABLE
U1800
PCH-PPT-MB-SFF-ES1
BGA
P80-MM915462
(6 OF 10)

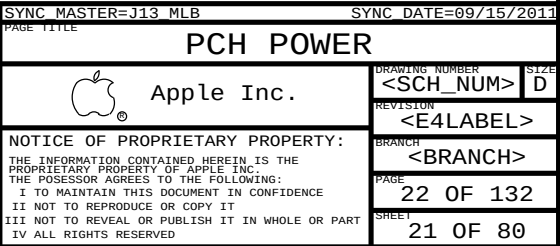
GPIO
GPIO/MISC
CPU/MISC
NCTF
VSS_NCTF
VSS_NCTF

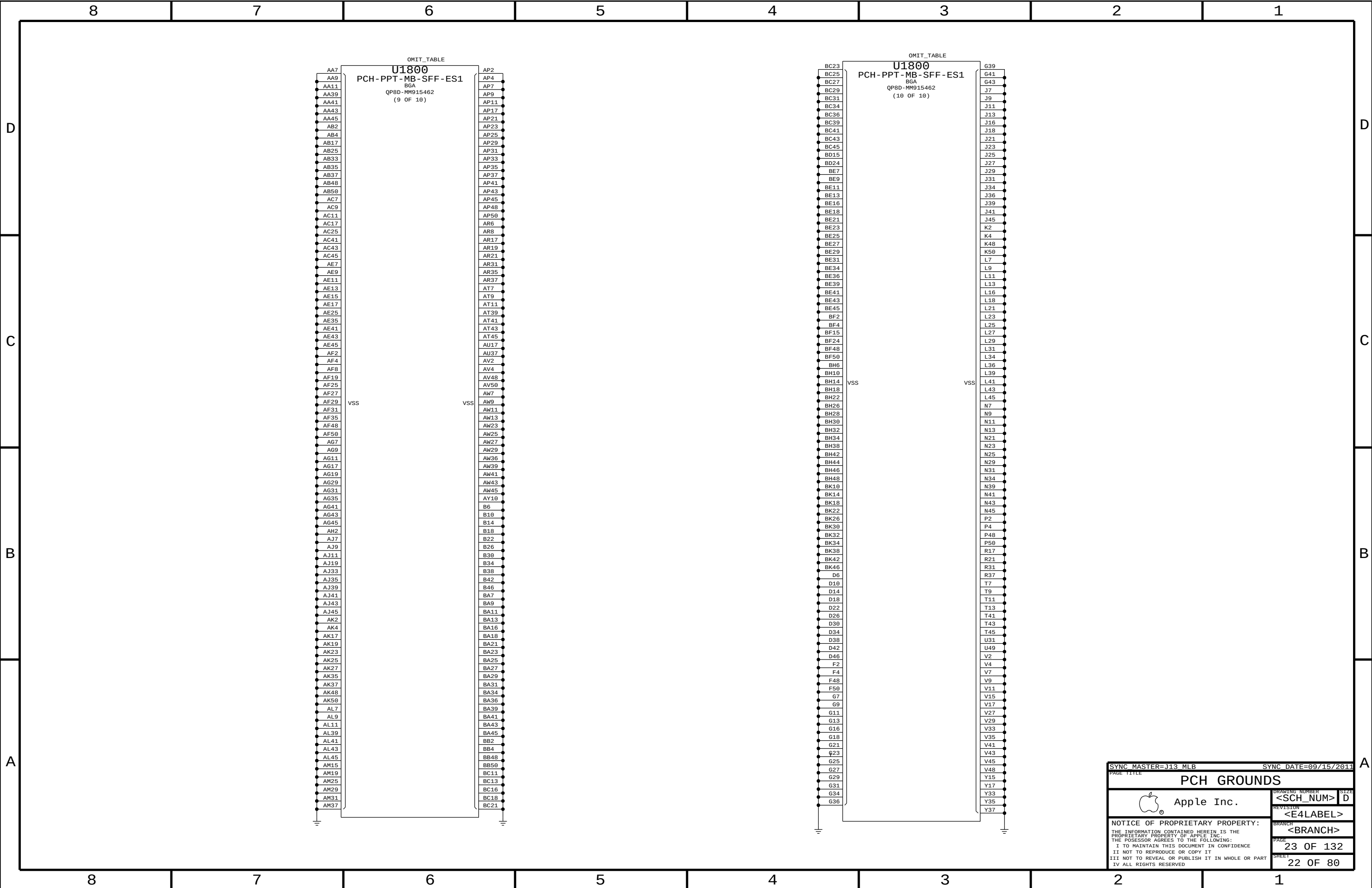
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U2100
TBT CIO PLUG EVENT
TBT CIO PLUG EVENT ISOL
R2160
C2113
X5R-CERM
SOT891

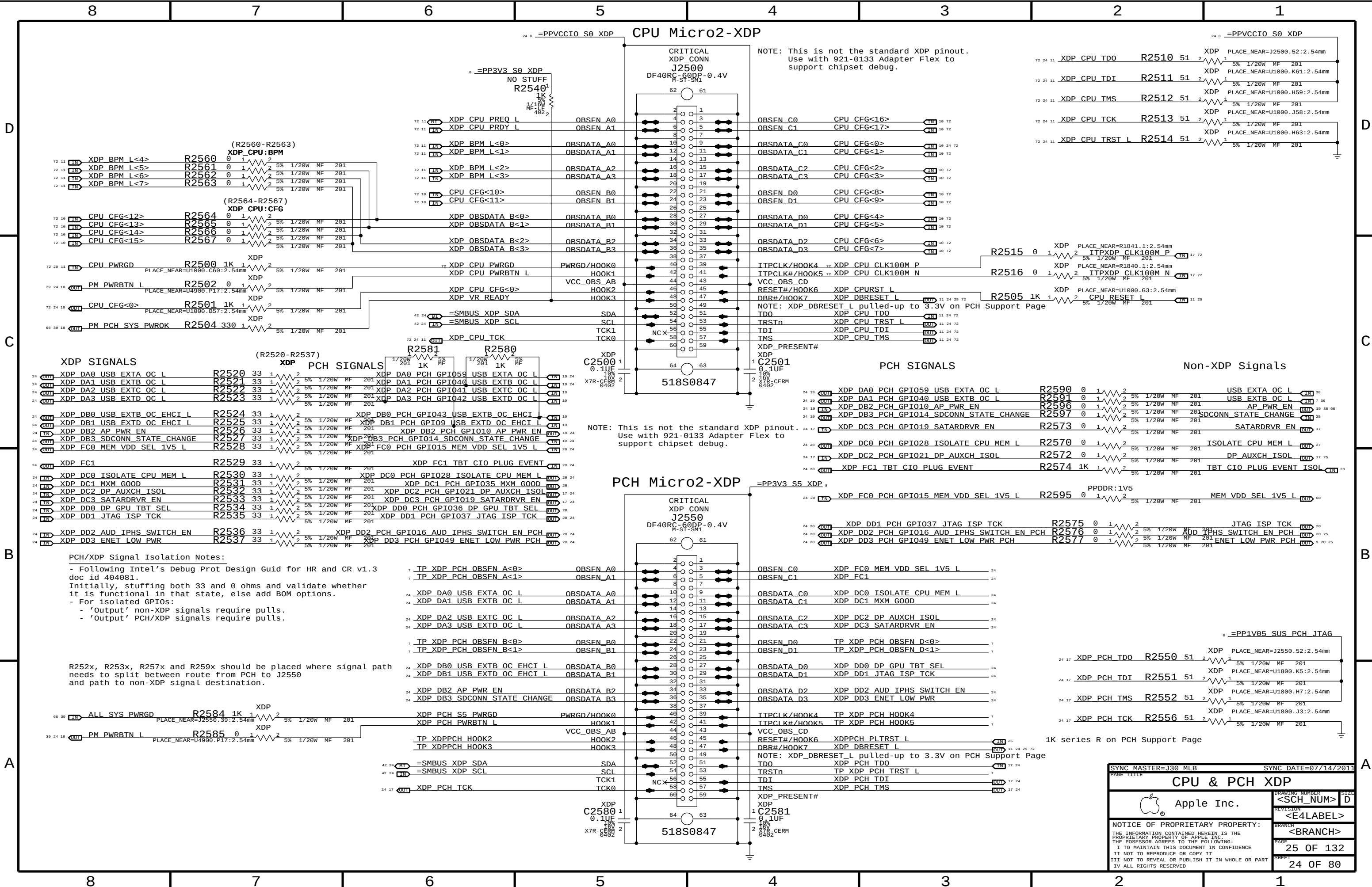
CRITICAL
Q2160
SM6N15AFE
SOT563
JTAG ISP TMS
JTAG TBT TMS
R2188
R2163
CRITICAL
Q2160
SM6N15AFE
SOT563
JTAG ISP TDI
JTAG TBT TDI
R2199
R2161

CRITICAL
Q2162
SM6N15AFE
SOT563
JTAG ISP TDO
JTAG TBT TDO
R2186
R2162
CRITICAL
Q2162
SM6N15AFE
SOT563
JTAG ISP TCK
JTAG TBT TCK
R2113
R2164

SYNC MASTER=J13 MLB		SYNC DATE=09/15/2011	
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D

C

B

A

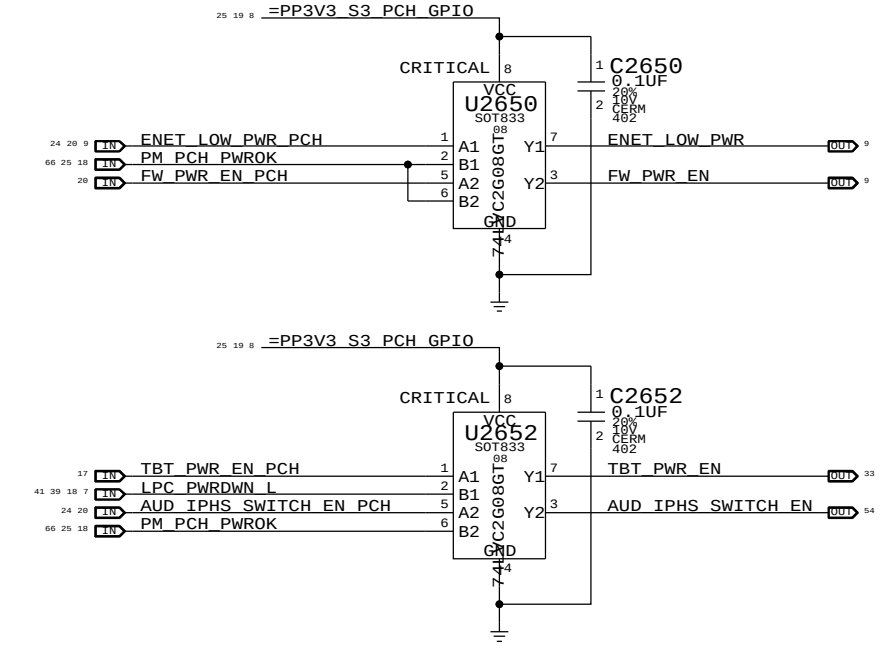
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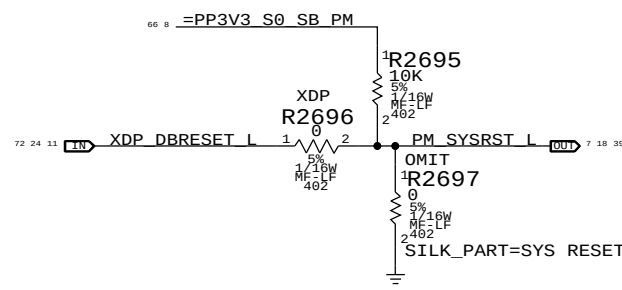
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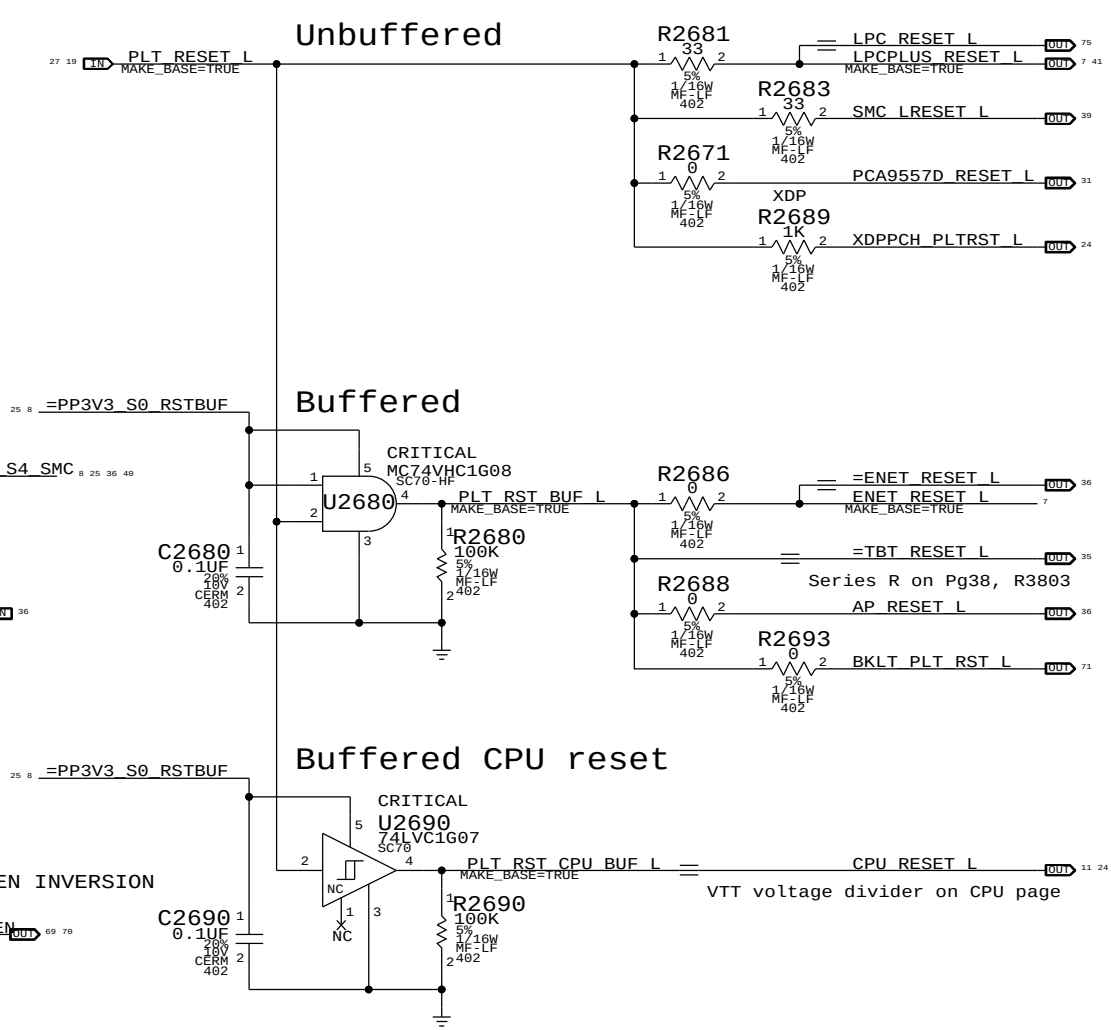
GPIO Glitch Prevention



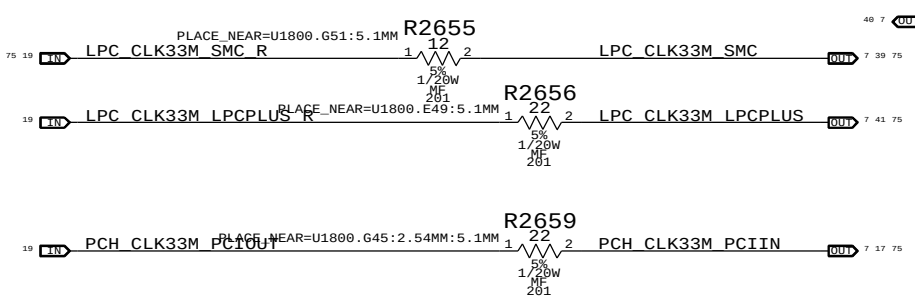
PCH Reset Button



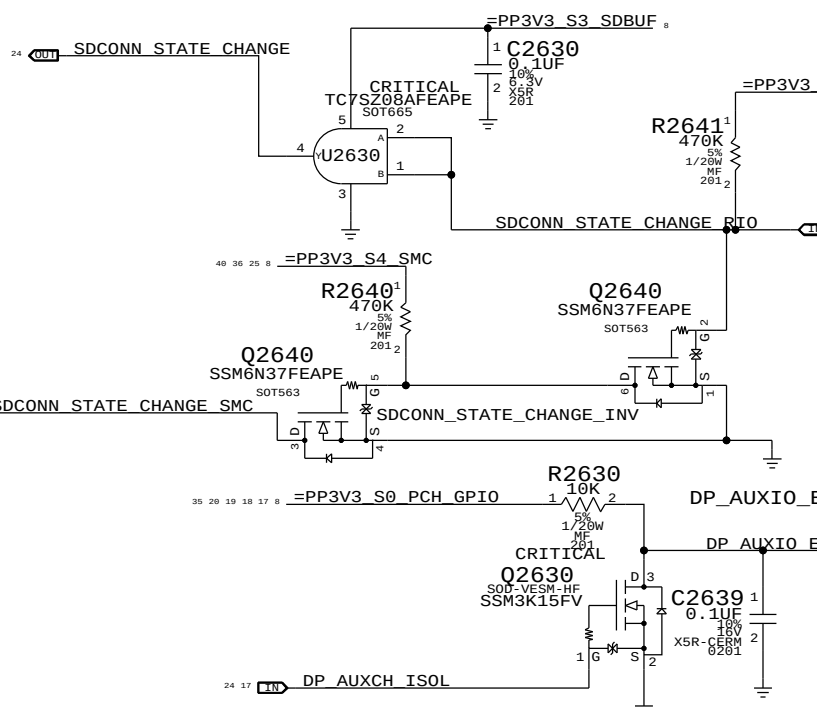
Platform Reset Connections



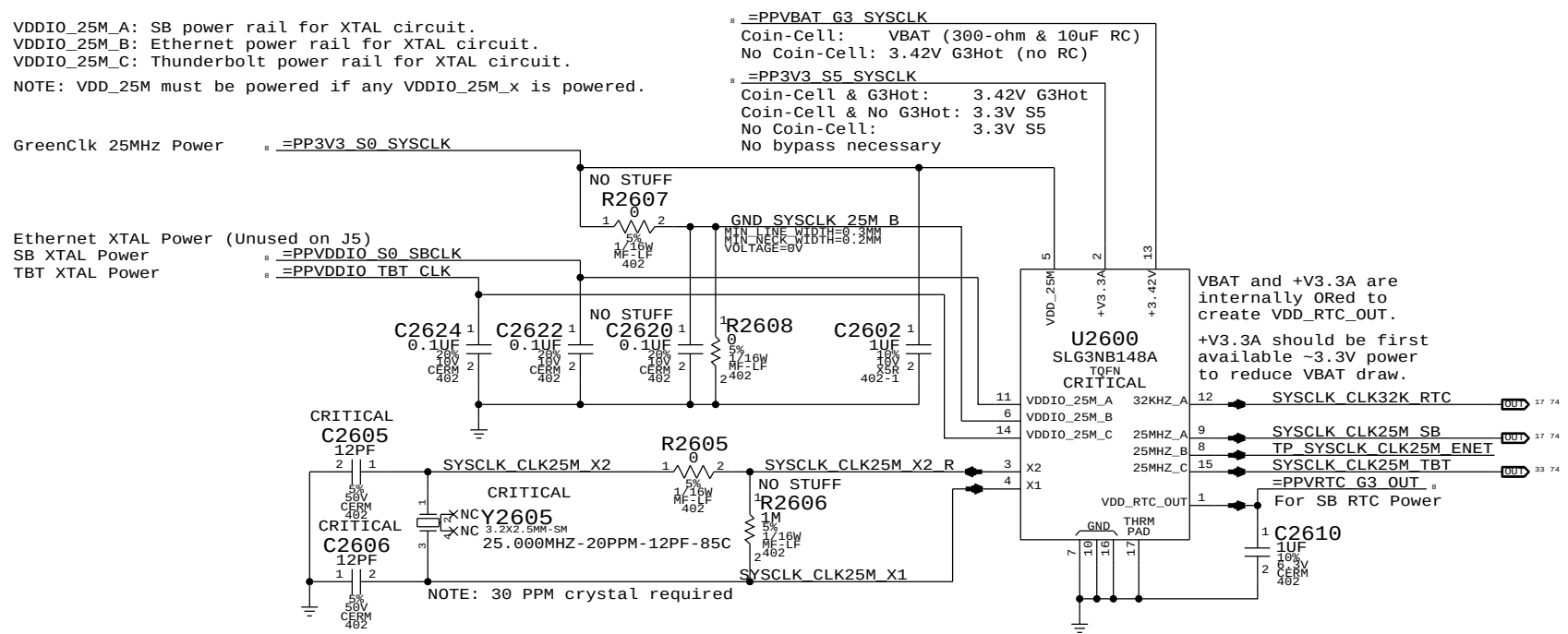
33 MHz Clock Series Termination



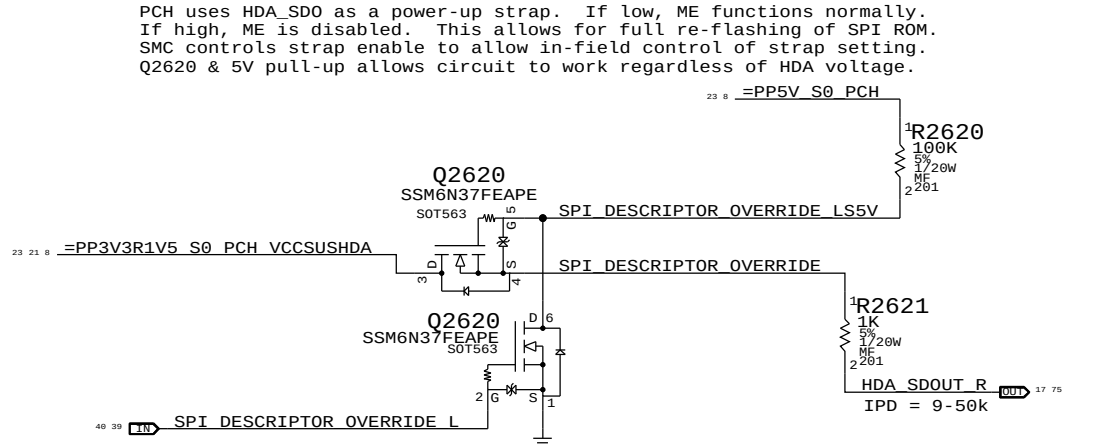
SDCONN_STATE_CHANGE ISOLATION



System RTC Power Source & 32kHz / 25MHz Clock Generator



PCH ME Disable Strap



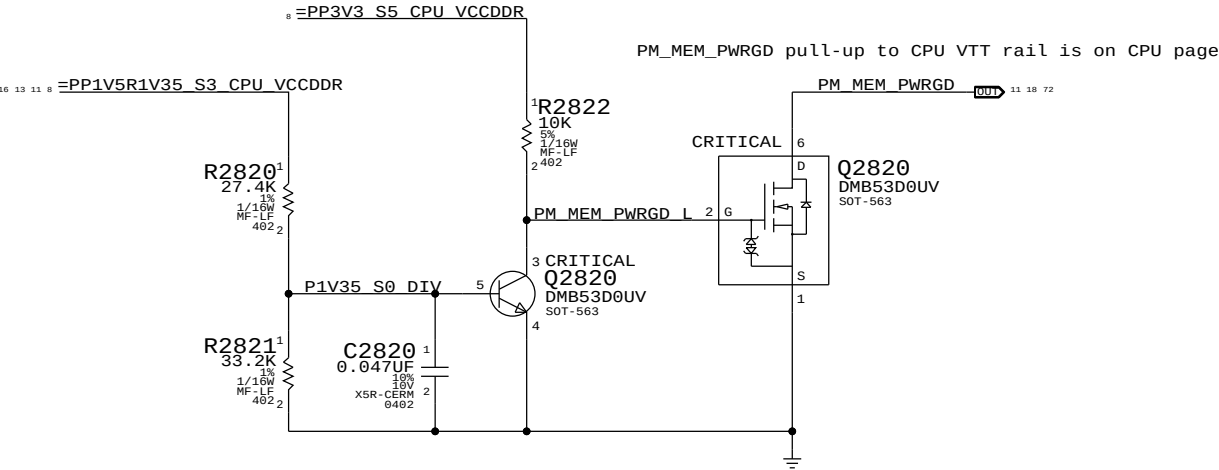
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

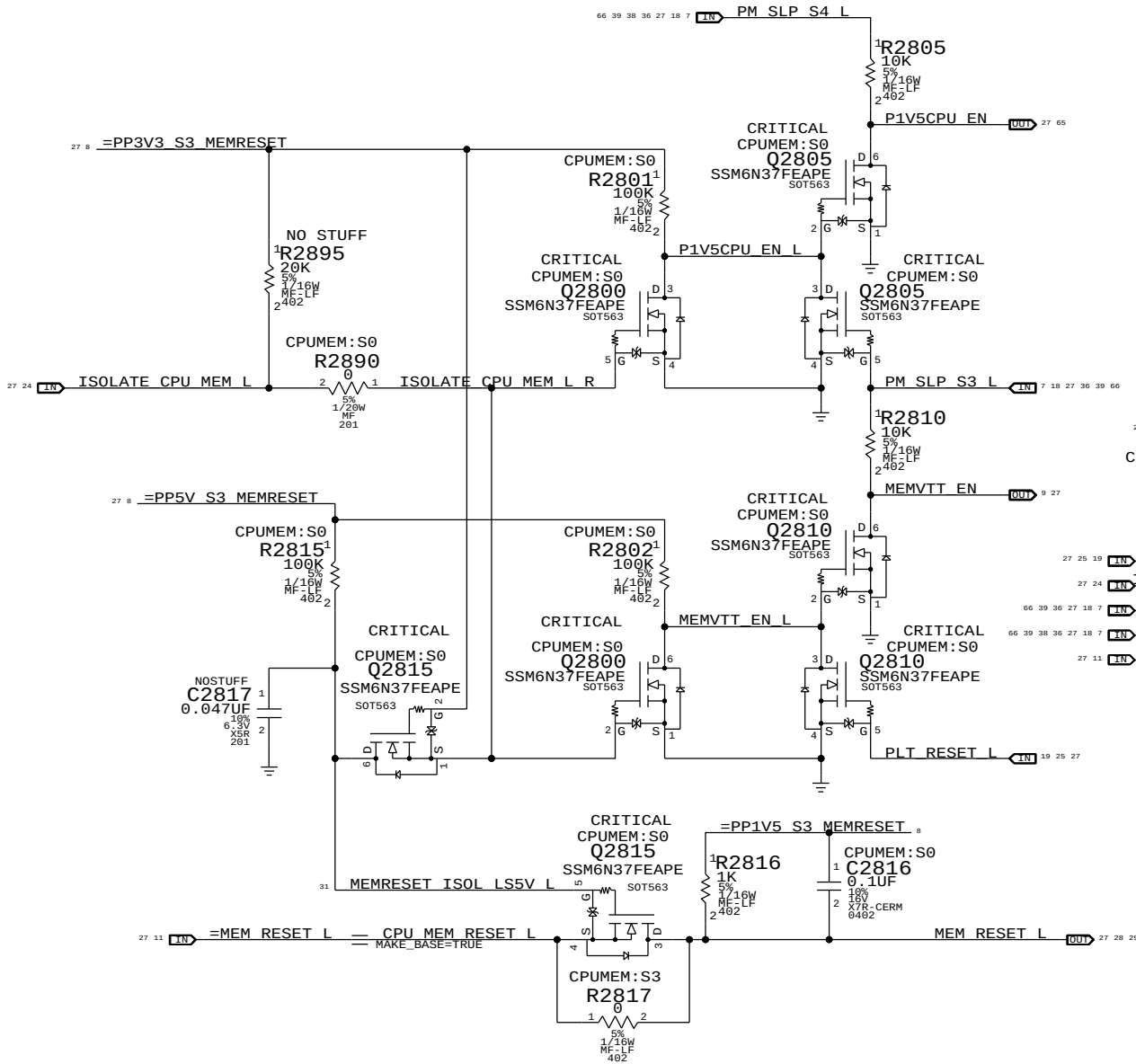
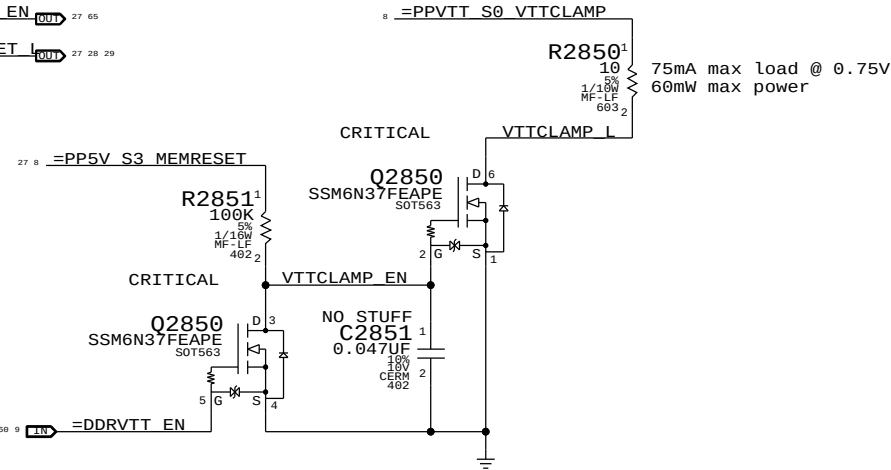
P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

1V35 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

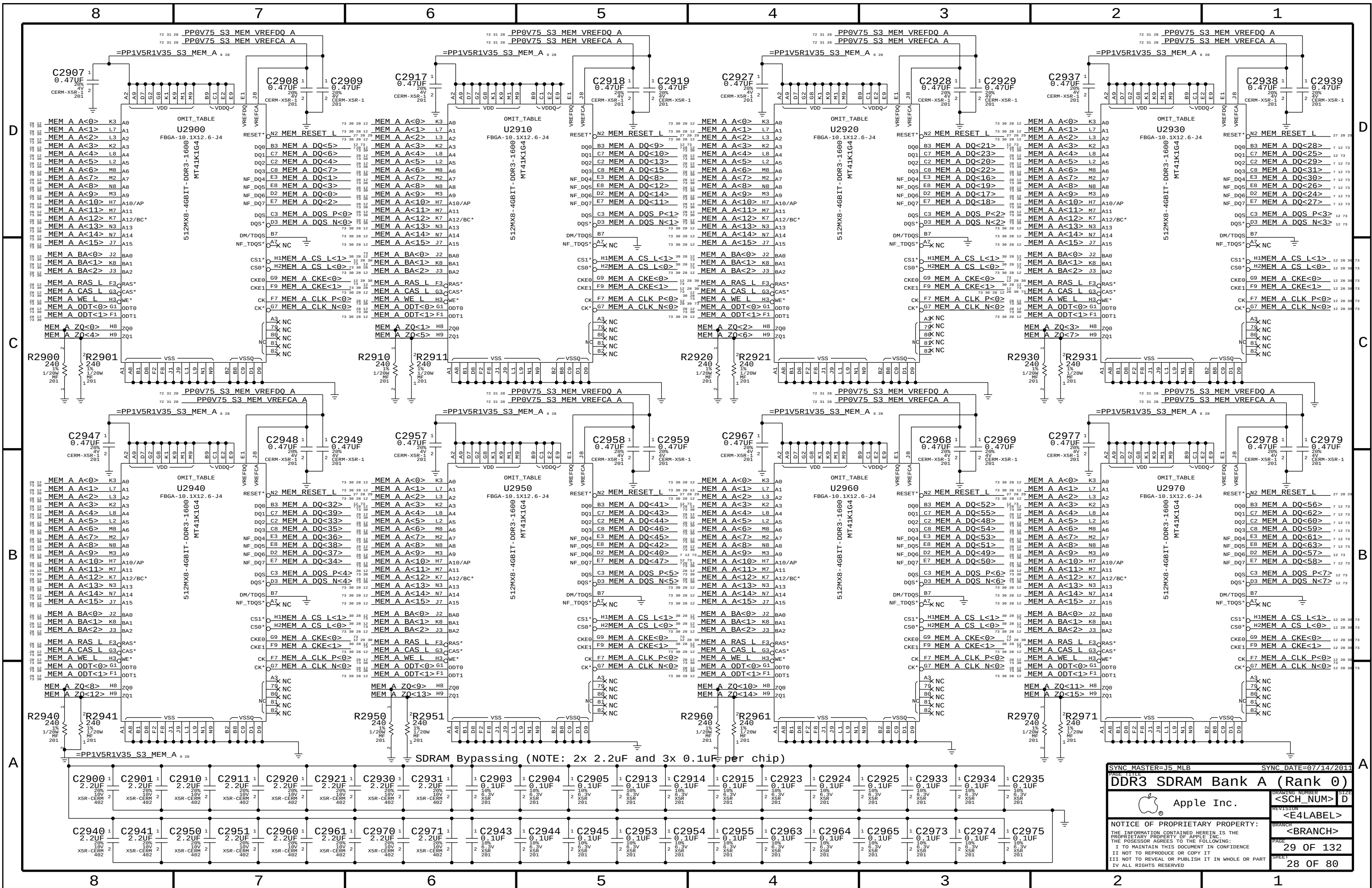


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	0
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

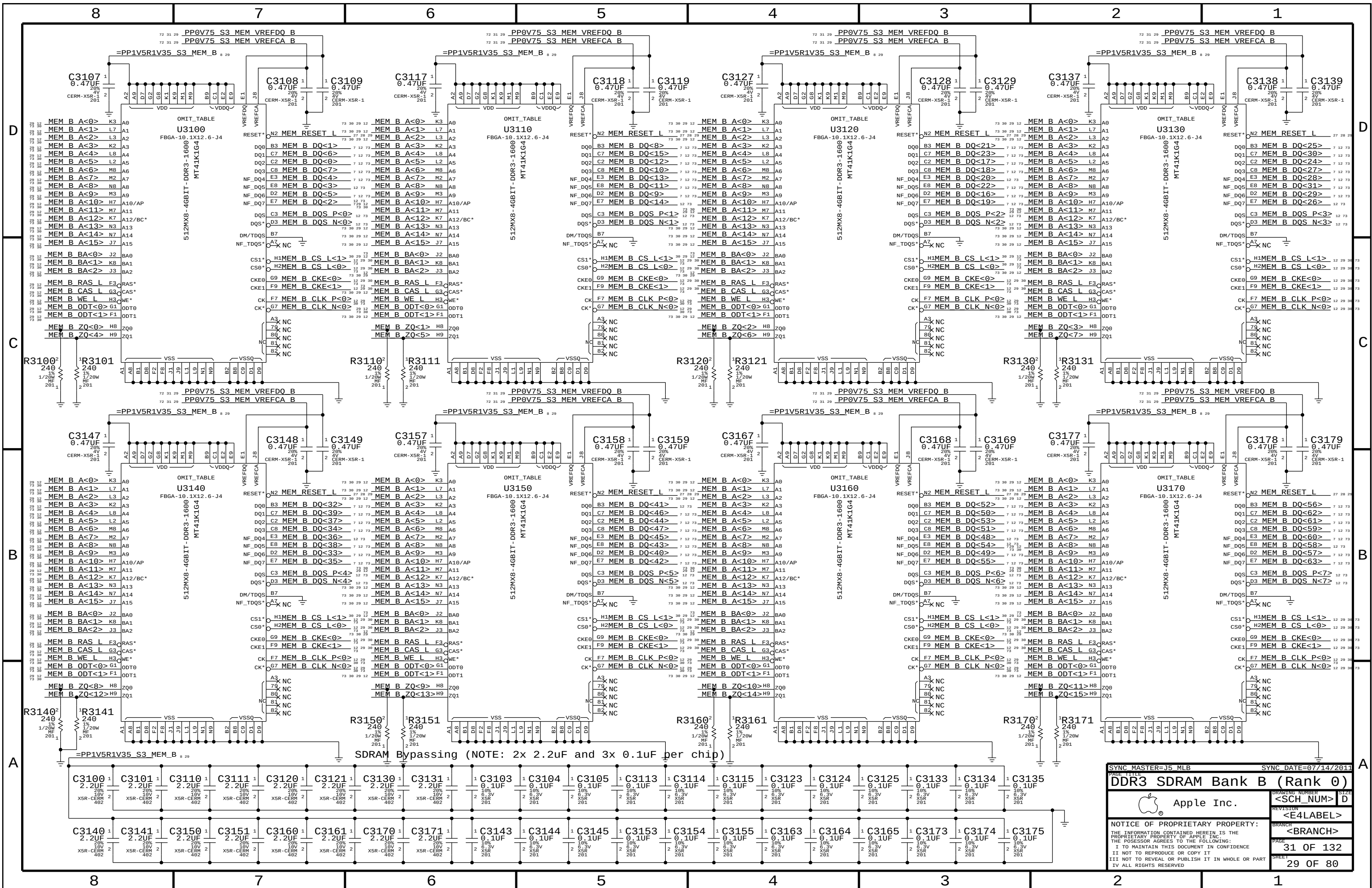
(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

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CPU Memory S3 Support			
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DDR3 SDRAM Bank A (Rank 0)		SIZE	
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SYNC MASTER=J5 MLB

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DDR3 SDRAM Bank B (Rank 0)

Apple Inc.

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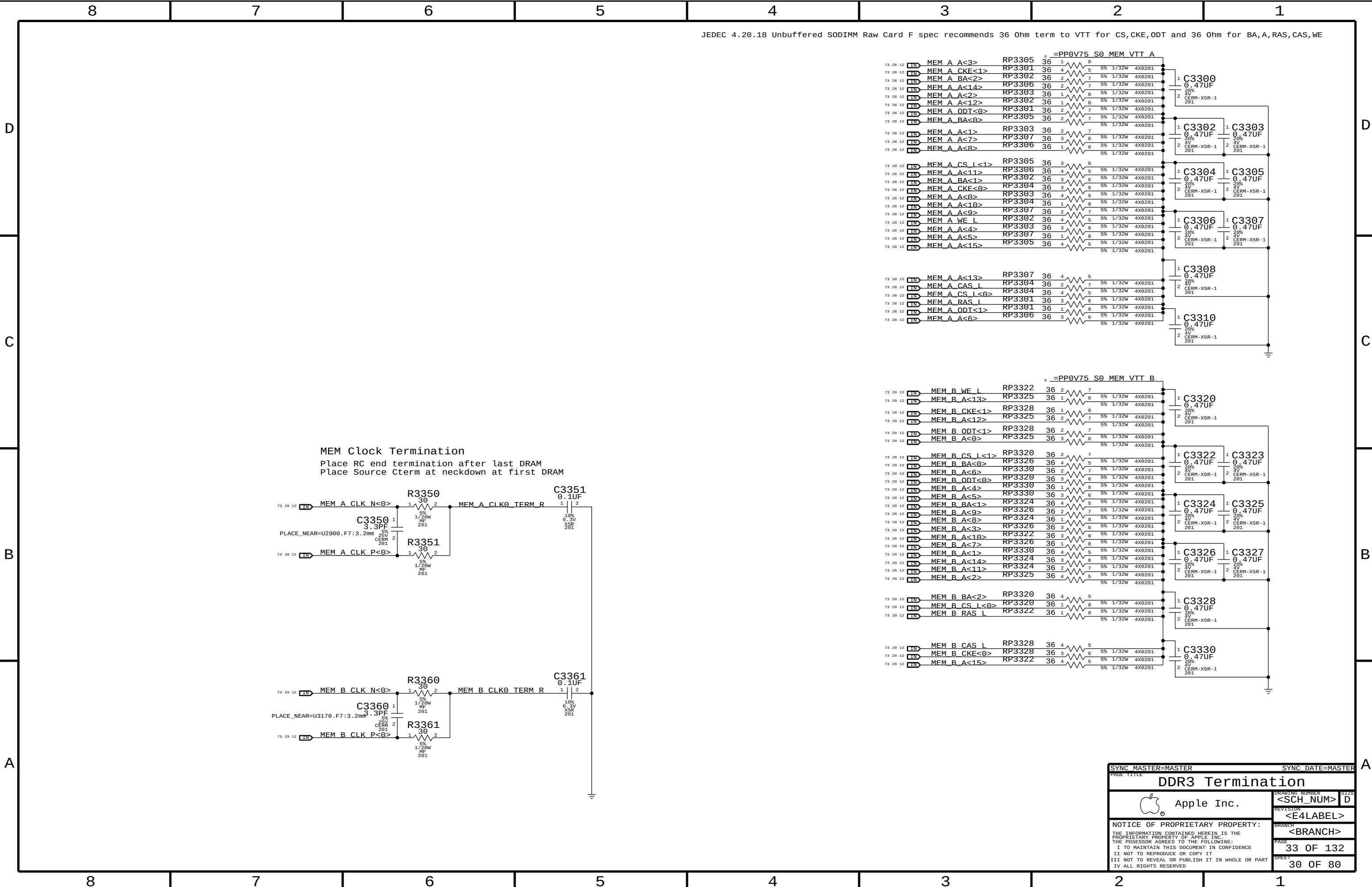
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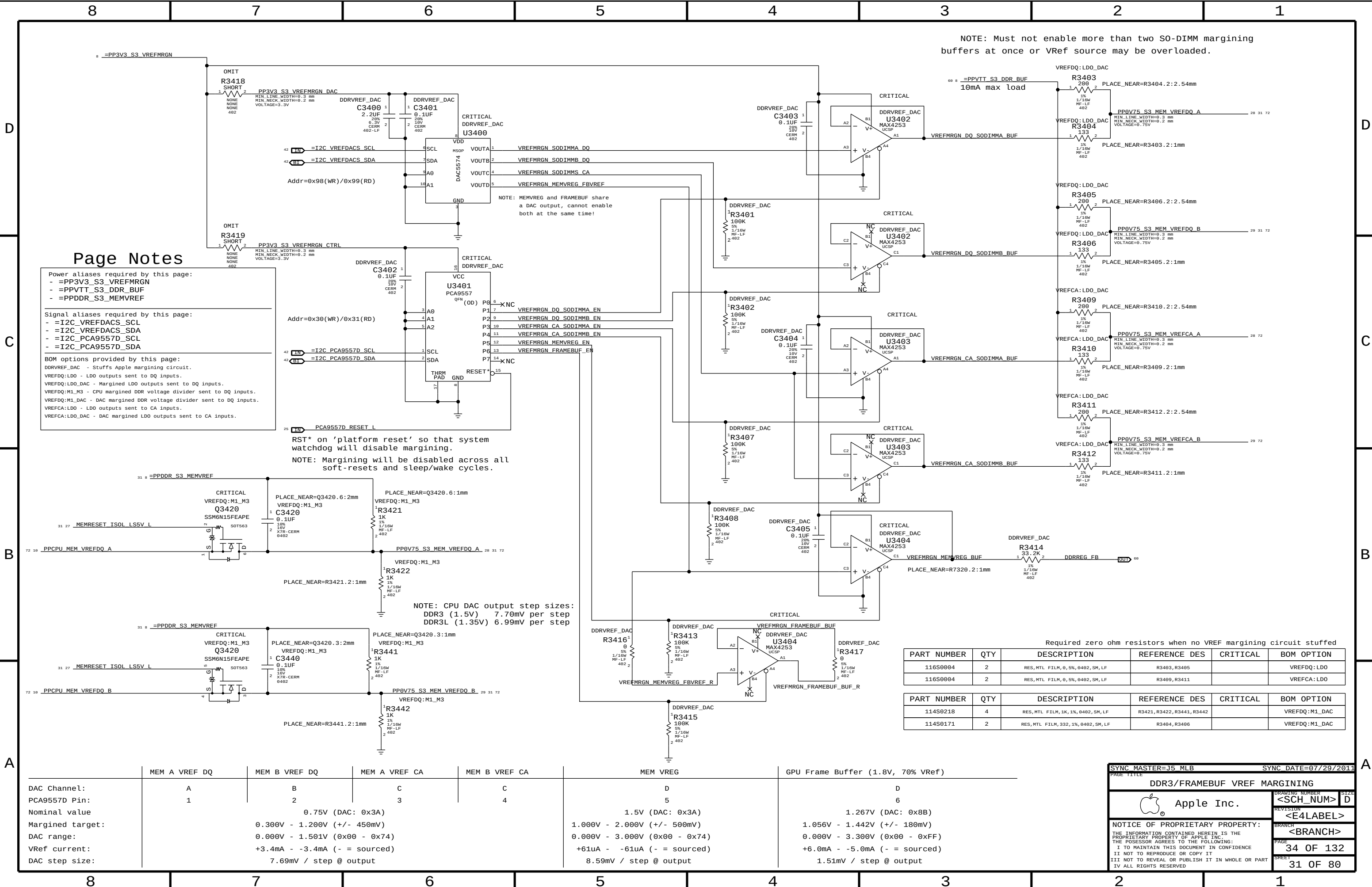
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Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACs_SCL
- =I2C_VREFDACs_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
DDRREF_DAC - Stuffs Apple margining circuit.
VREFDQ:LDO - LDO outputs sent to DQ inputs.
VREFDQ:LDO_DAC - Margined LDO outputs sent to DQ inputs.
VREFDQ:M1_M3 - CPU margined DDR voltage divider sent to DQ inputs.
VREFDQ:M1_DAC - DAC margined DDR voltage divider sent to DQ inputs.
VREFCA:LDO - LDO outputs sent to CA inputs.
VREFCA:LDO_DAC - DAC margined LDO outputs sent to CA inputs.

RST* on 'platform reset' so that system watchdog will disable margining.
NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3403, R3405		VREFDQ:LDO
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3409, R3411		VREFCA:LDO

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0218	4	RES, MTL FILM, 1K, 1%, 0402, SM, LF	R3421, R3422, R3441, R3442		VREFDQ:M1_DAC
114S0171	2	RES, MTL FILM, 332, 1%, 0402, SM, LF	R3404, R3406		VREFDQ:M1_DAC

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

SYNC MASTER=J5 MLB

SYNC DATE=07/29/2011

DDR3/FRAMEBUF VREF MARGINING

Apple Inc.

<SCH_NUM>

D

<E4LABEL>

REVISION

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SHEET

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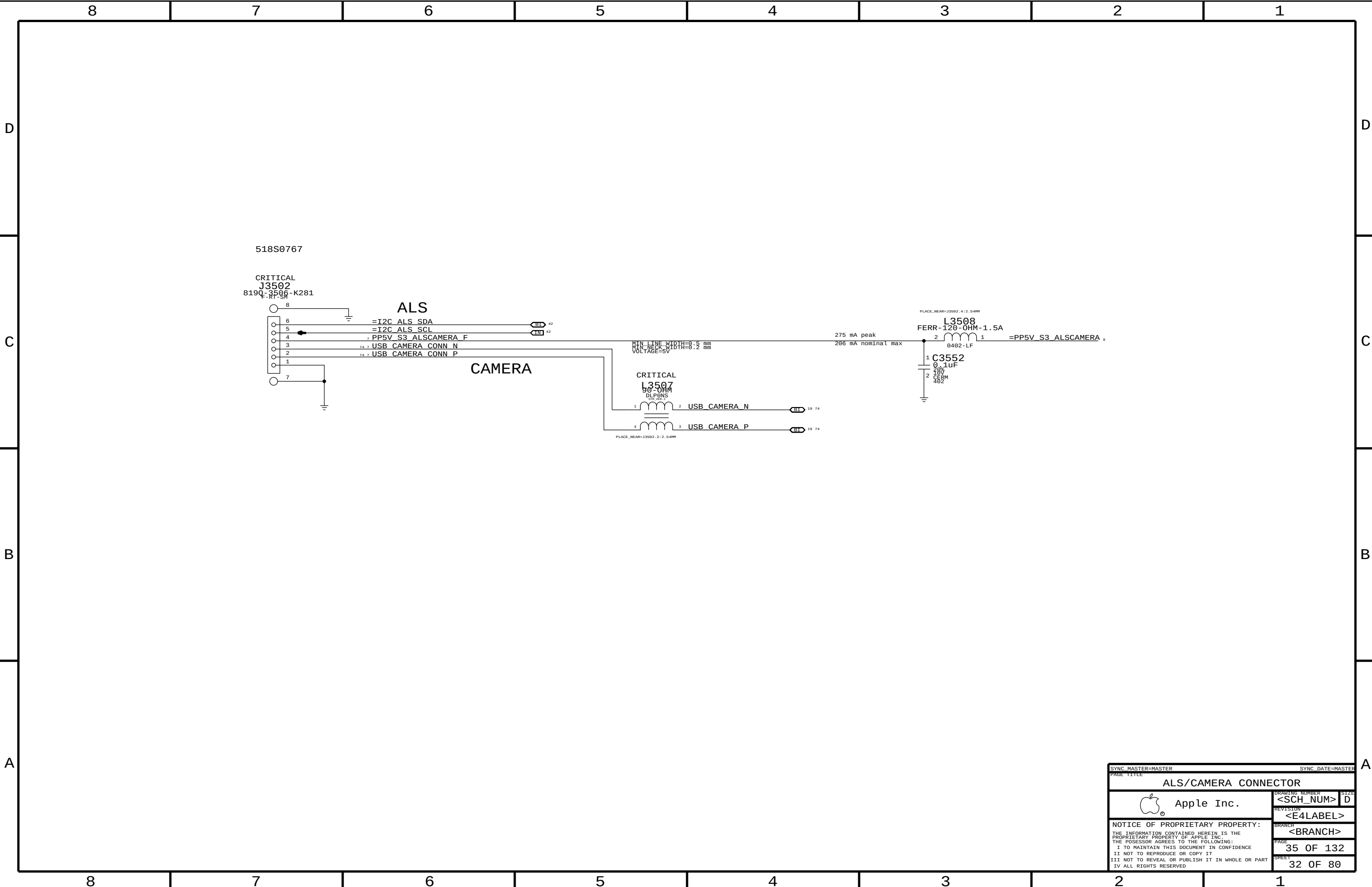
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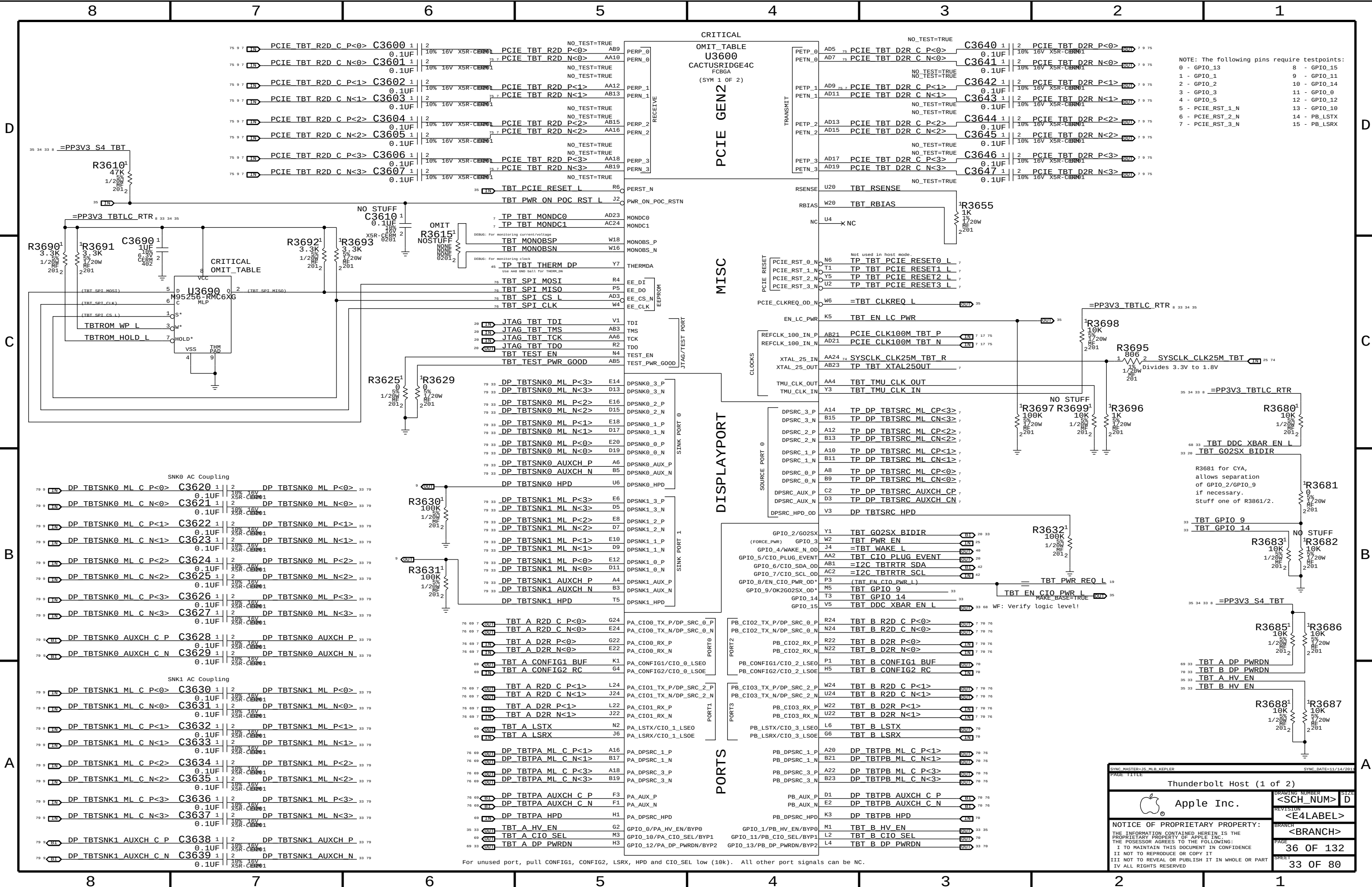
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PAGE TITLE			
ALS/CAMERA CONNECTOR			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
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		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	35 OF 132
		SHEET	32 OF 80



NOTE: The following pins require testpoints:

0 - GPIO_13	8 - GPIO_15
1 - GPIO_1	9 - GPIO_11
2 - GPIO_2	10 - GPIO_14
3 - GPIO_3	11 - GPIO_0
4 - GPIO_5	12 - GPIO_12
5 - PCIE_RST_1_N	13 - GPIO_10
6 - PCIE_RST_2_N	14 - PB_LSTX
7 - PCIE_RST_3_N	15 - PB_LSRX

SYNC MASTER=J5 MLB KEPLER

SYNC DATE=11/14/2011

Thunderbolt Host (1 of 2)

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For unused port, pull CONFIG1, CONFIG2, HPD and CIO_SEL low (10k). All other port signals can be NC.

Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBTLC_FET (3.3V FET Output)
- =PP3V3_S0_TBT_PWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBTLC_FET (1.05V FET Output)

Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

Thunderbolt 15V Boost Regulator

SI8409DB:

Vds(max): -30V
Vgs(max): +/-12V
Vgs(th): -1.4V
Rds(on): 46mOhm @ 4.5V Vgs
Id(max): 3.7A @ 70C

CRITICAL
TBTBST:Y
Q3880
SI8409DB

=PPVIN_SW_TBTBST
8-13V Input
Changes required
for 2S.

CRITICAL
TBTBST:Y
L3895
3.3uH-6.5A

TBTBST BOOST
MIN_LINE_WIDTH=0.5 mm
MIN_NECK_WIDTH=0.25 mm
SWITCH_NOISE=TRUE
DIDT=TRUE

CRITICAL
TBTBST:Y
D3895
PDS540XF
PWRD15

=PP15V_TBT_REG_
Vout = 15.47V
Max Current = 2A?
Freq = 500KHz

UVLO(falling) = 1.22 * (R1 + R2) / R2
UVLO(rising) = UVLO(falling) + (2uA * R1)
UVLO = 4.55V (falling), 4.95 (rising)

Vout = 1.6V * (1 + Ra / Rb)

SGND shorted to
GND inside package,
no XW necessary.

Supervisor & CLKREQ# Isolation

TBT "POC" Power-up Reset

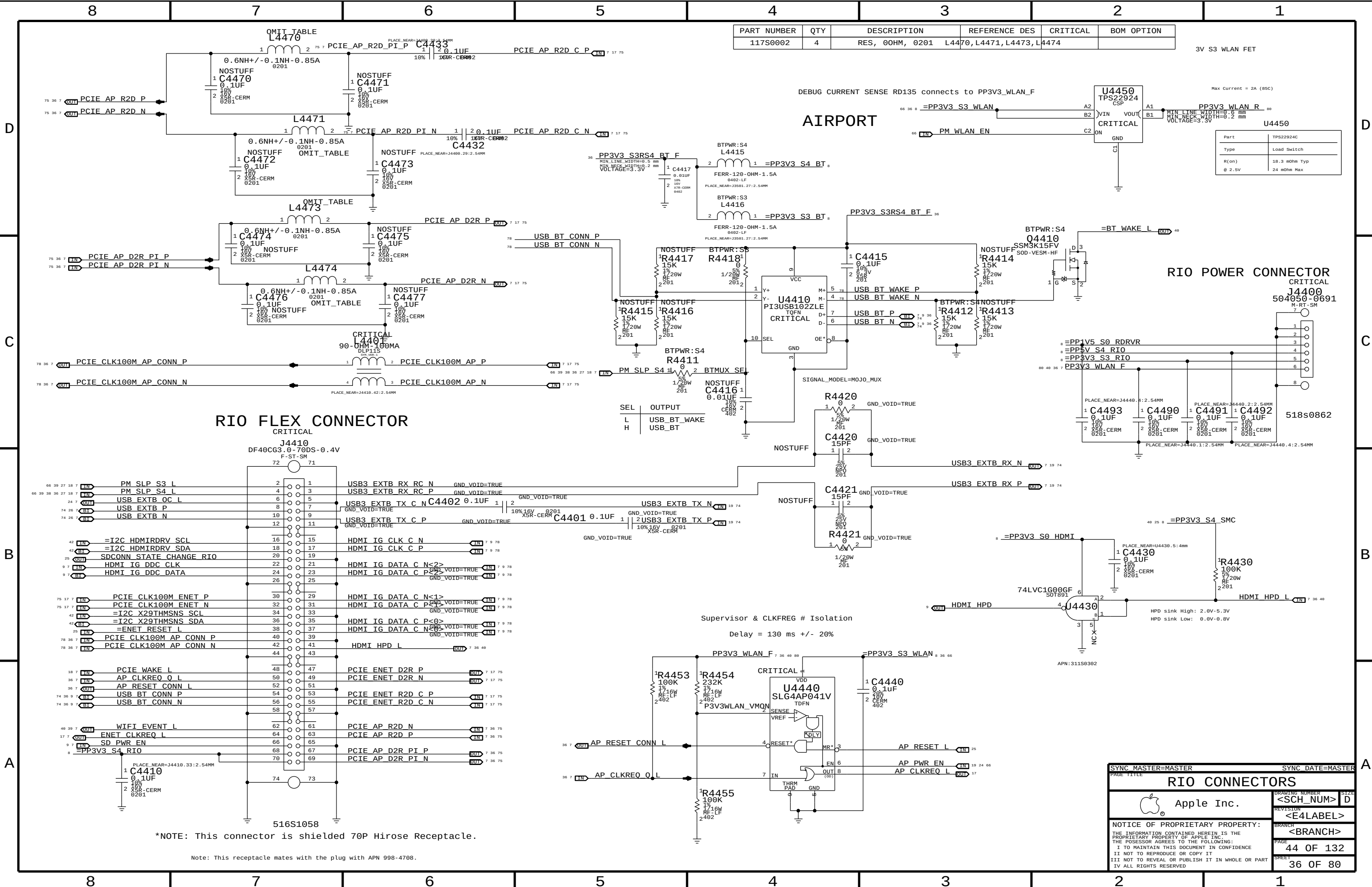
Intel investigating whether RC is sufficient.

1.05V TBT "CIO" Switch

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
13850811	4	CAP, CER, 4.7UF, 16%, 25V, X6S, 0603	C3895, C3897, C3898, C389A	CRITICAL	TBTBST:Y

SYNC MASTER=J5, M.B. KEPLER SYNC DATE=11/14/2011

Thunderbolt Power Support		SIZE
Apple Inc.	DRAWING NUMBER	D
	REVISION	<E4LABEL>
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	PAGE	38 OF 132
	SHEET	35 OF 80



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0002	4	RES, 00HM, 0201	L4470,L4471,L4473,L4474		

3V S3 WLAN FET	
Max Current = 2A (85C)	
Part	TPS22924C
Type	Load Switch
R(on)	18.3 mOhm Typ
@ 2.5V	24 mOhm Max

RIO POWER CONNECTOR	
CRITICAL	
J4400	
504050-0691	
M-RT-SM	
1	
2	
3	
4	
5	
6	
7	
8	

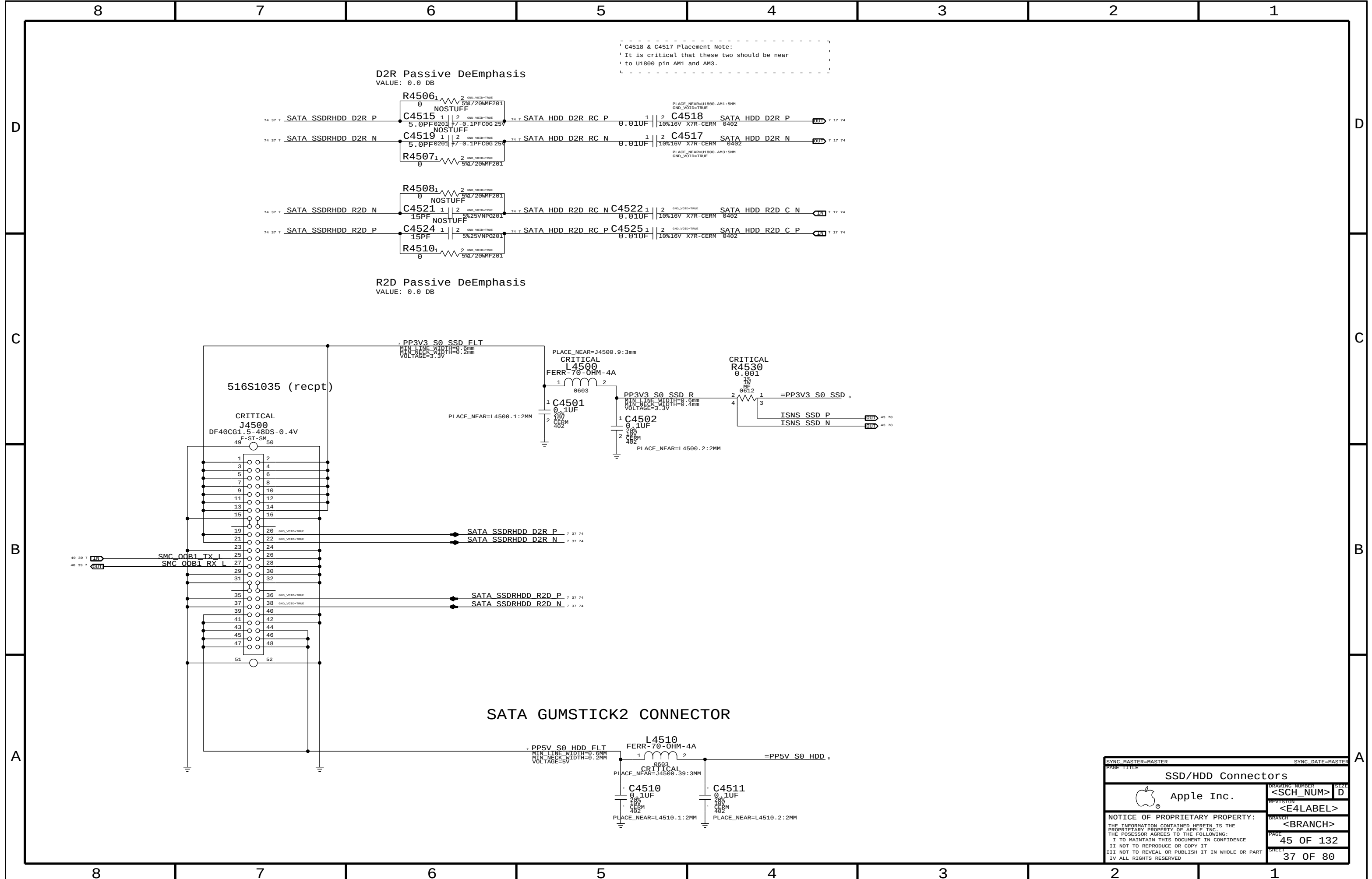
1	C4493	0.1UF	18V	XSR-CERM	0201
2	C4490	0.1UF	18V	XSR-CERM	0201
3	C4491	0.1UF	18V	XSR-CERM	0201
4	C4492	0.1UF	18V	XSR-CERM	0201

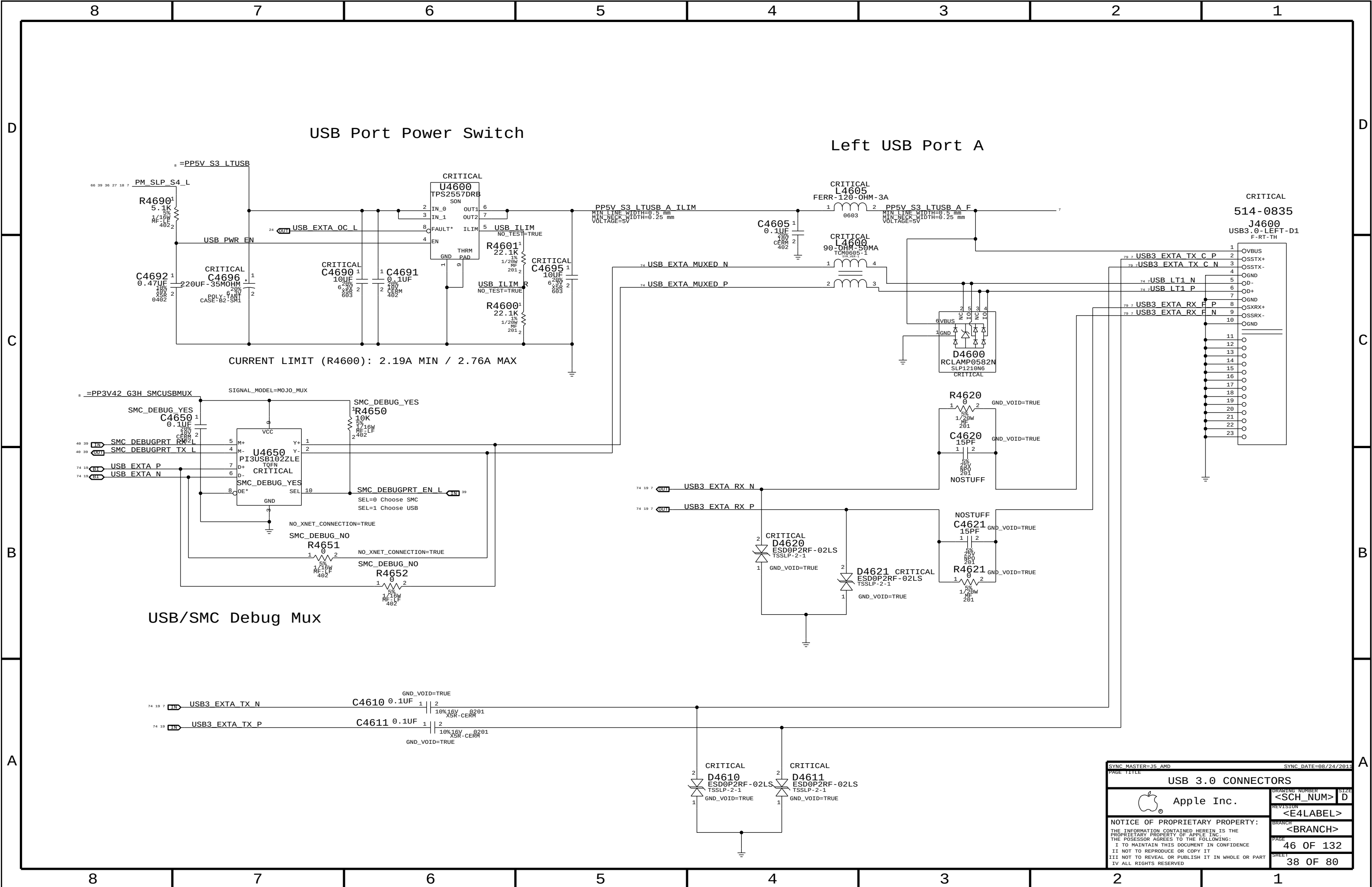
1	C4430	0.1UF	18V	XSR-CERM	0201
2	R4430	100K	1/20W	MF	201

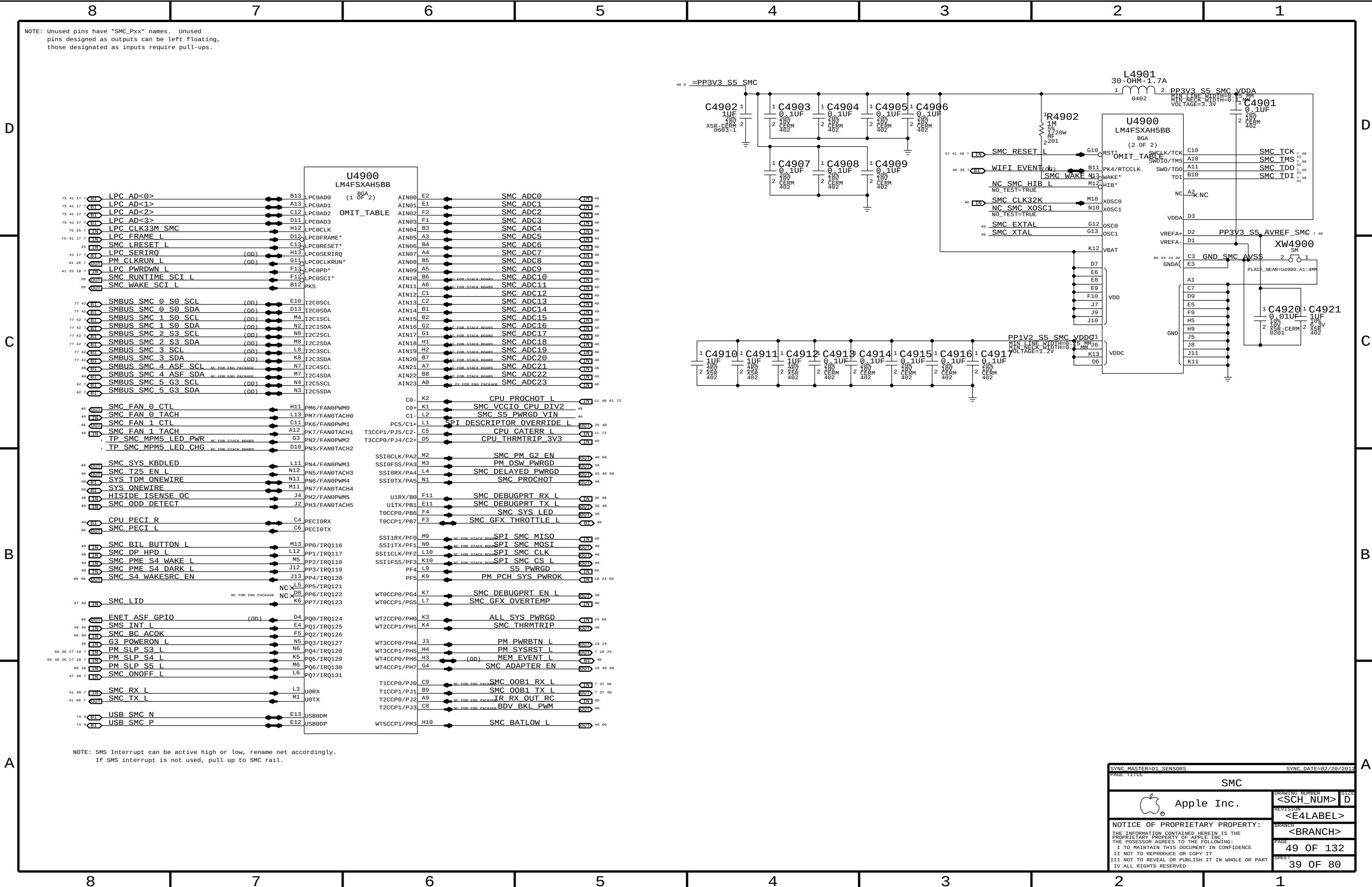
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PAGE TITLE		PAGE	
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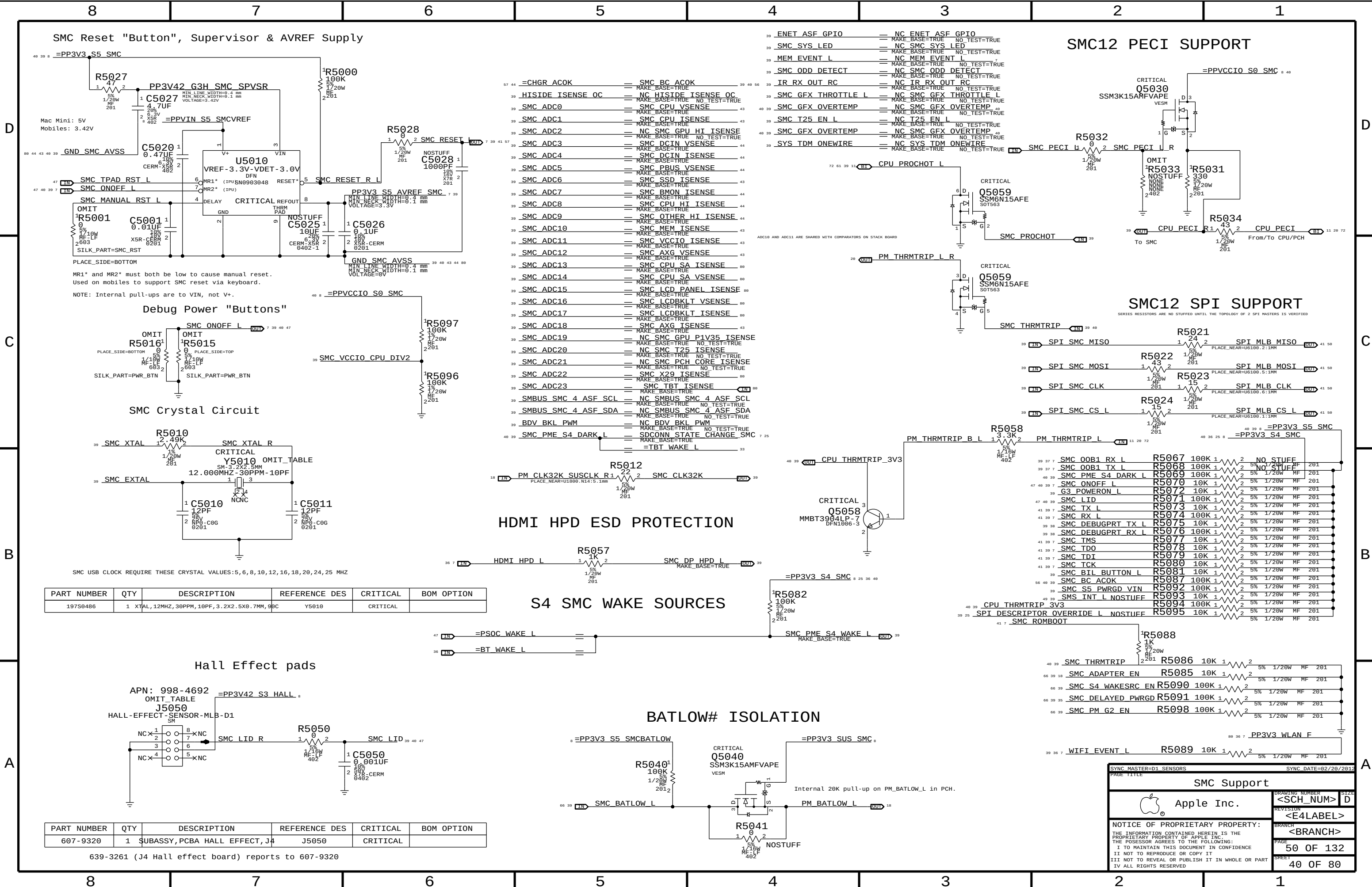
*NOTE: This connector is shielded 70P Hirose Receptacle.

Note: This receptacle mates with the plug with APN 998-4708.

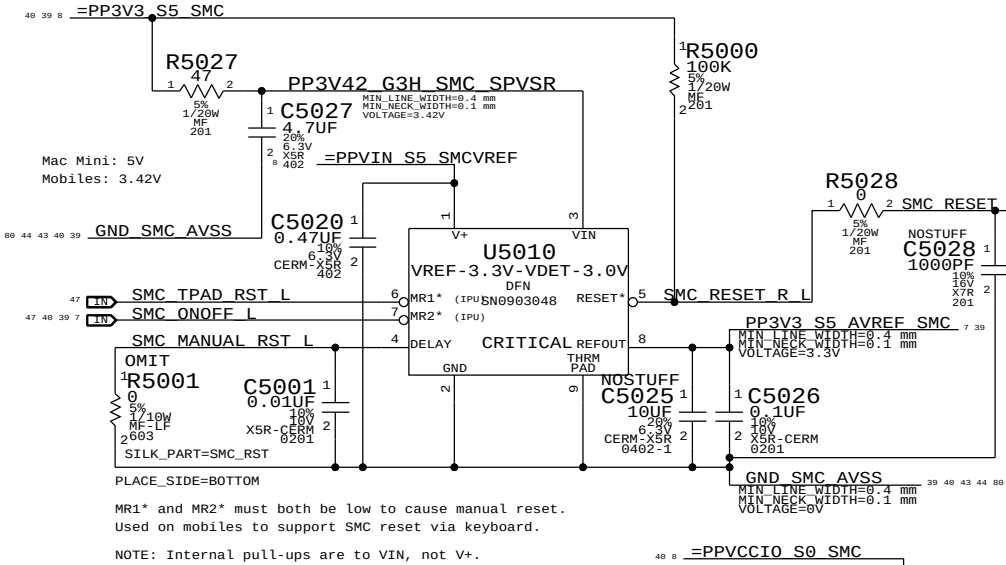




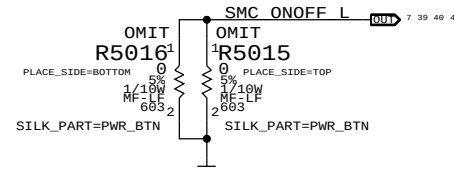




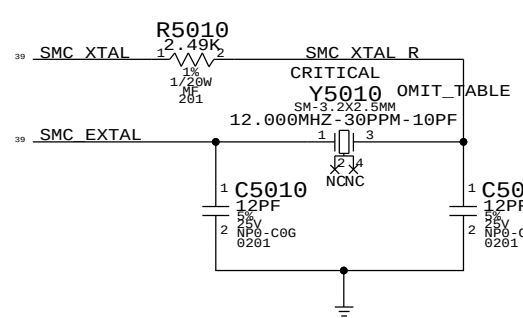
SMC Reset "Button", Supervisor & AVREF Supply



Debug Power "Buttons"



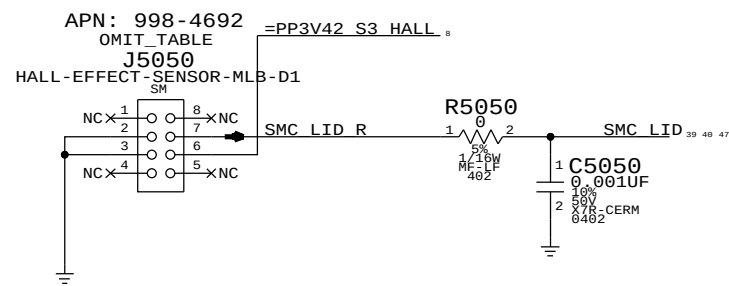
SMC Crystal Circuit



SMC USB CLOCK REQUIRE THESE CRYSTAL VALUES: 5, 6, 8, 10, 12, 16, 18, 20, 24, 25 MHZ

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
197S0486	1	XTAL, 12MHZ, 36PPM, 10PF, 3.2X2.5X0.7MM, 99C	Y5010	CRITICAL	

Hall Effect pads

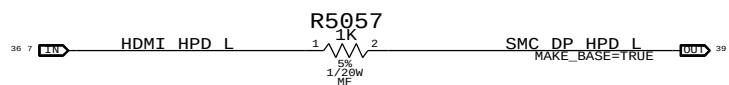


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-9320	1	SUBASSY, PCBA HALL EFFECT, J4	J5050	CRITICAL	

639-3261 (J4 Hall effect board) reports to 607-9320

- =CHGR ACOK
- HISIDE ISENSE OC
- SMC ADC0
- SMC ADC1
- SMC ADC2
- SMC ADC3
- SMC ADC4
- SMC ADC5
- SMC ADC6
- SMC ADC7
- SMC ADC8
- SMC ADC9
- SMC ADC10
- SMC ADC11
- SMC ADC12
- SMC ADC13
- SMC ADC14
- SMC ADC15
- SMC ADC16
- SMC ADC17
- SMC ADC18
- SMC ADC19
- SMC ADC20
- SMC ADC21
- SMC ADC22
- SMC ADC23
- SMBUS SMC 4 ASF SCL
- SMBUS SMC 4 ASF SDA
- BDV BKL PWM
- SMC PME S4 DARK L
- SMC BC ACOK
- NC HISIDE ISENSE OC
- SMC CPU VSENSE
- SMC CPU ISENSE
- NC SMC GPU HI ISENSE
- SMC DCIN VSENSE
- SMC DCIN ISENSE
- SMC PBUS VSENSE
- SMC SSD ISENSE
- SMC CPU HI ISENSE
- SMC OTHER HI ISENSE
- SMC MEM ISENSE
- SMC VCCIO ISENSE
- SMC AXG VSENSE
- SMC CPU SA ISENSE
- SMC CPU SA VSENSE
- SMC LCD PANEL ISENSE
- SMC LCDPKLT VSENSE
- SMC LCDPKLT ISENSE
- SMC AXG ISENSE
- NC SMC GPU P1V35 ISENSE
- NC SMC T25 ISENSE
- NC SMC PCH CORE ISENSE
- SMC X29 ISENSE
- SMC TBT ISENSE
- NC SMBUS SMC 4 ASF SCL
- NC SMBUS SMC 4 ASF SDA
- NC BDV BKL PWM
- SDCONN STATE CHANGE SMC
- =TBT WAKE L

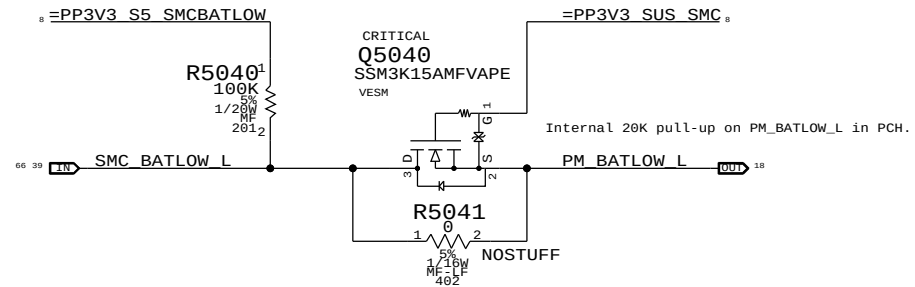
HDMI HPD ESD PROTECTION



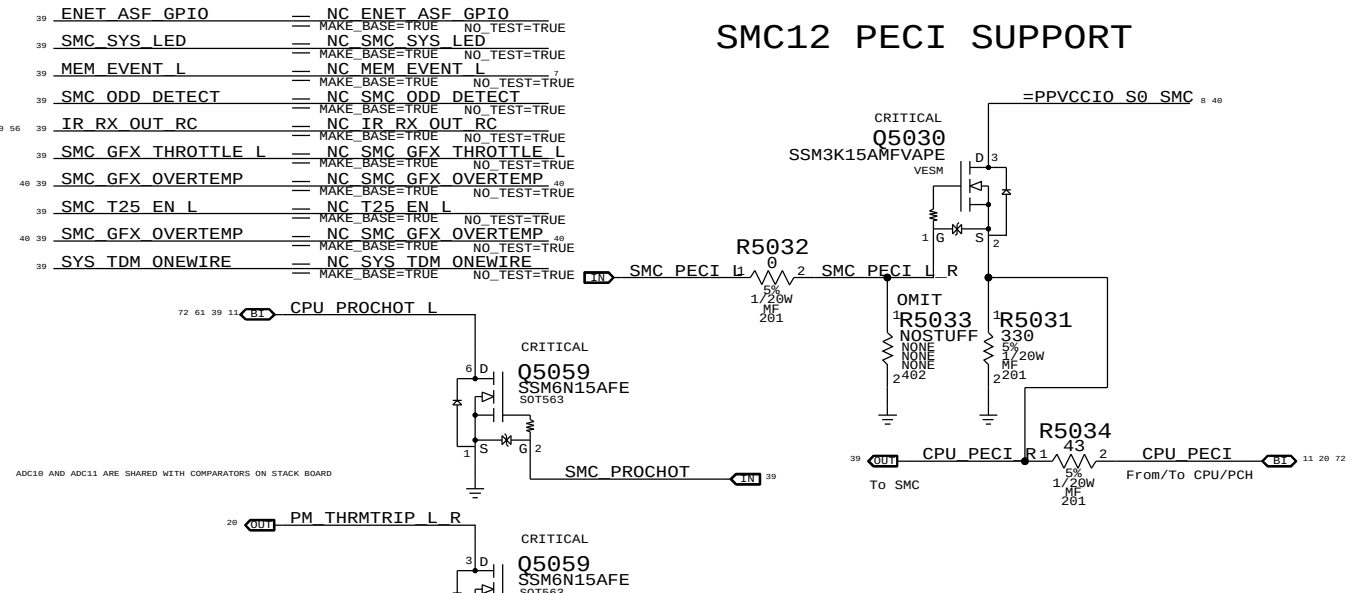
S4 SMC WAKE SOURCES



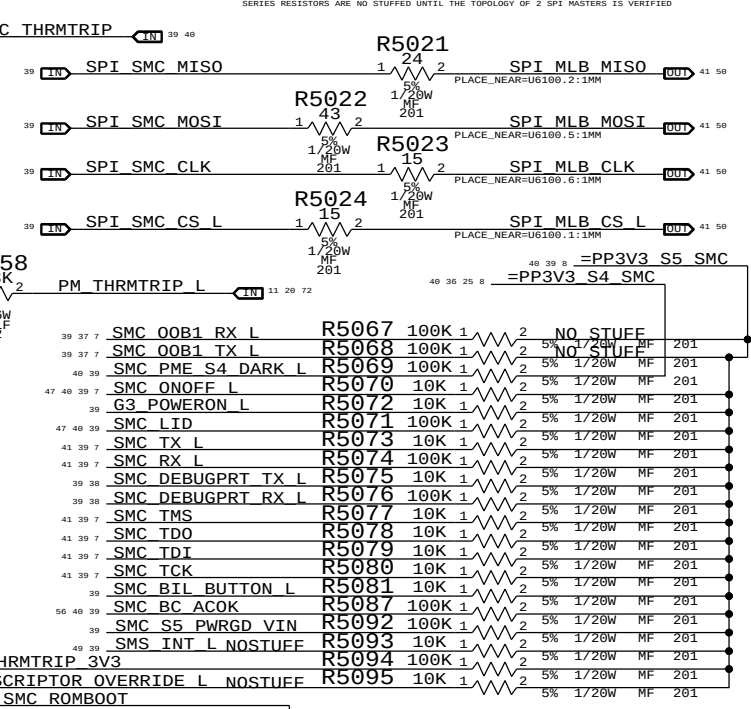
BATLOW# ISOLATION



SMC12 PECE SUPPORT



SMC12 SPI SUPPORT



SYNC MASTER=D1 SENSORS

SYNC DATE=02/28/2012

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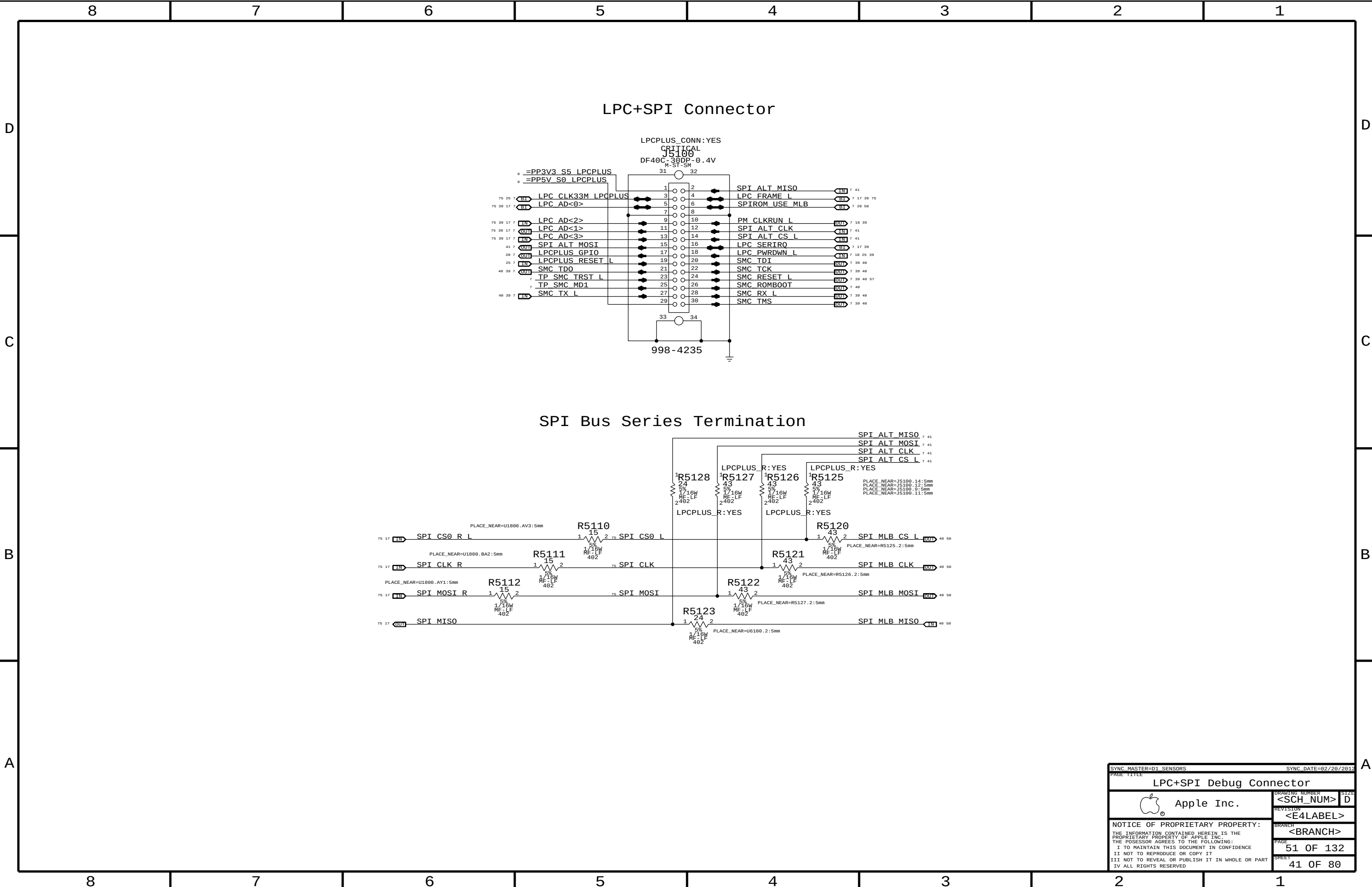
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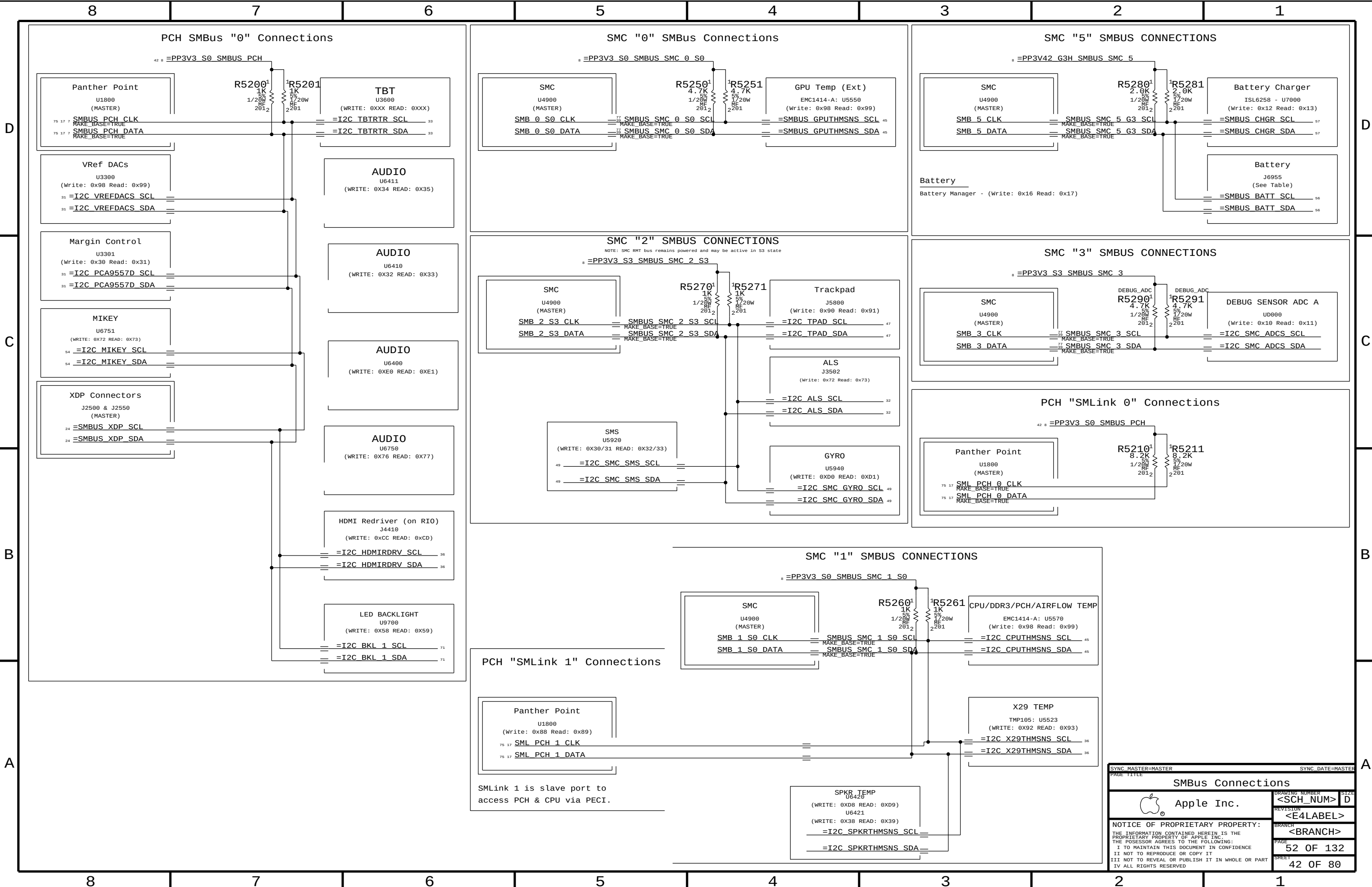
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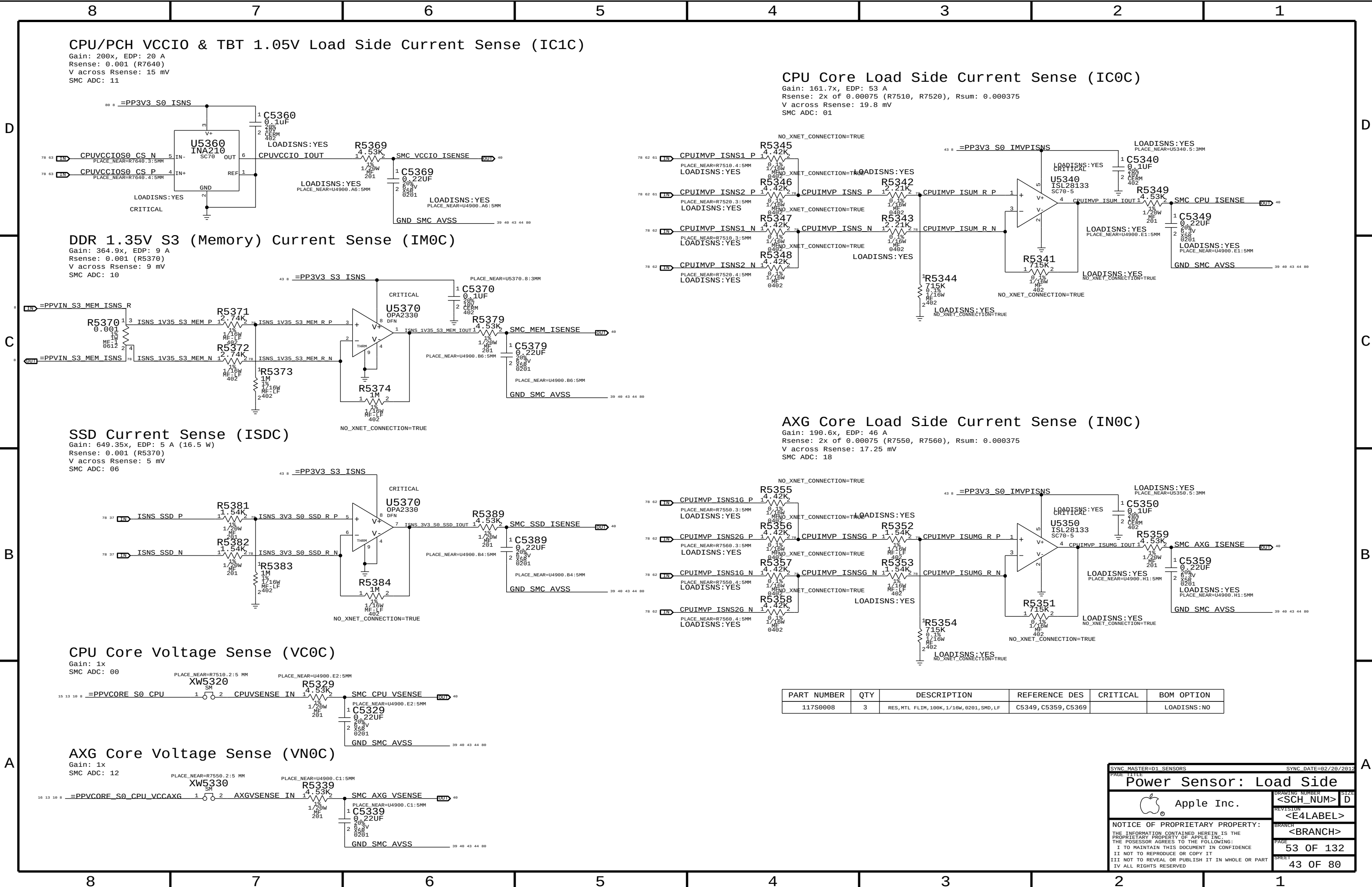
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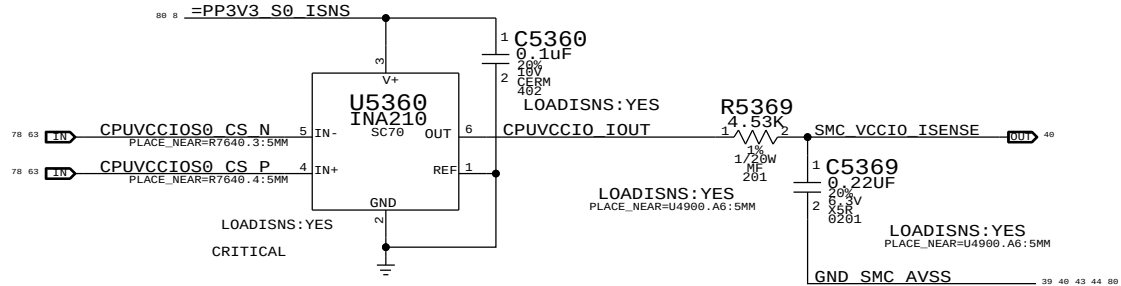






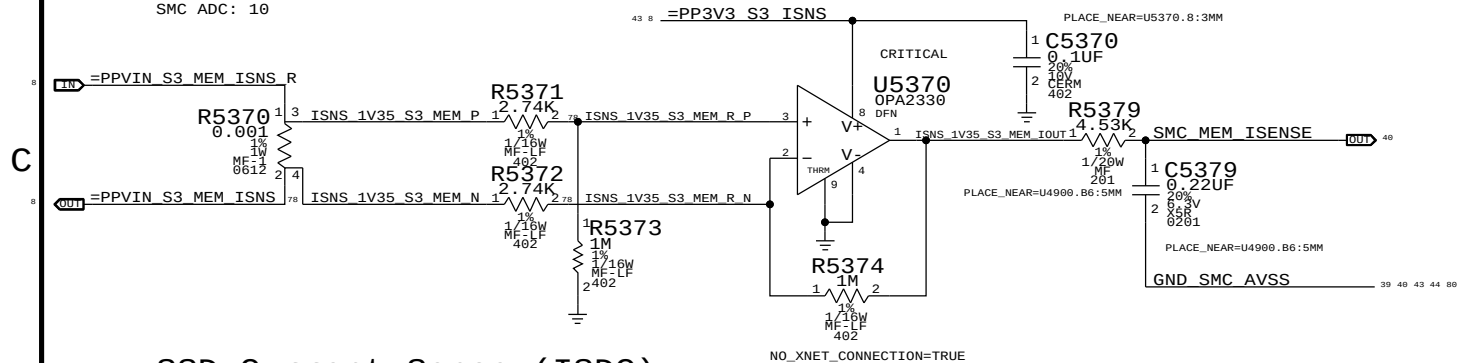
CPU/PCH VCCIO & TBT 1.05V Load Side Current Sense (IC1C)

Gain: 200x, EDP: 20 A
Rsense: 0.001 (R7640)
V across Rsense: 15 mV
SMC ADC: 11



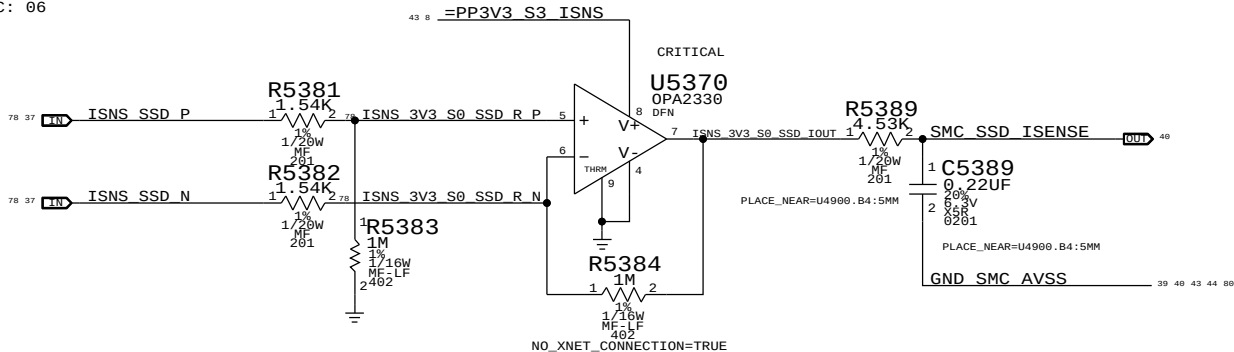
DDR 1.35V S3 (Memory) Current Sense (IM0C)

Gain: 364.9x, EDP: 9 A
Rsense: 0.001 (R5370)
V across Rsense: 9 mV
SMC ADC: 10



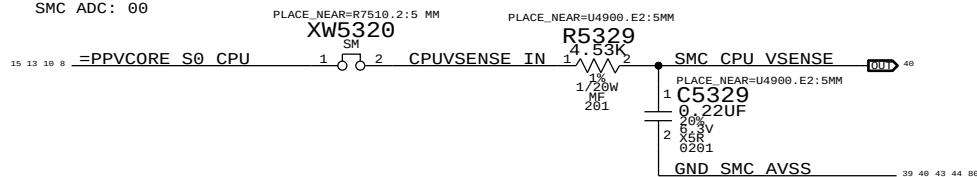
SSD Current Sense (ISDC)

Gain: 649.35x, EDP: 5 A (16.5 W)
Rsense: 0.001 (R5370)
V across Rsense: 5 mV
SMC ADC: 06



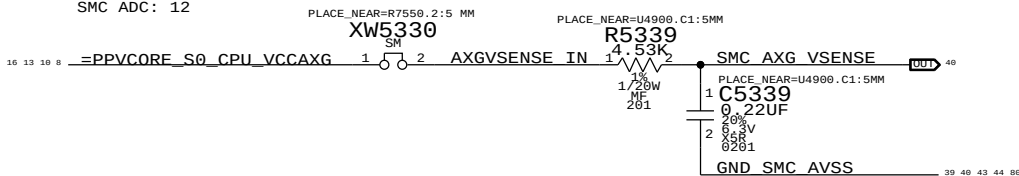
CPU Core Voltage Sense (VC0C)

Gain: 1x
SMC ADC: 00



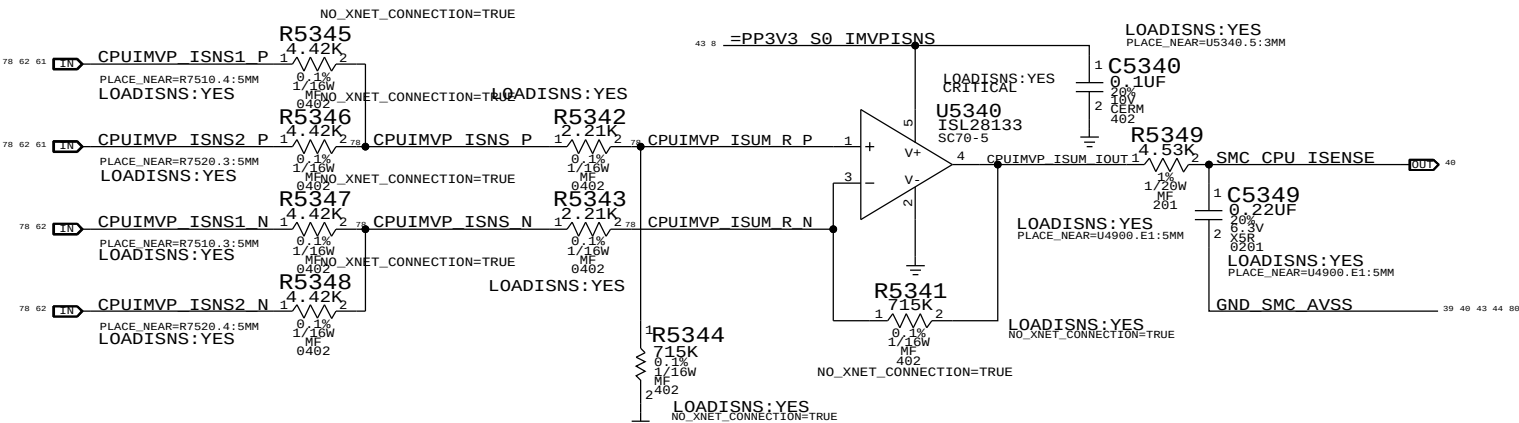
AXG Core Voltage Sense (VN0C)

Gain: 1x
SMC ADC: 12



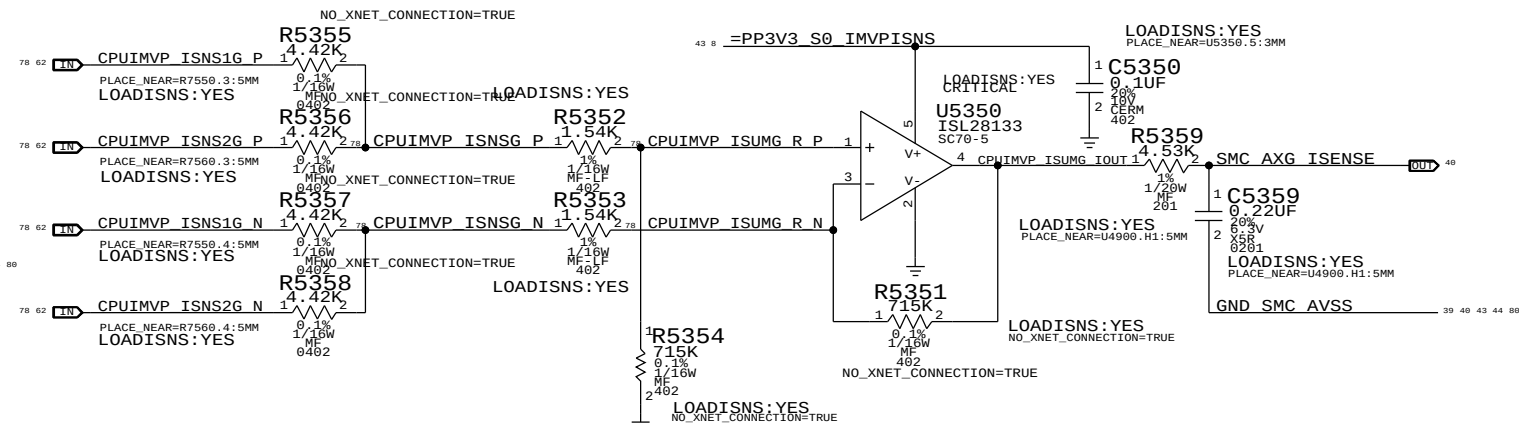
CPU Core Load Side Current Sense (IC0C)

Gain: 161.7x, EDP: 53 A
Rsense: 2x of 0.00075 (R7510, R7520), Rsum: 0.000375
V across Rsense: 19.8 mV
SMC ADC: 01



AXG Core Load Side Current Sense (IN0C)

Gain: 190.6x, EDP: 46 A
Rsense: 2x of 0.00075 (R7550, R7560), Rsum: 0.000375
V across Rsense: 17.25 mV
SMC ADC: 18

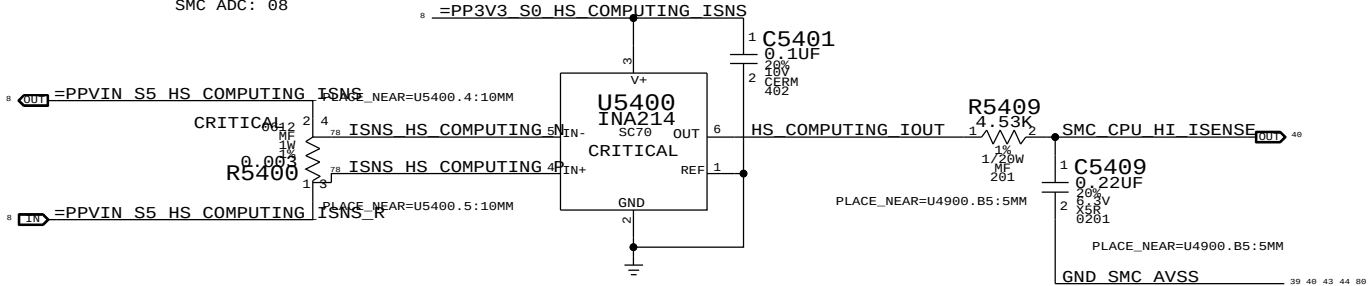


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	3	RES, MTL FLIM, 100K, 1/16W, 0201, SMD, LF	C5349, C5359, C5369		LOADISNS:NO

SYNC MASTER=D1 SENSORS		SYNC DATE=02/28/2012	
PAGE TITLE			
Power Sensor: Load Side			
Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
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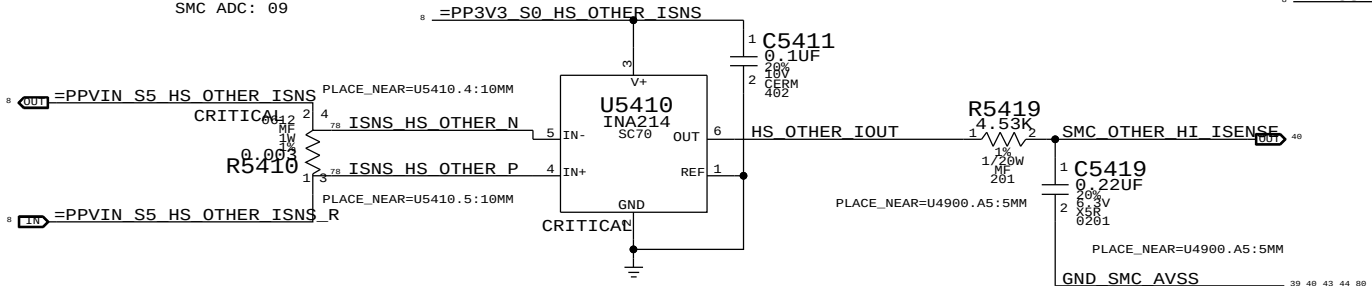
CPU High Side Current Sense (IC0R)

Gain: 50x, EDP: 17.4 A
Rsense: 0.003 (R5400)
V across Rsense: 52.2 mV
SMC ADC: 08



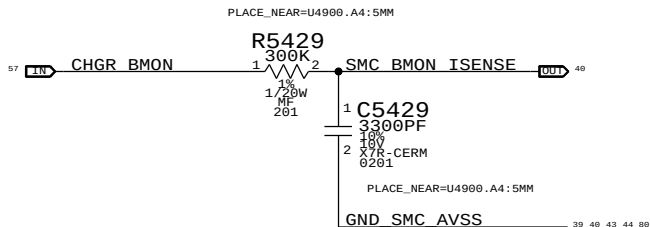
OTHER High Side Current Sense (IO0R)

Gain: 100x, EDP: 8.8 A
Rsense: 0.003 (R5410)
V across Rsense: 26.4 mV
SMC ADC: 09



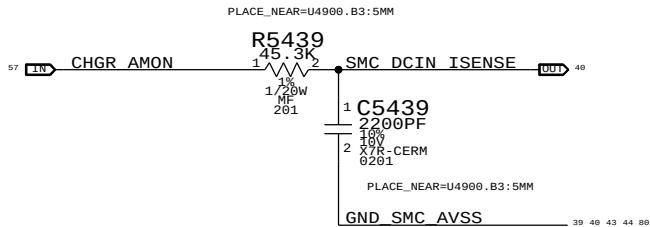
Charger (BMON Production) Current Sense (IPBR)

Charger Gain: 36x, EDP: 6.6 A
Rsense: 0.010 (R7050)
SMC ADC: 07



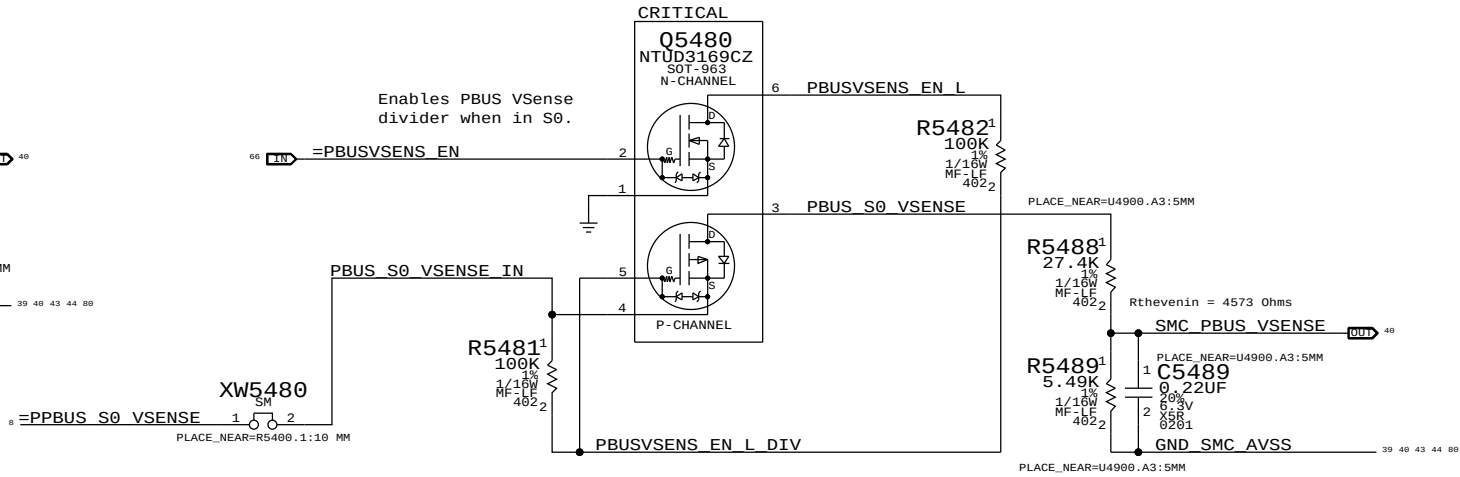
DC-In (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
Rsense: 0.020 (R7020)
SMC ADC: 04



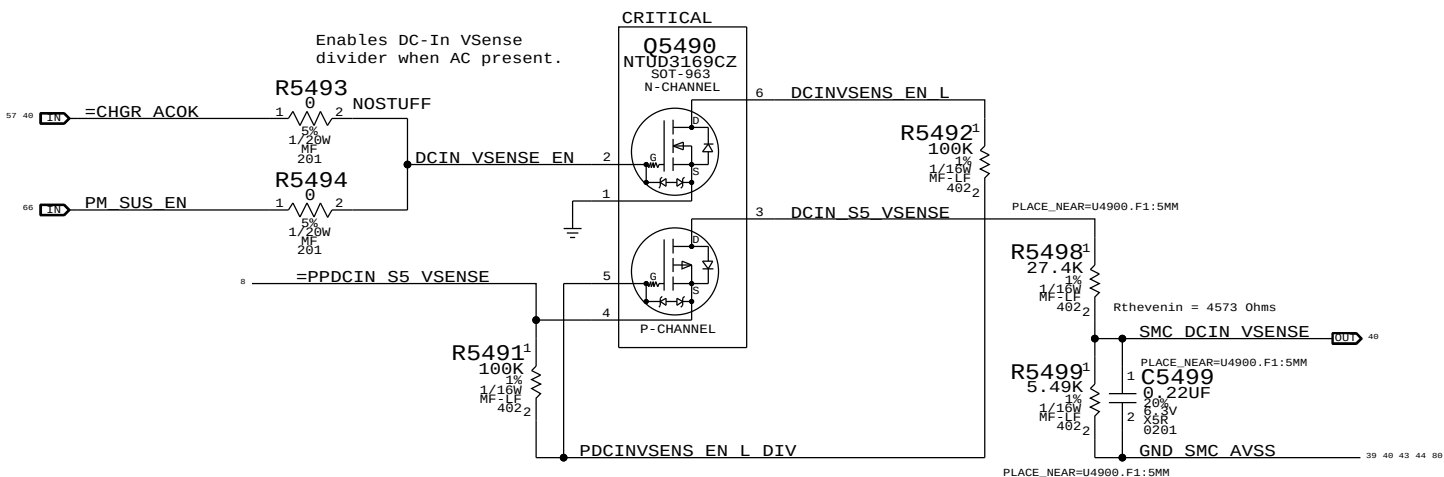
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
SMC ADC: 05



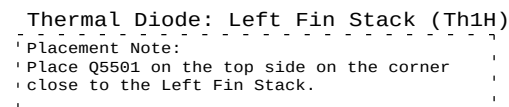
DC In Voltage Sense & Enable (VD0R)

Gain: 0.167x
SMC ADC: 03

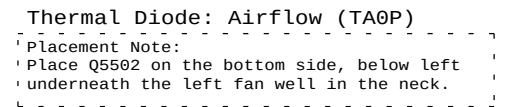


SYNC MASTER=D1 SENSORS		SYNC DATE=02/20/2012	
PAGE TITLE			
Power Sensor: High Side			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	54 OF 132
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I2C Write: 0x98, I2C Read: 0x99



I2C Write: 0x98, I2C Read: 0x99



35 TP TBT THERM DP 5V

78 TBT THERMD P MAKE_BASE=TRUE

1 TBT THERMD N

2 R5520 10K 5V

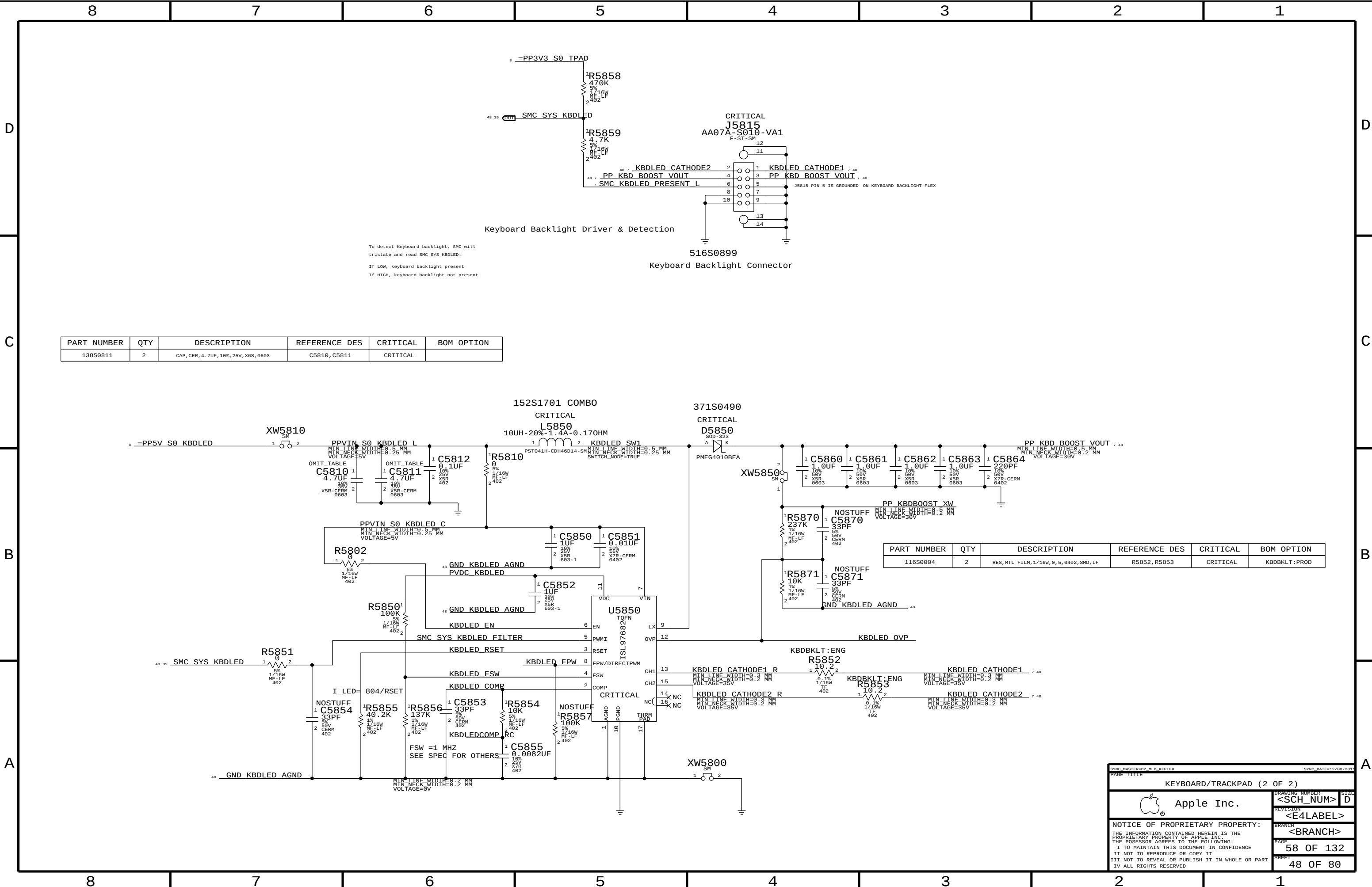
XW5520 SM

PLACE_NEAR=U3600. B1:2mm

PLACE_SIDE=TOP

Note: Use GND pin B1 on U3600 for N leg.

SYNC MASTER=D1 SENSORS		SYNC DATE=02/20/2012	
PAGE TITLE			
Thermal Sensors			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	SIZE D
		REVISION <E4LABEL>	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0811	2	CAP, CER, 4.7UF, 16%, 25V, X6S, 0603	C5810, C5811	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 1/16W, 0, 5, 0402, SMD, LF	R5852, R5853	CRITICAL	KBDBKLT:PROD

SYNC MASTER=02_MLB_KEPLER

SYNC DATE=12/08/2011

KEYBOARD/TRACKPAD (2 OF 2)

 Apple Inc.

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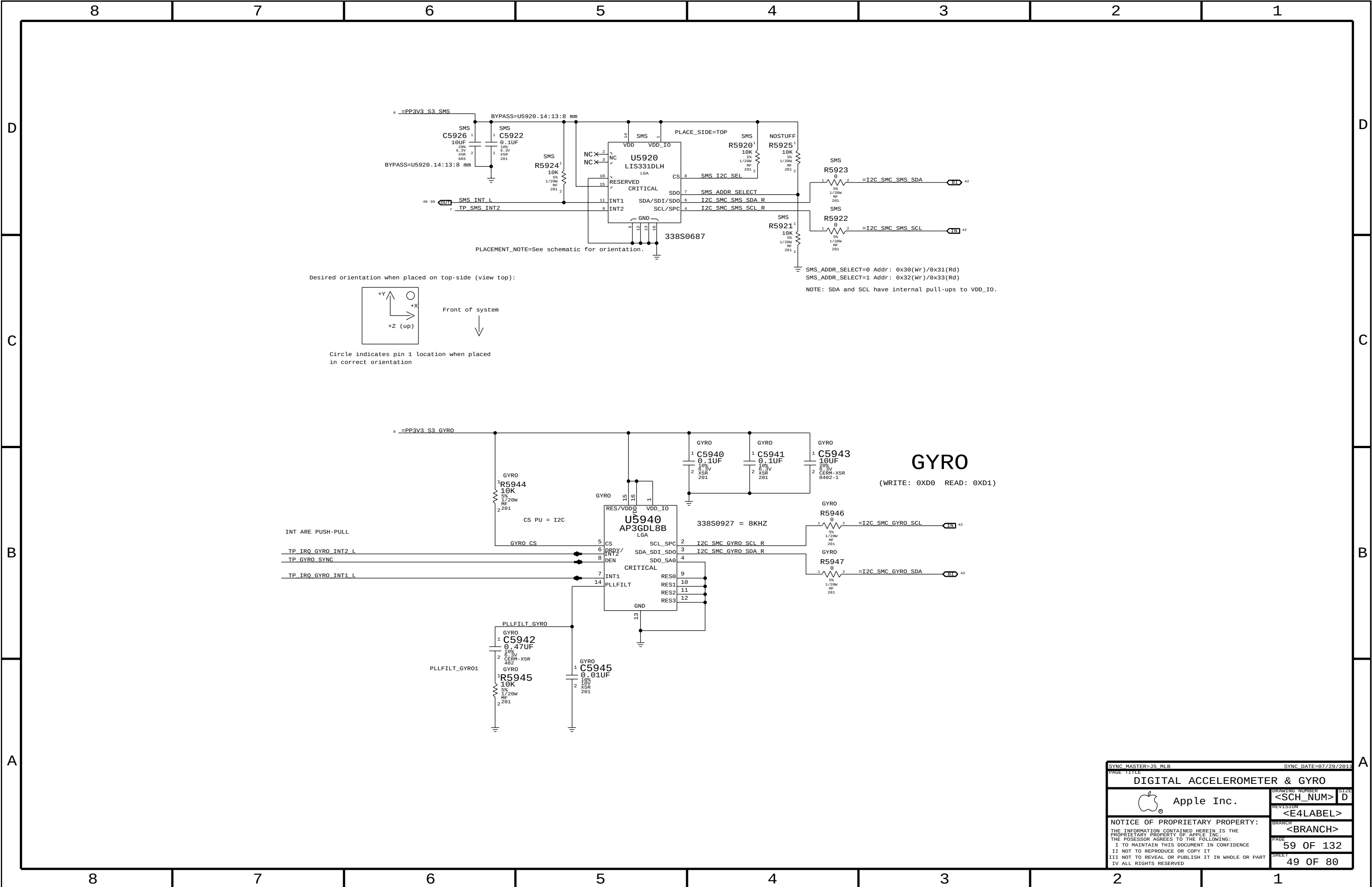
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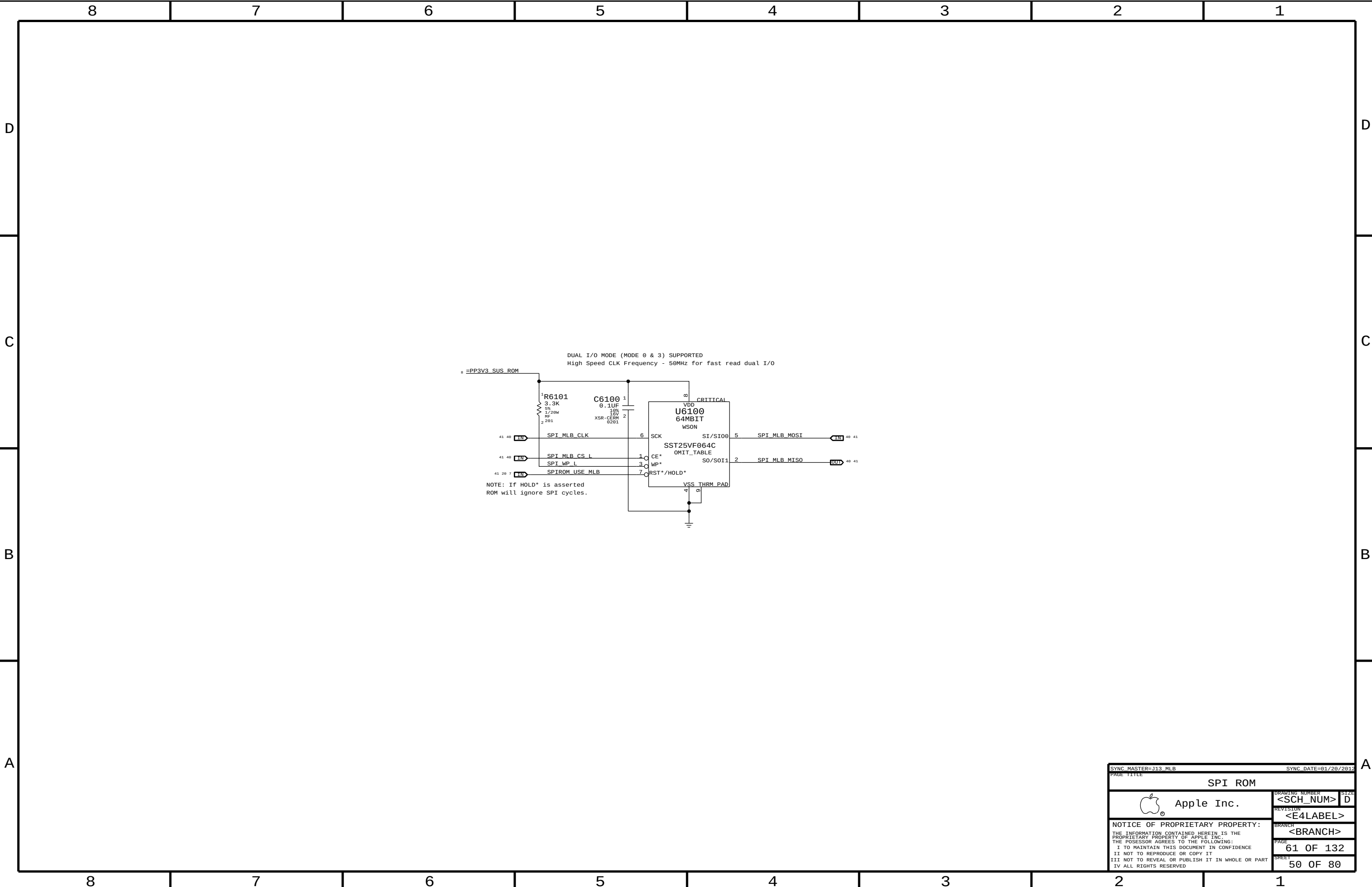
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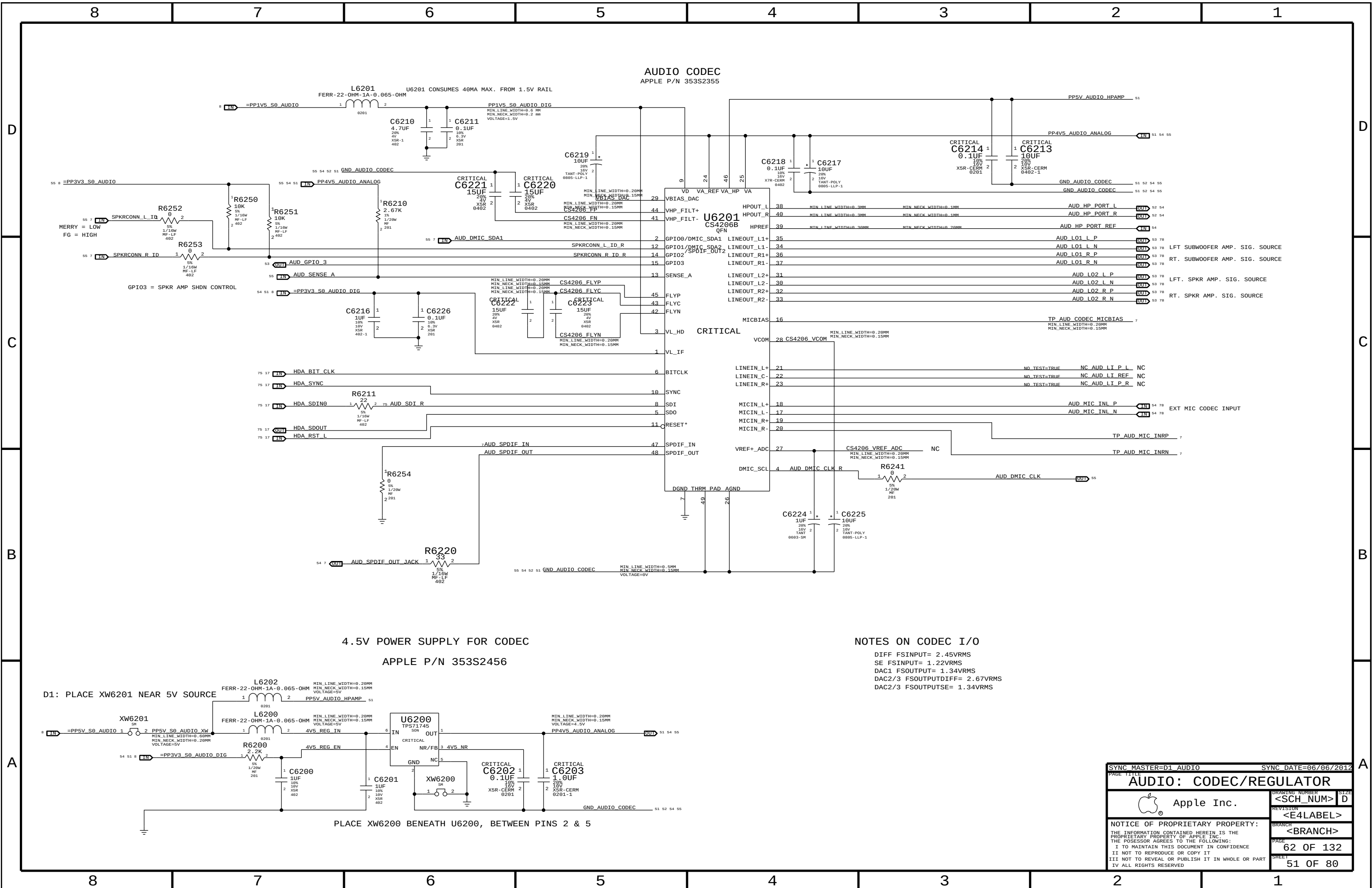
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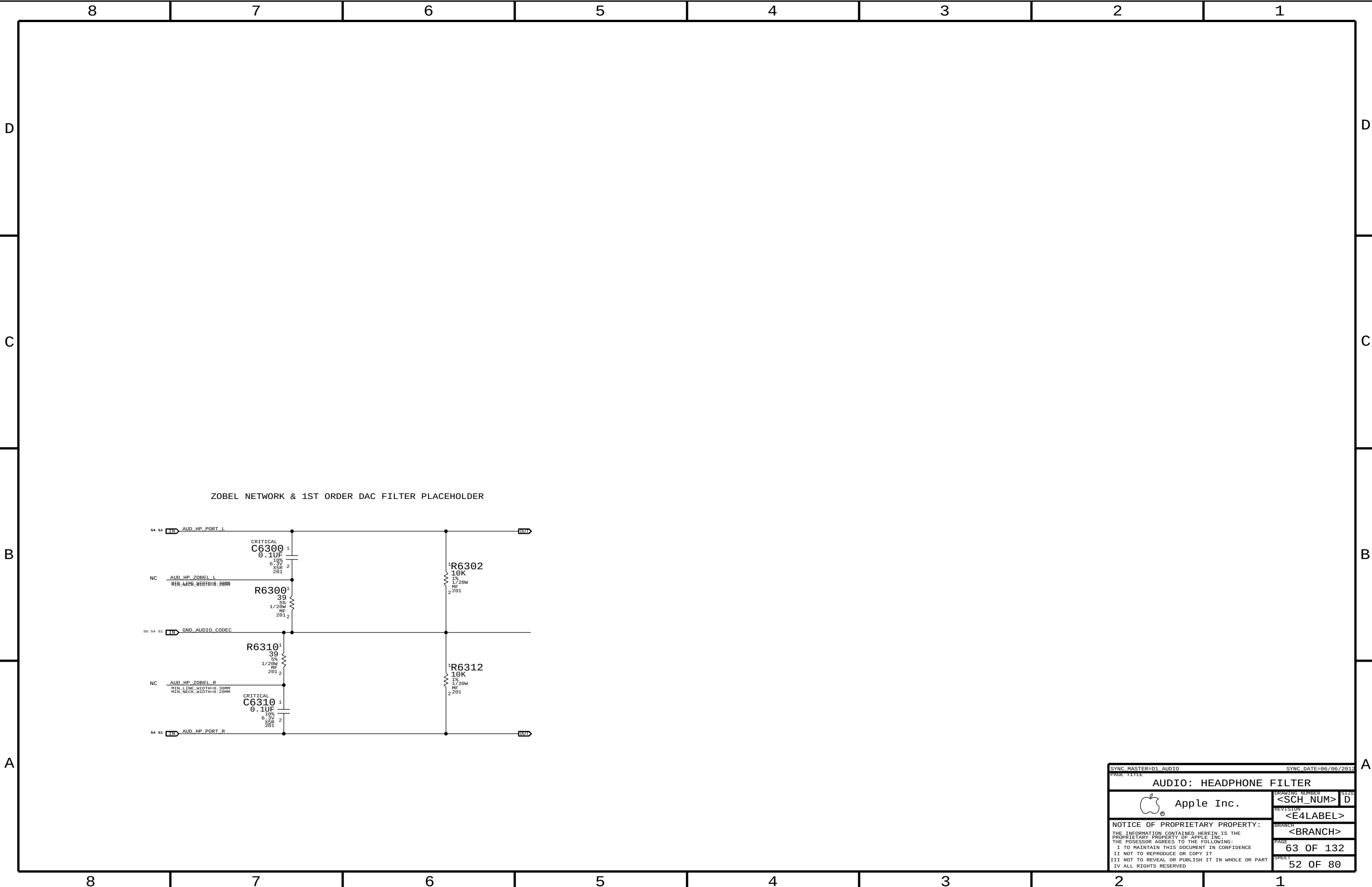
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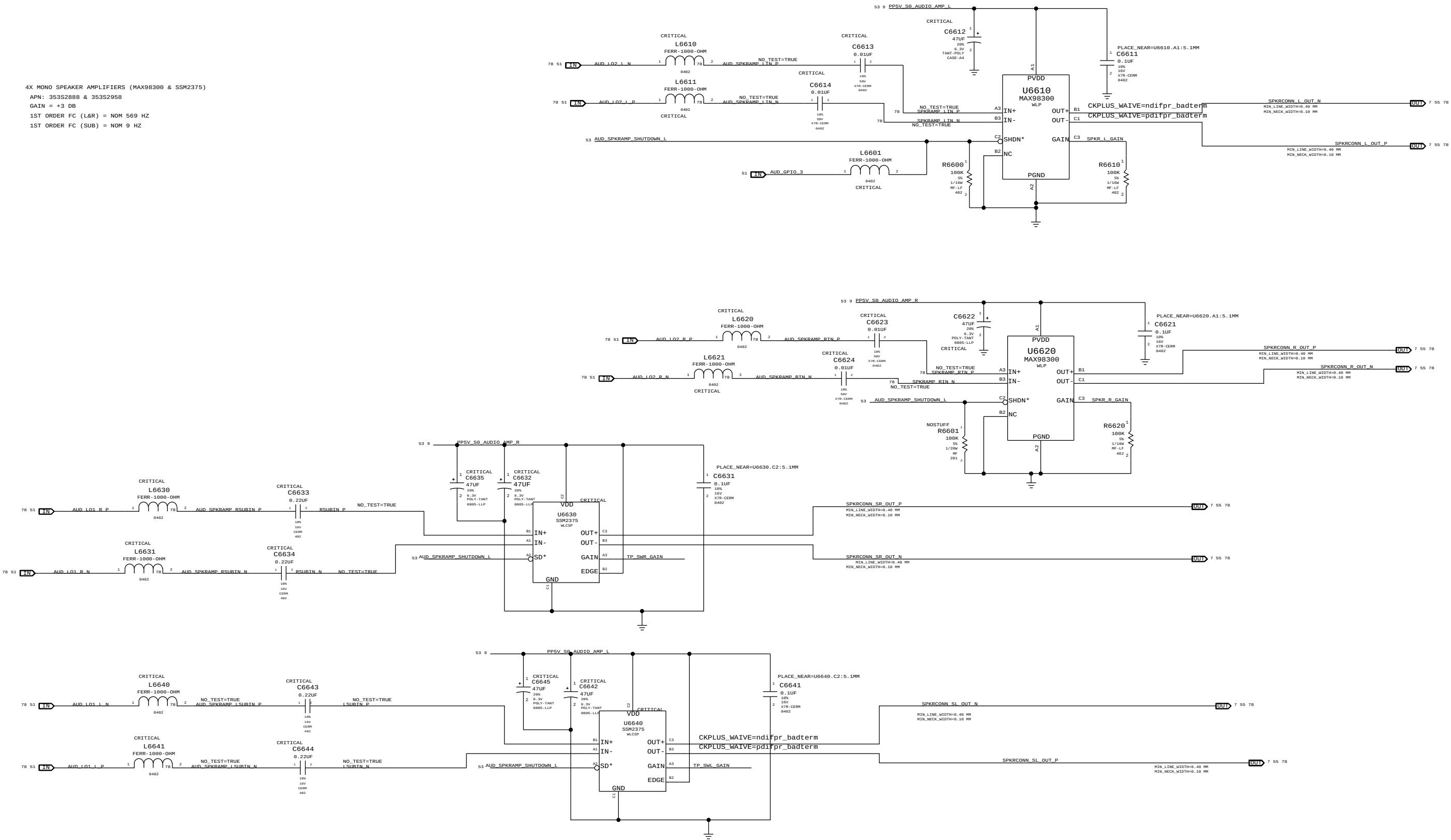




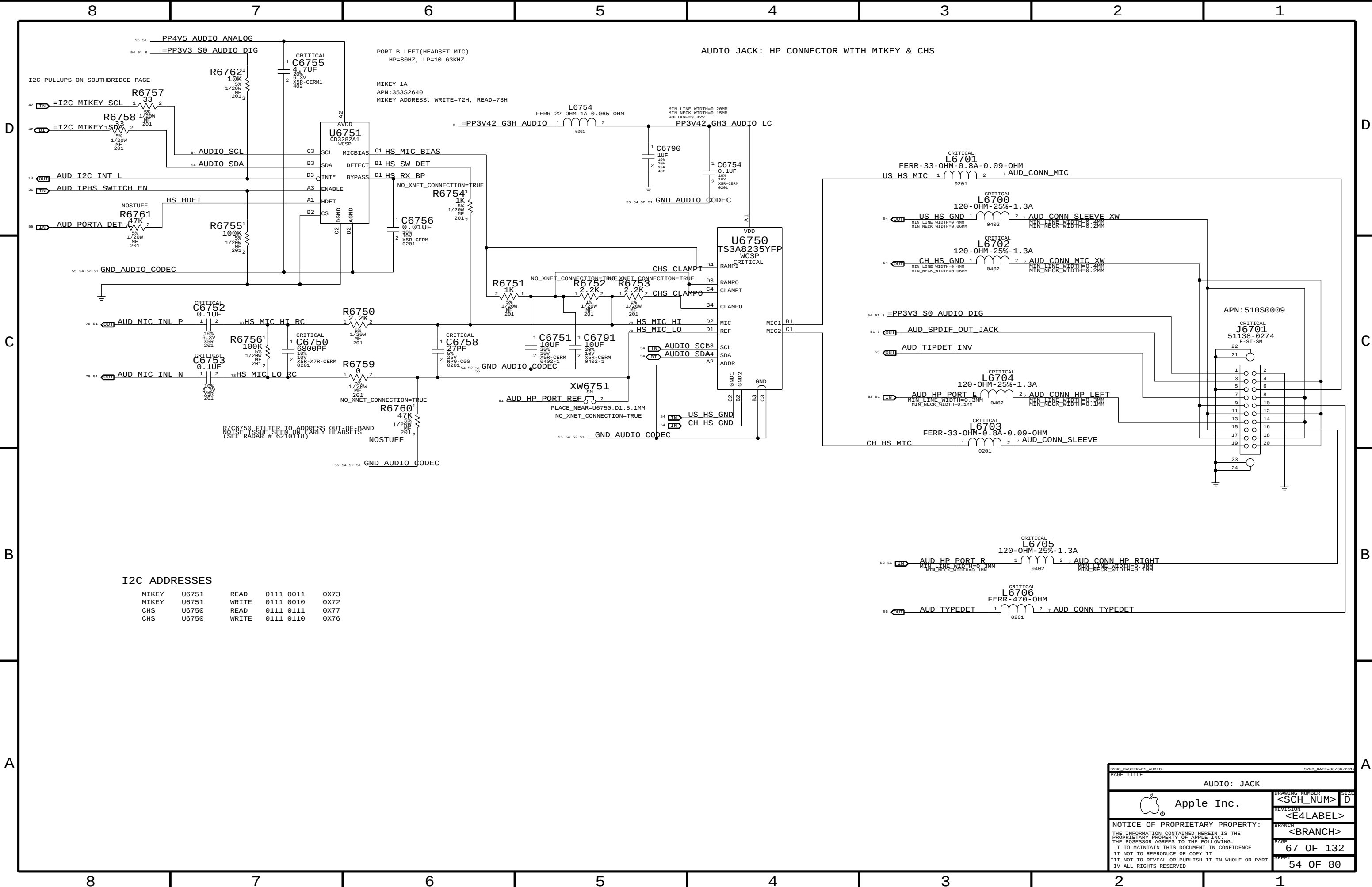


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 Apple Inc.		DRAWING NUMBER	SIZE
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		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	63 OF 132
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4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)
APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



SYNC MASTER=D1 AUDIO		SYNC DATE=06/06/2015	
PAGE TITLE			
AUDIO: SPEAKER AMP			
 Apple Inc.	DRAWING NUMBER		SIZE
	<SCH_NUM>		D
	REVISION		
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	BRANCH		
	<BRANCH>		
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I2C ADDRESSES				
MIKEY	U6751	READ	0111 0011	0X73
MIKEY	U6751	WRITE	0111 0010	0X72
CHS	U6750	READ	0111 0111	0X77
CHS	U6750	WRITE	0111 0110	0X76

SYNC MASTER=D1 AUDIO

SYNC DATE=06/06/2015

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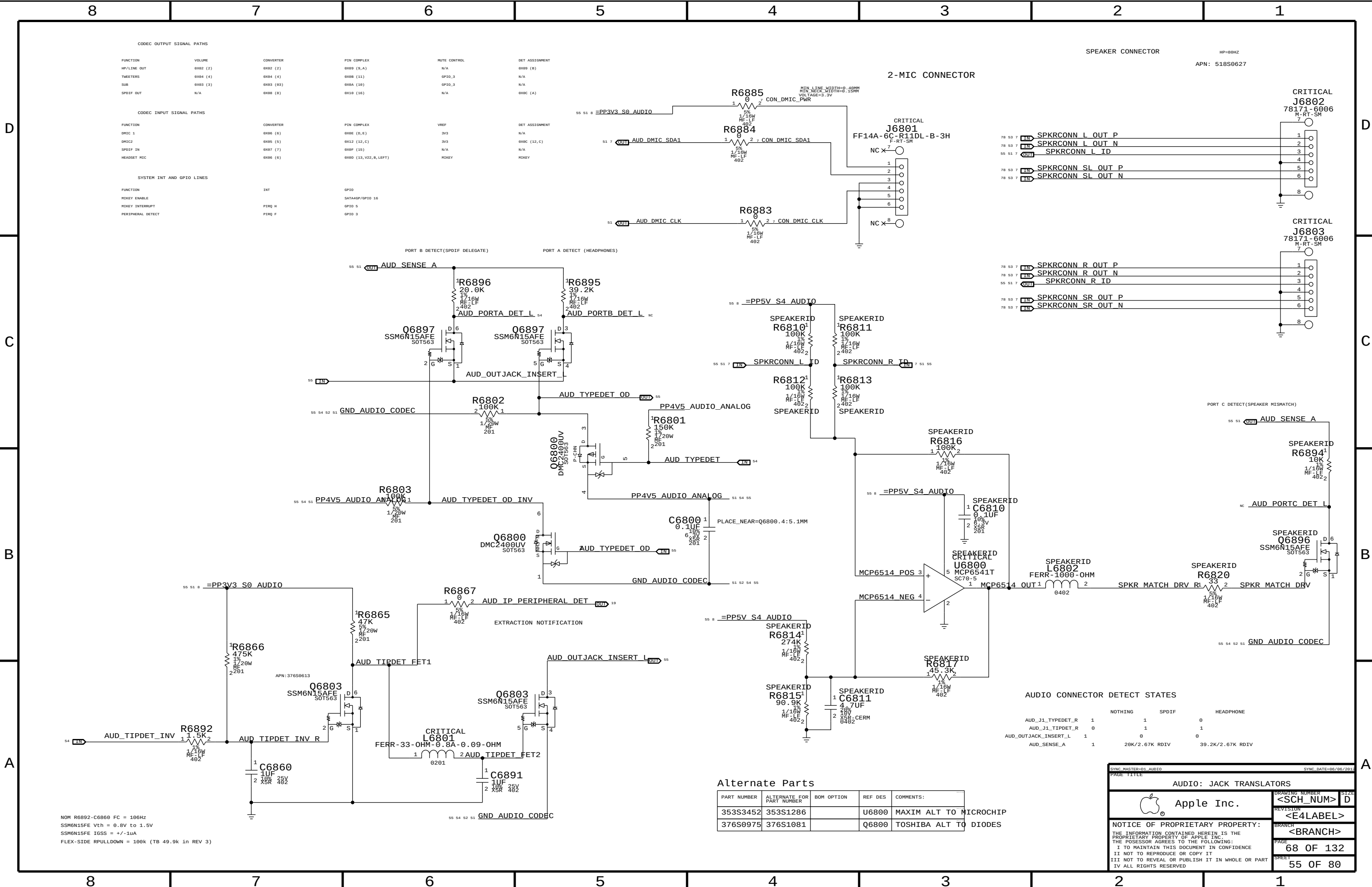
AUDIO: JACK

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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)	N/A	0X09 (8)
TWEETERS	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SUB	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (10)	N/A	0X0C (A)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
DMIC 1	0X06 (6)	0X0E (0,E)	3V3	N/A
DMIC2	0X05 (5)	0X12 (12,C)	3V3	0X0C (12,C)
SPDIF IN	0X07 (7)	0X0F (15)	N/A	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY

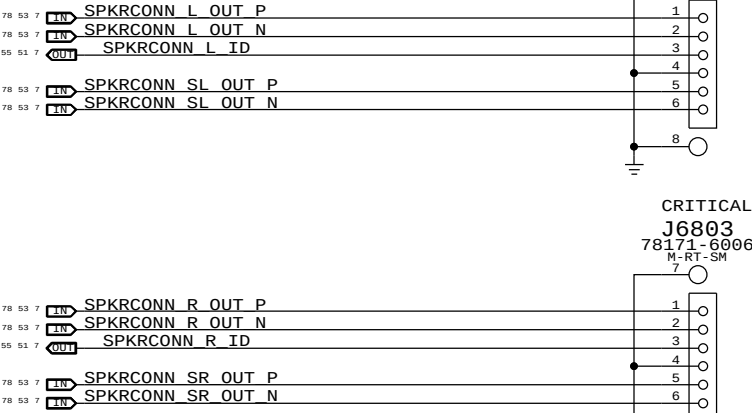
SYSTEM INT AND GPIO LINES

FUNCTION	INT	GPIO
MIKEY ENABLE	PIRQ H	GPIO 5
MIKEY INTERRUPT	PIRQ F	GPIO 3
PERIPHERAL DETECT		

SPEAKER CONNECTOR

HP=80KHZ
APN: 518S0627

2-MIC CONNECTOR



AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_J1_TYDEDET_R	1	1	0
AUD_J1_TIPDET_R	0	1	1
AUD_OUTJACK_INSERT_L	1	0	0
AUD_SENSE_A	1	20K/2.67K RDIV	39.2K/2.67K RDIV

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S3452	353S1286		U6800	MAXIM ALT TO MICROCHIP
376S0975	376S1081		Q6800	TOSHIBA ALT TO DIODES

NOM R6892-C6860 FC = 106KHZ
SSM6N15FE Vth = 0.8V to 1.5V
SSM6N15FE IGSS = +/-1uA
FLEX-SIDE RPULLDOWN = 100k (TB 49.9k in REV 3)

SYNC MASTER=01 AUDIO

SYNC DATE=05/05/2013

AUDIO: JACK TRANSLATORS

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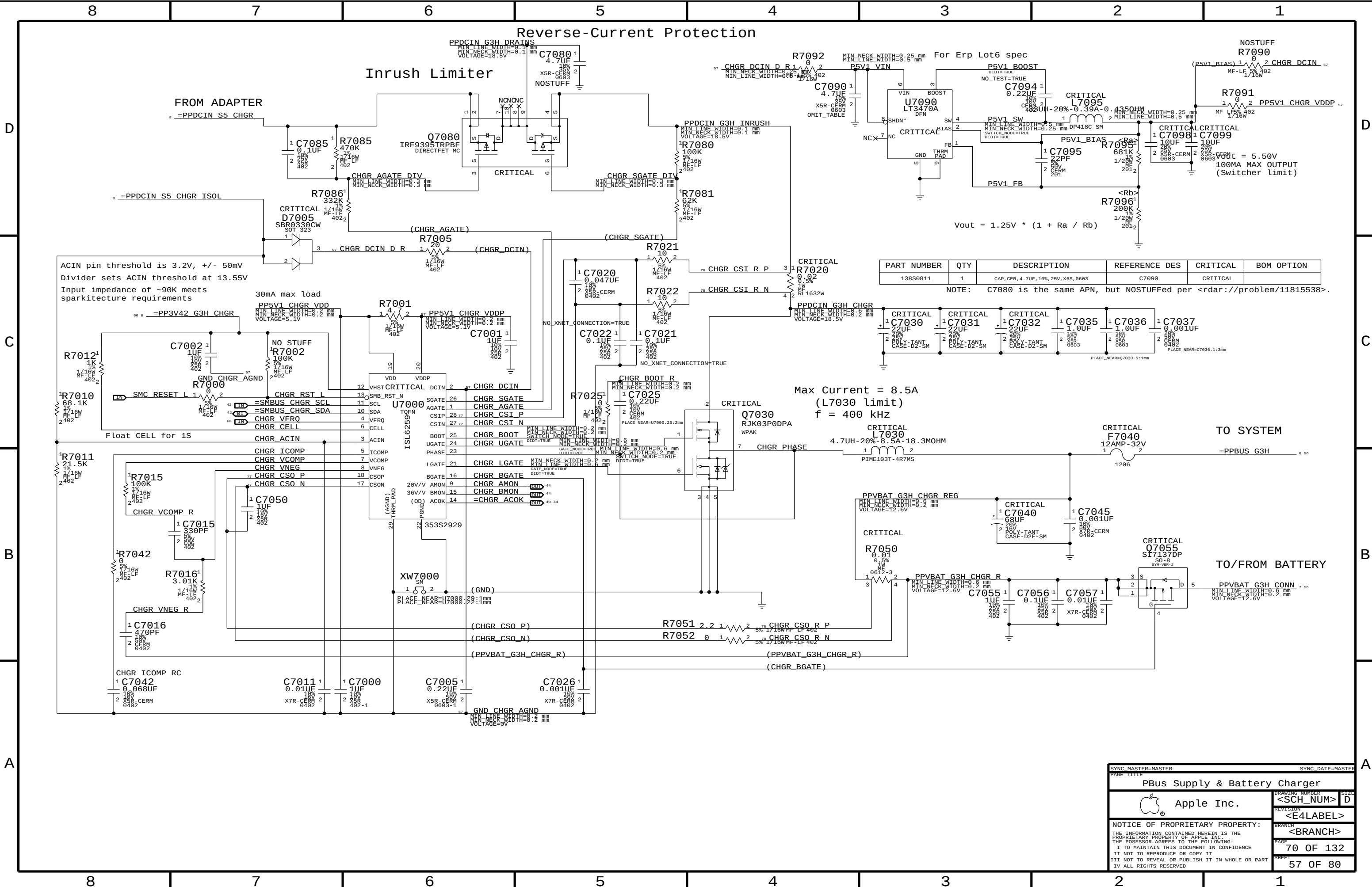
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0811	1	CAP, CER, 4.7UF, 10%, 25V, X6S, 0603	C7090	CRITICAL	

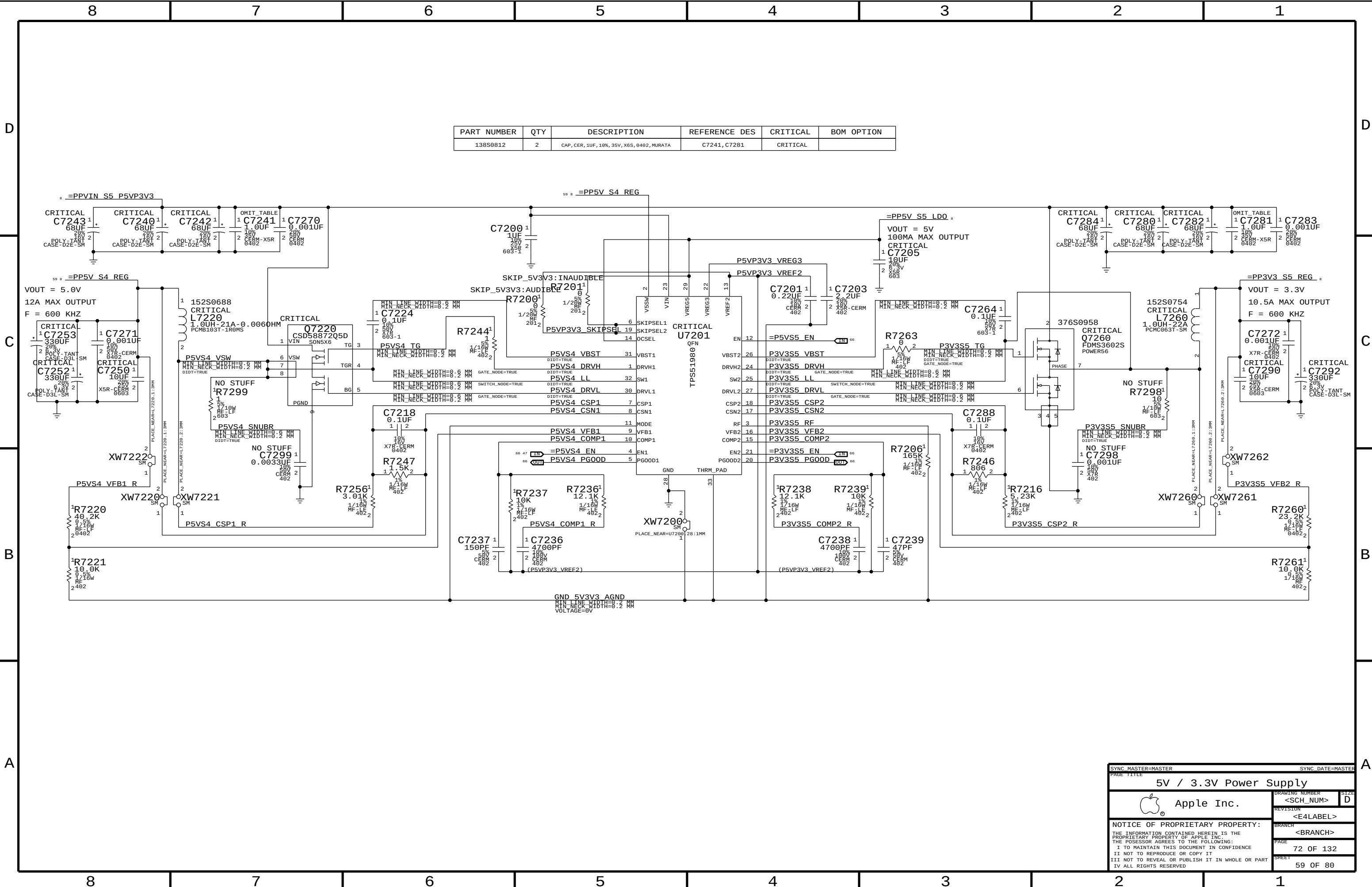
NOTE: C7080 is the same APN, but NOSTUFFed per <rdar://problem/11815538>.

PAGE TITLE		PAGE TITLE	
PBus Supply & Battery Charger		PBus Supply & Battery Charger	
Apple Inc.		Apple Inc.	
DRAWING NUMBER		DRAWING NUMBER	
<SCH_NUM>		<SCH_NUM>	
REVISION		REVISION	
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BRANCH		BRANCH	
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PAGE		PAGE	
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SHEET		SHEET	
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D

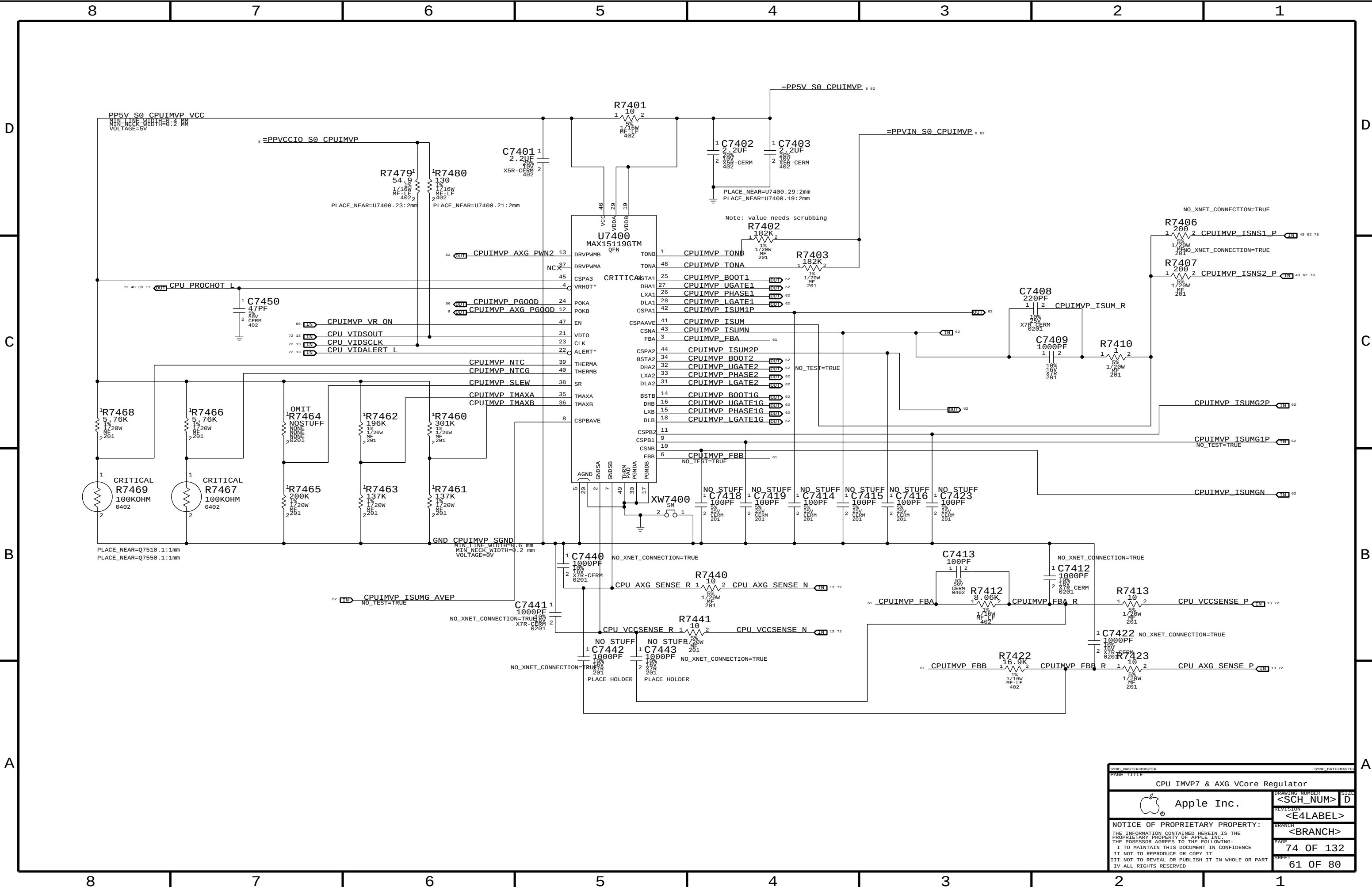
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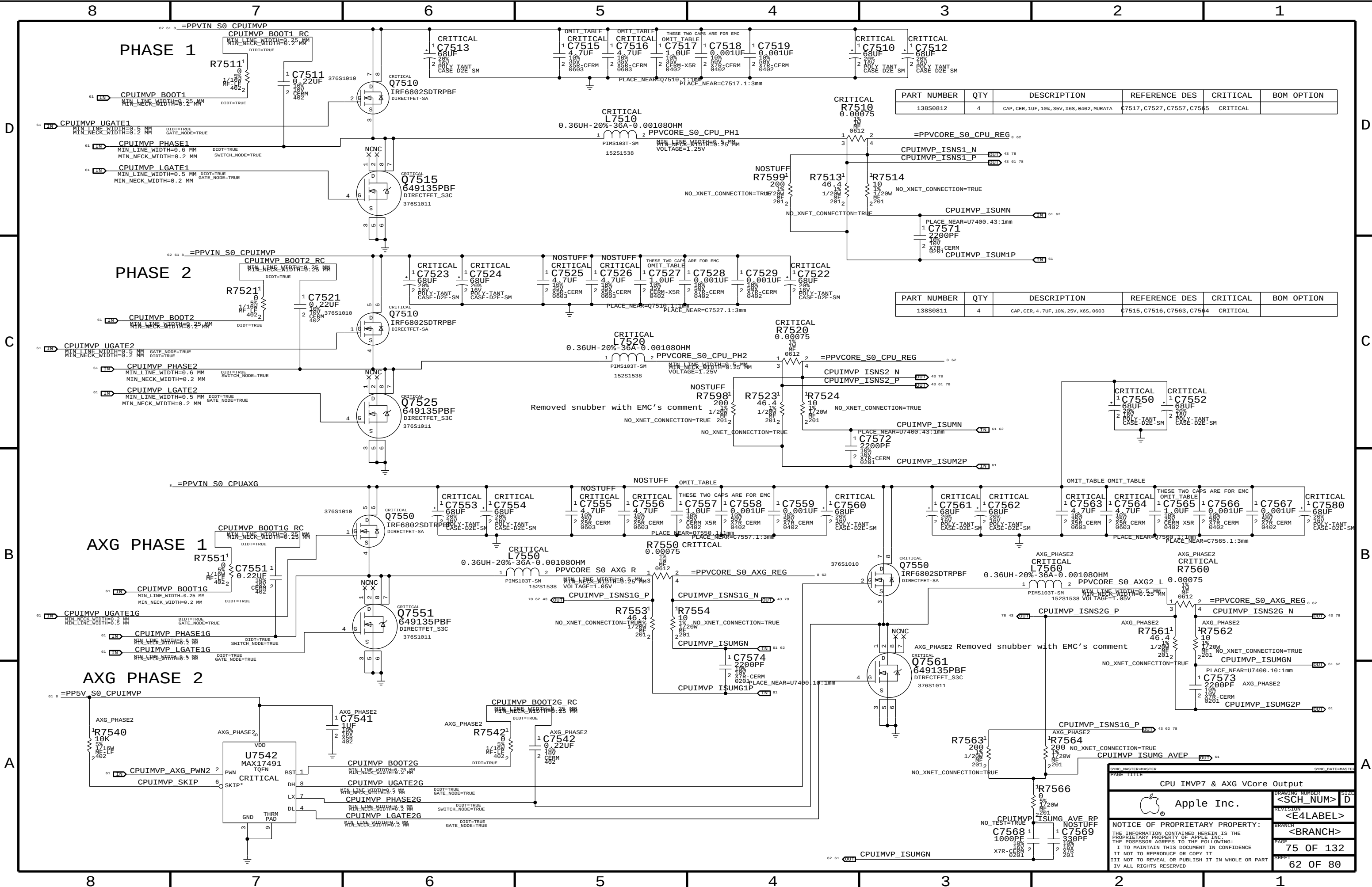
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0812	2	CAP, CER, 1UF, 16%, 35V, X6S, 0402, MURATA	C7241, C7281	CRITICAL	

SYNC MASTER=MASTER		SYNC DATE=MASTER	
PAGE TITLE			
5V / 3.3V Power Supply			
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
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		<BRANCH>	
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		SHEET	59 OF 80
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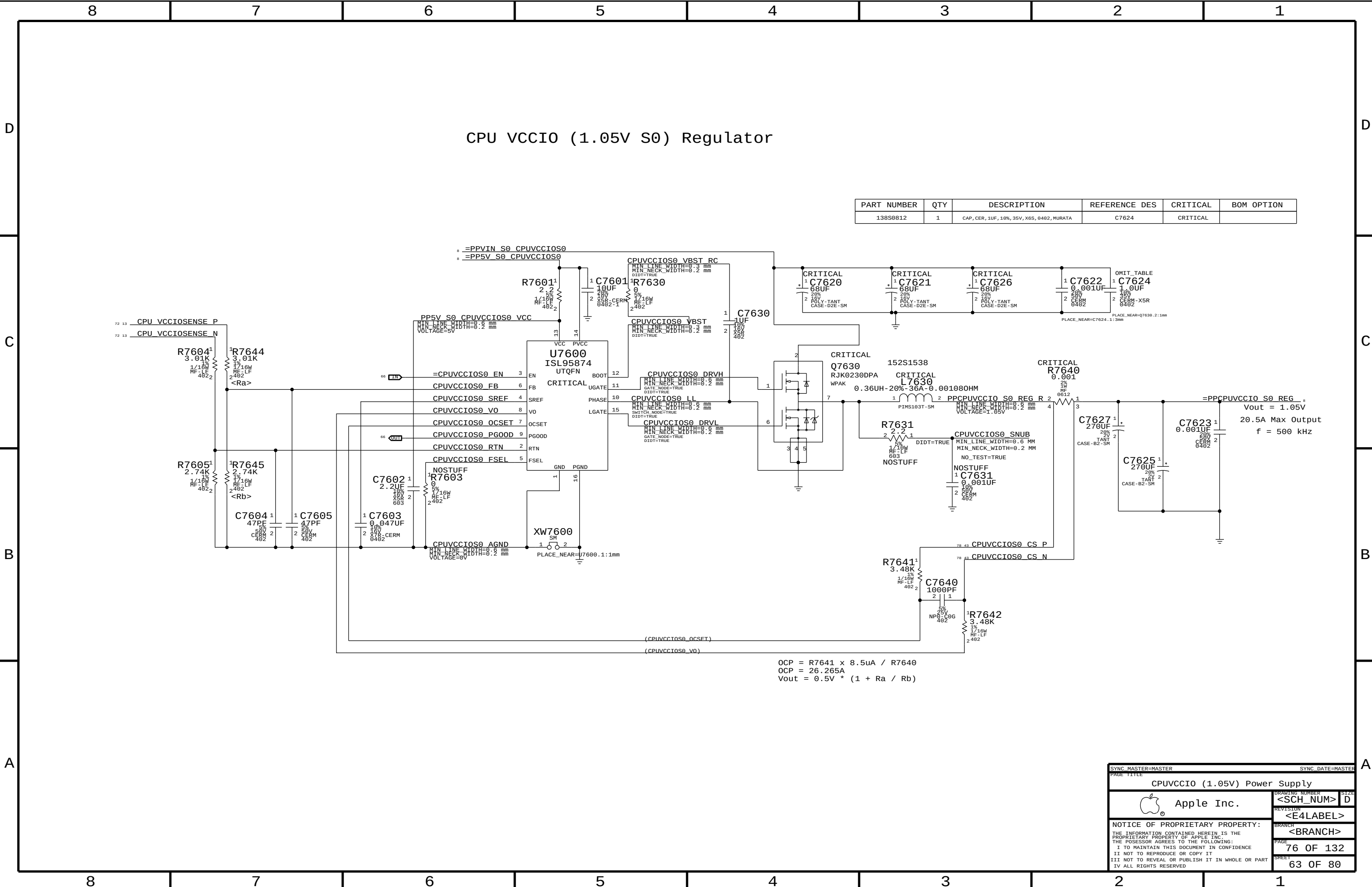




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0812	4	CAP, CER, 1UF, 10%, 35V, X6S, 0402, MURATA	C7517, C7527, C7557, C7565	CRITICAL	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0811	4	CAP, CER, 4.7UF, 10%, 25V, X6S, 0603	C7515, C7516, C7563, C7564	CRITICAL	

CPU IMV7 & AXG VCore Output		DRAWING NUMBER		SIZE
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CPU VCCIO (1.05V S0) Regulator

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0812	1	CAP, CER, 1UF, 16%, 35V, X6S, 0402, MURATA	C7624	CRITICAL	

OCP = $R7641 \times 8.5\mu A / R7640$
OCP = 26.265A
 $V_{out} = 0.5V * (1 + R_a / R_b)$

SYNC_MASTER=MASTER

SYNC_DATE=MASTER

CPUVCCIO (1.05V) Power Supply

 Apple Inc.

<SCH_NUM>

SIZE

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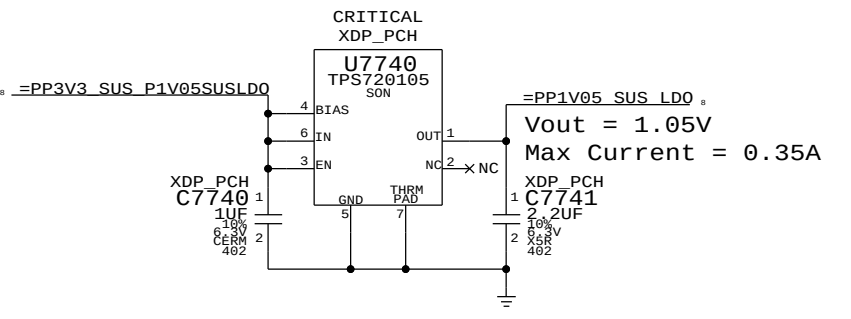
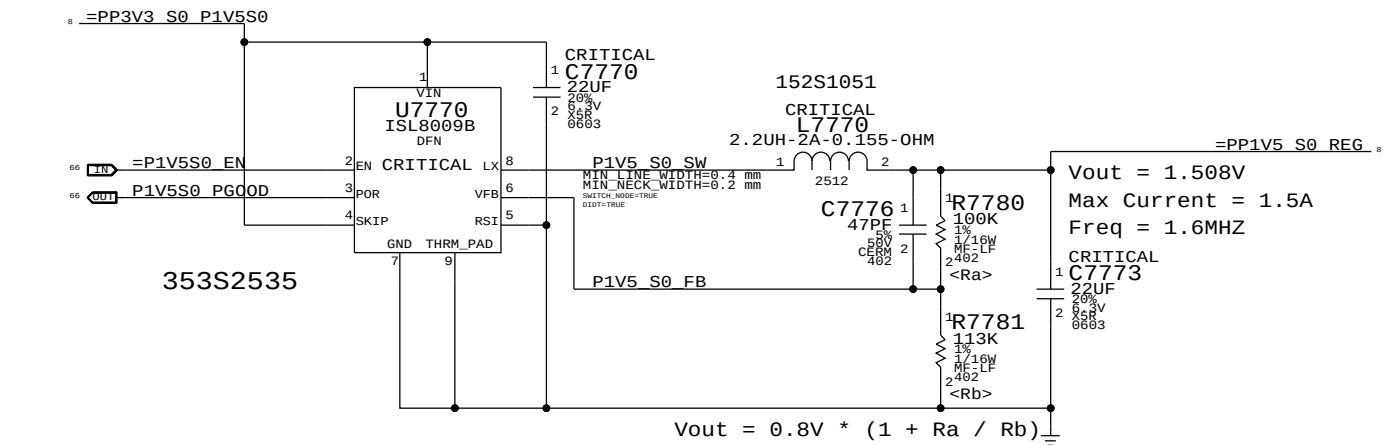
SHEET

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1.5V S0 Switcher

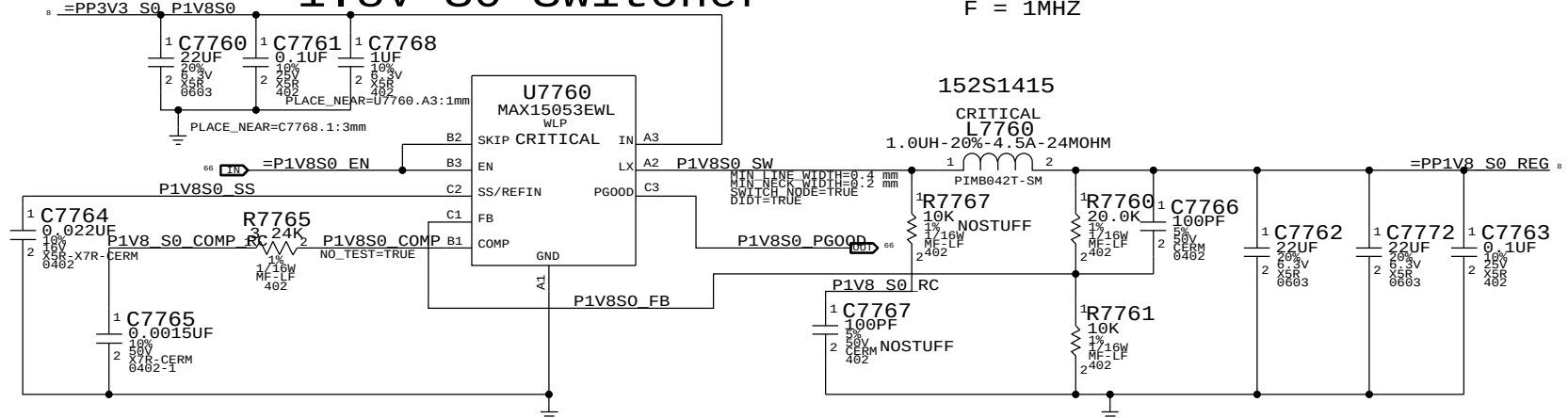
1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V in SUS.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



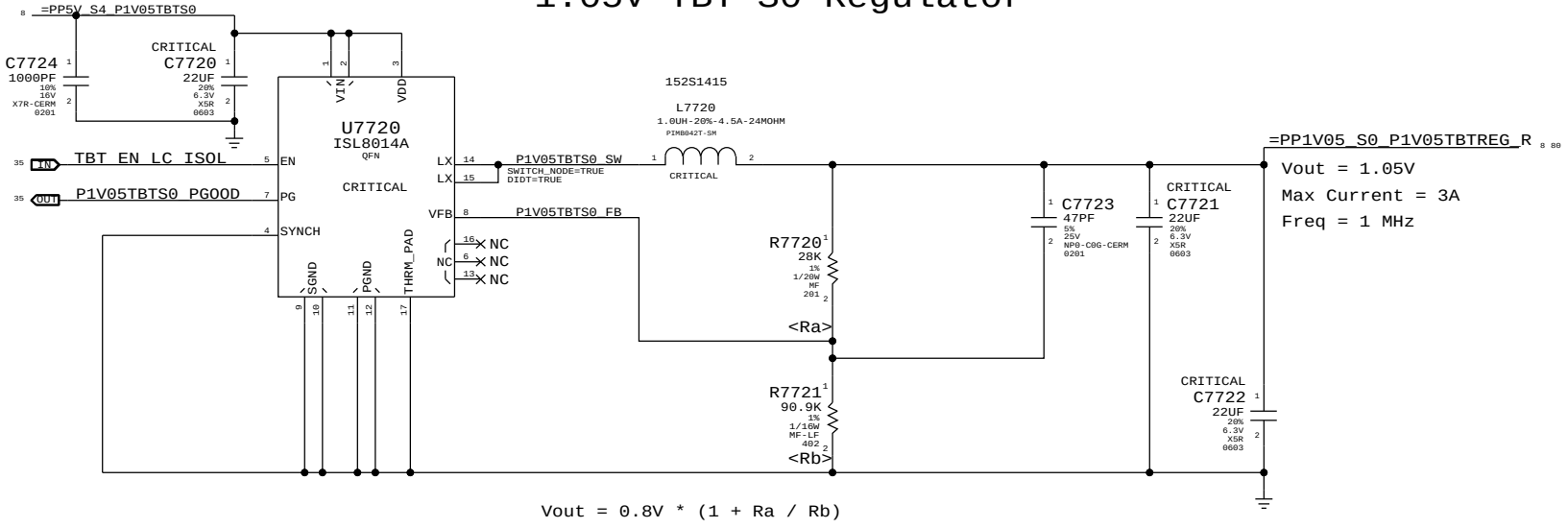
1.8V S0 Switcher

Vout = 1.8V
MAX CURRENT = 2A
F = 1MHZ

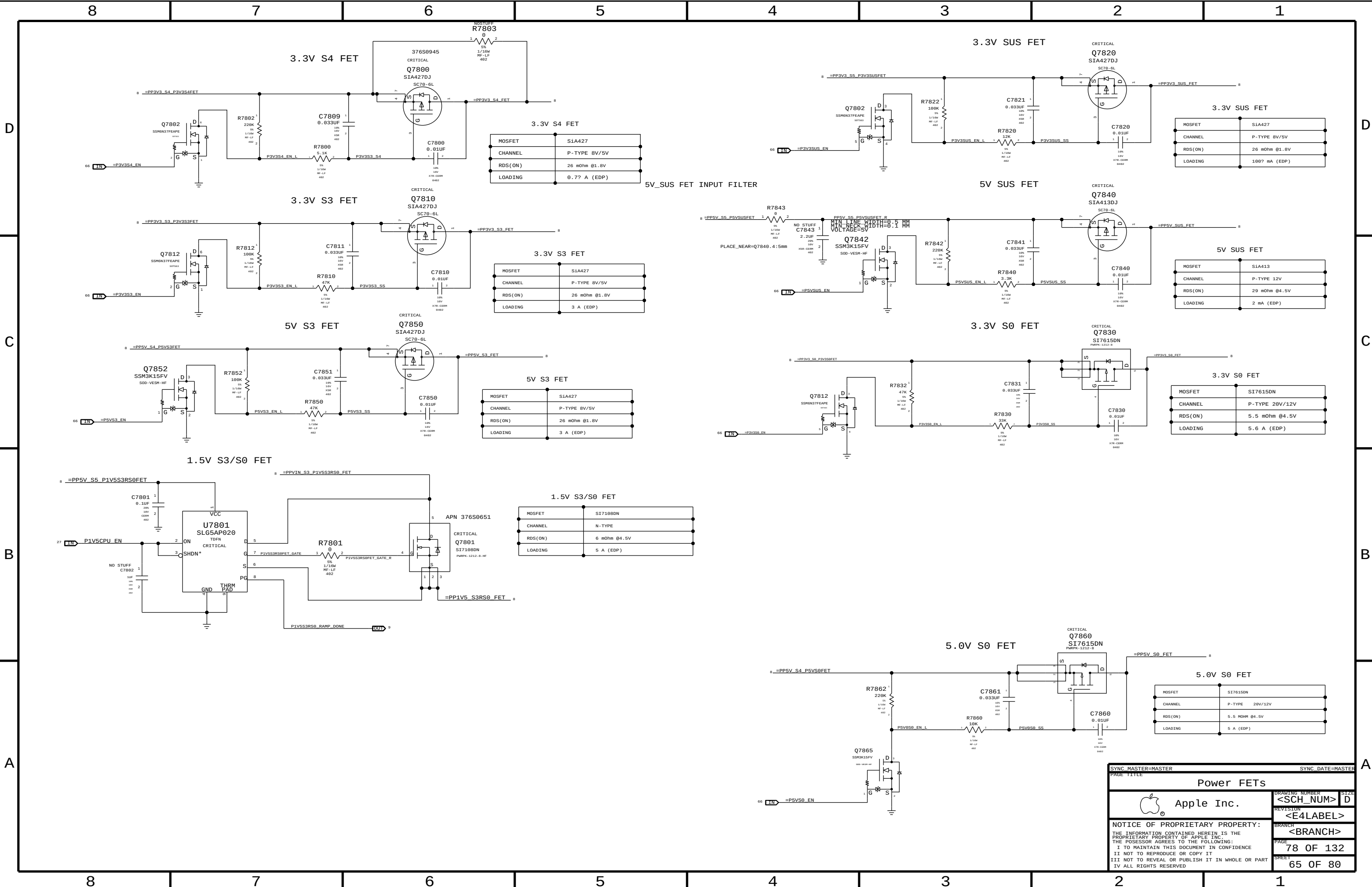


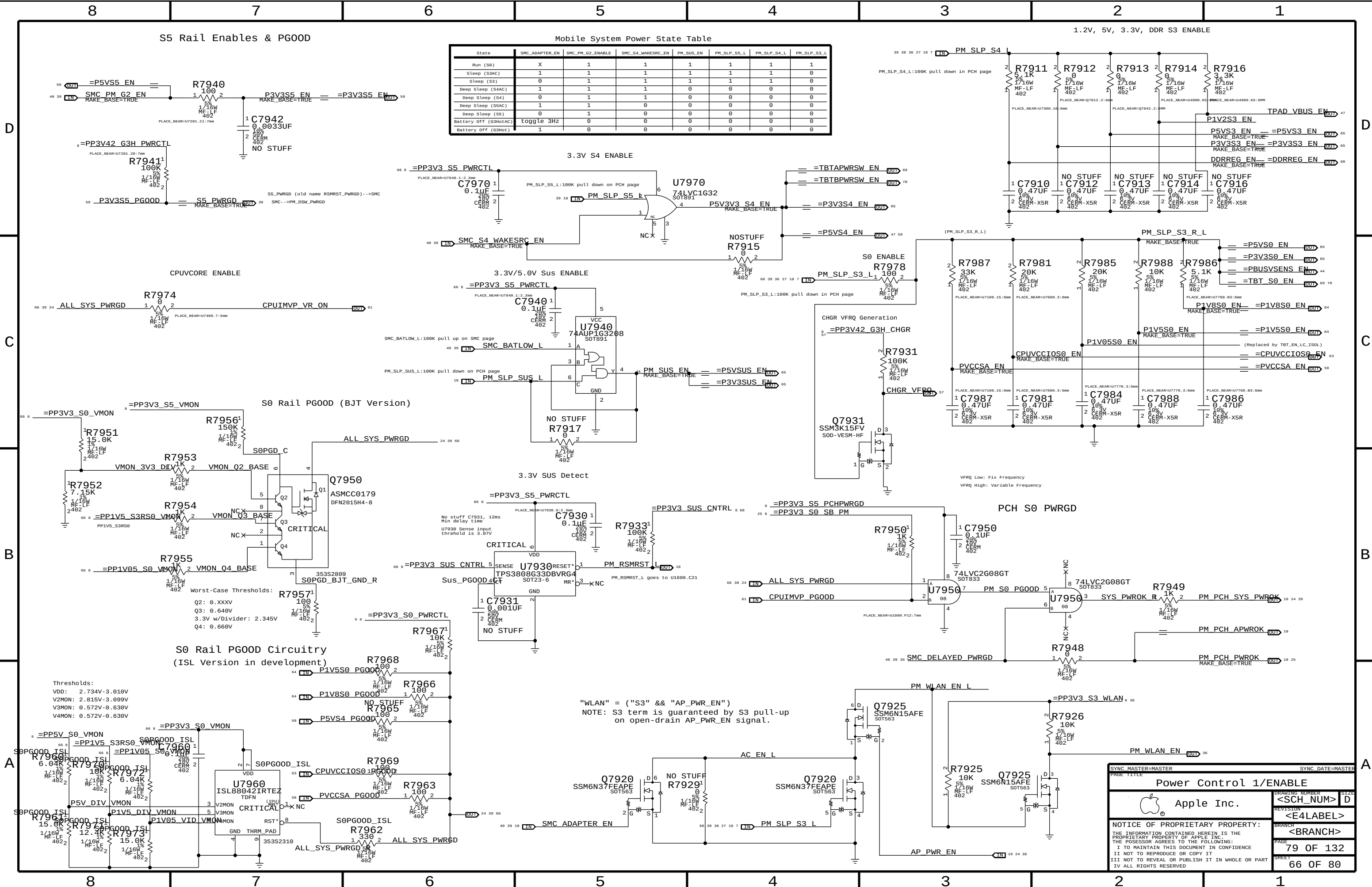
1.05V TBT S0 Regulator

Vout = 1.05V
Max Current = 3A
Freq = 1 MHz



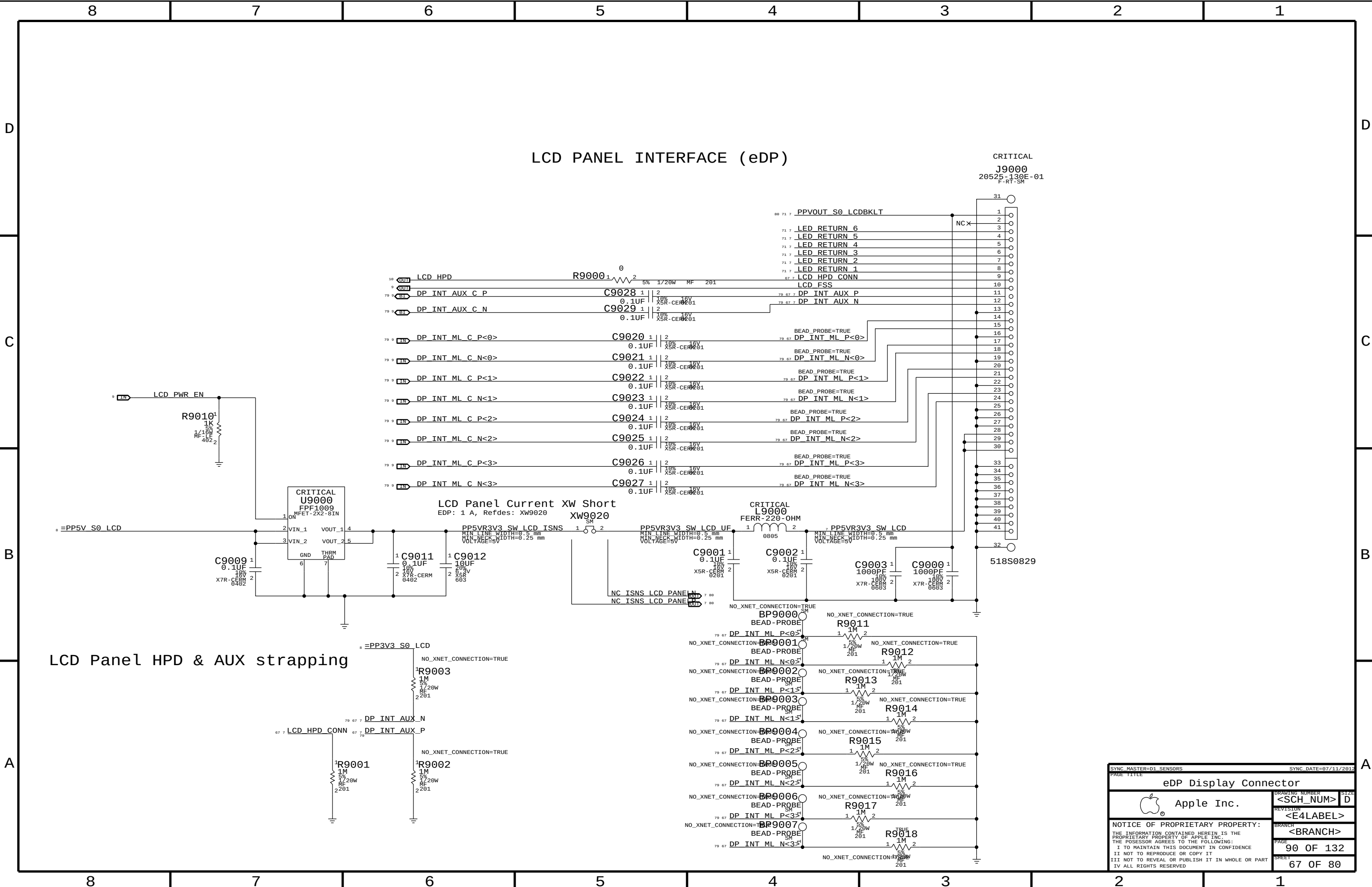
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Misc Power Supplies		DRAWING NUMBER	
Apple Inc.		<SCH_NUM> D	
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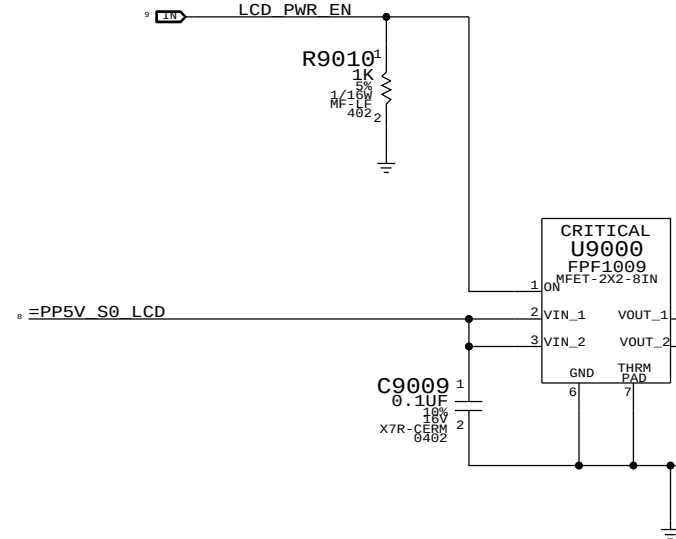
Mobile System Power State Table							
State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (S4AC)	1	1	1	0	0	0	0
Deep Sleep (S4)	0	1	1	0	0	0	0
Deep Sleep (S5AC)	1	1	0	0	0	0	0
Deep Sleep (S5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

S0 Rail PG00D Circuitry (ISL Version in development)							
Thresholds:	VDD	V2MON	V3MON	V4MON	V5MON	V6MON	V7MON
	2.734V-3.010V	2.815V-3.099V	0.572V-0.630V	0.572V-0.630V	0.572V-0.630V	0.572V-0.630V	0.572V-0.630V

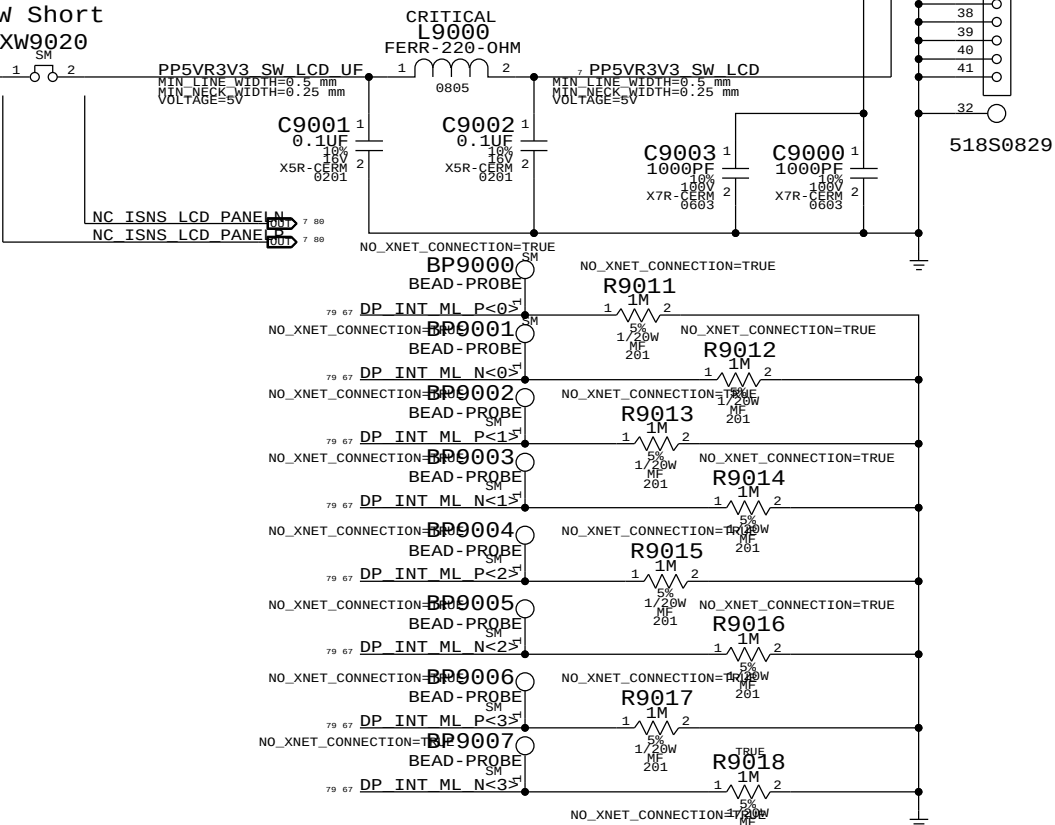
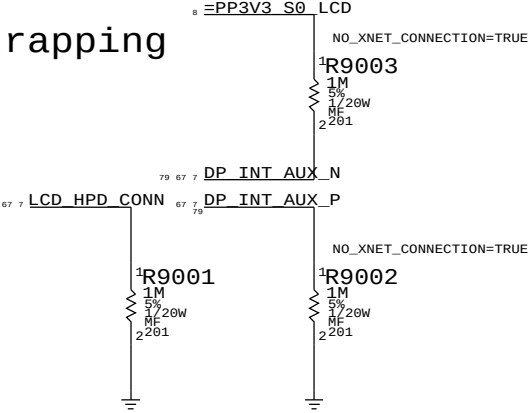


LCD PANEL INTERFACE (eDP)

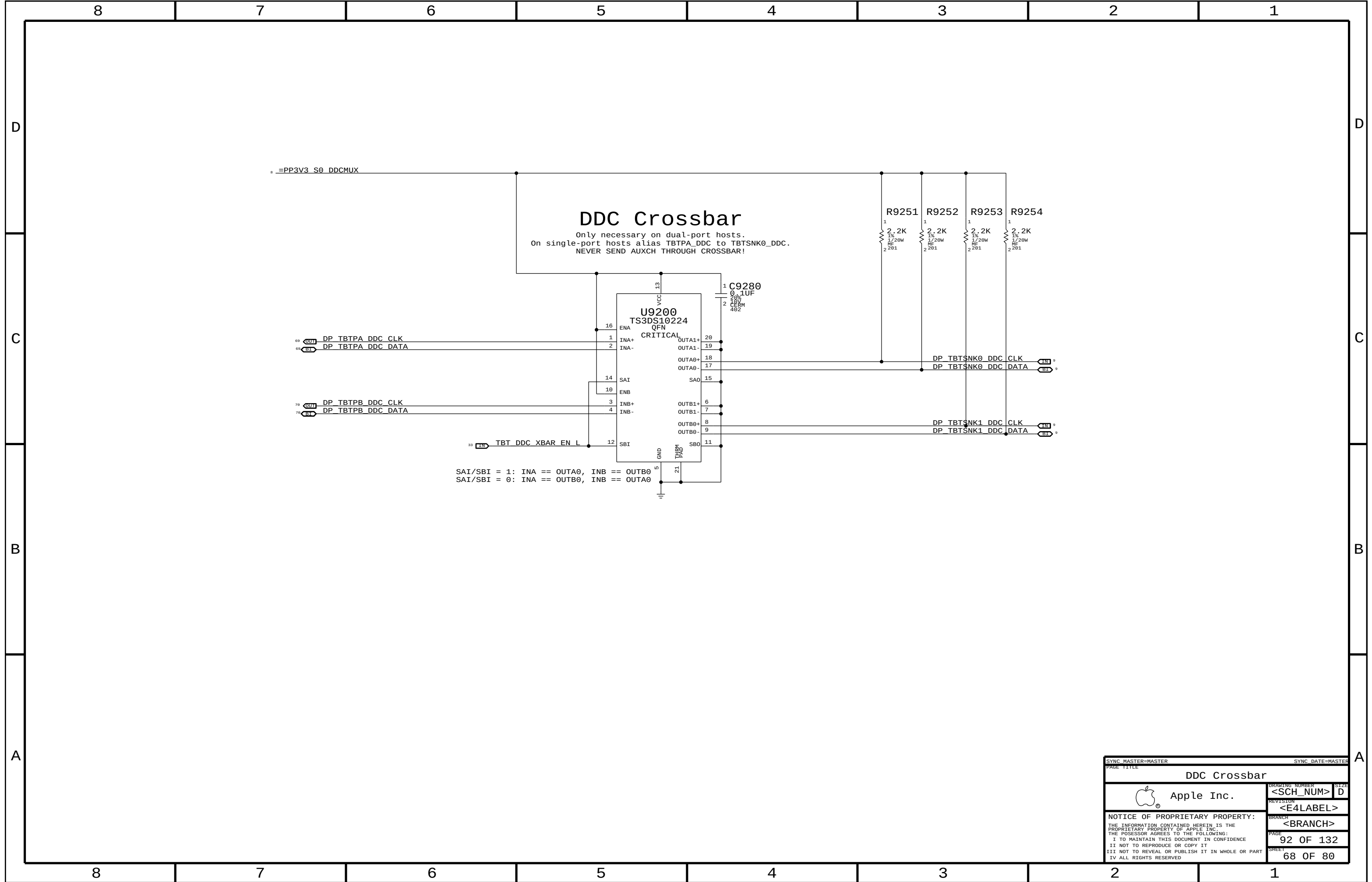
CRITICAL
J9000
20525-130E-01
F-RT-SM

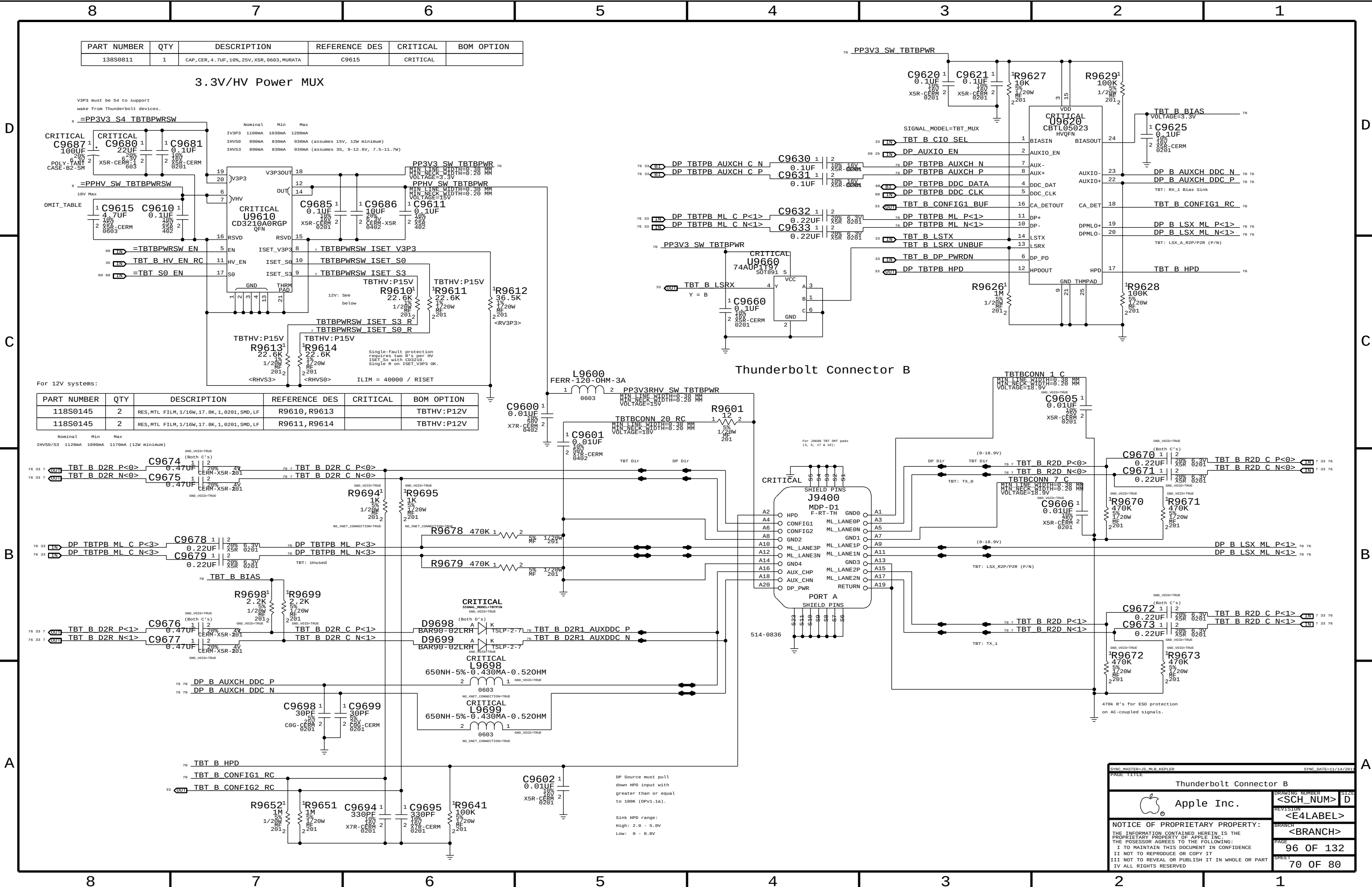


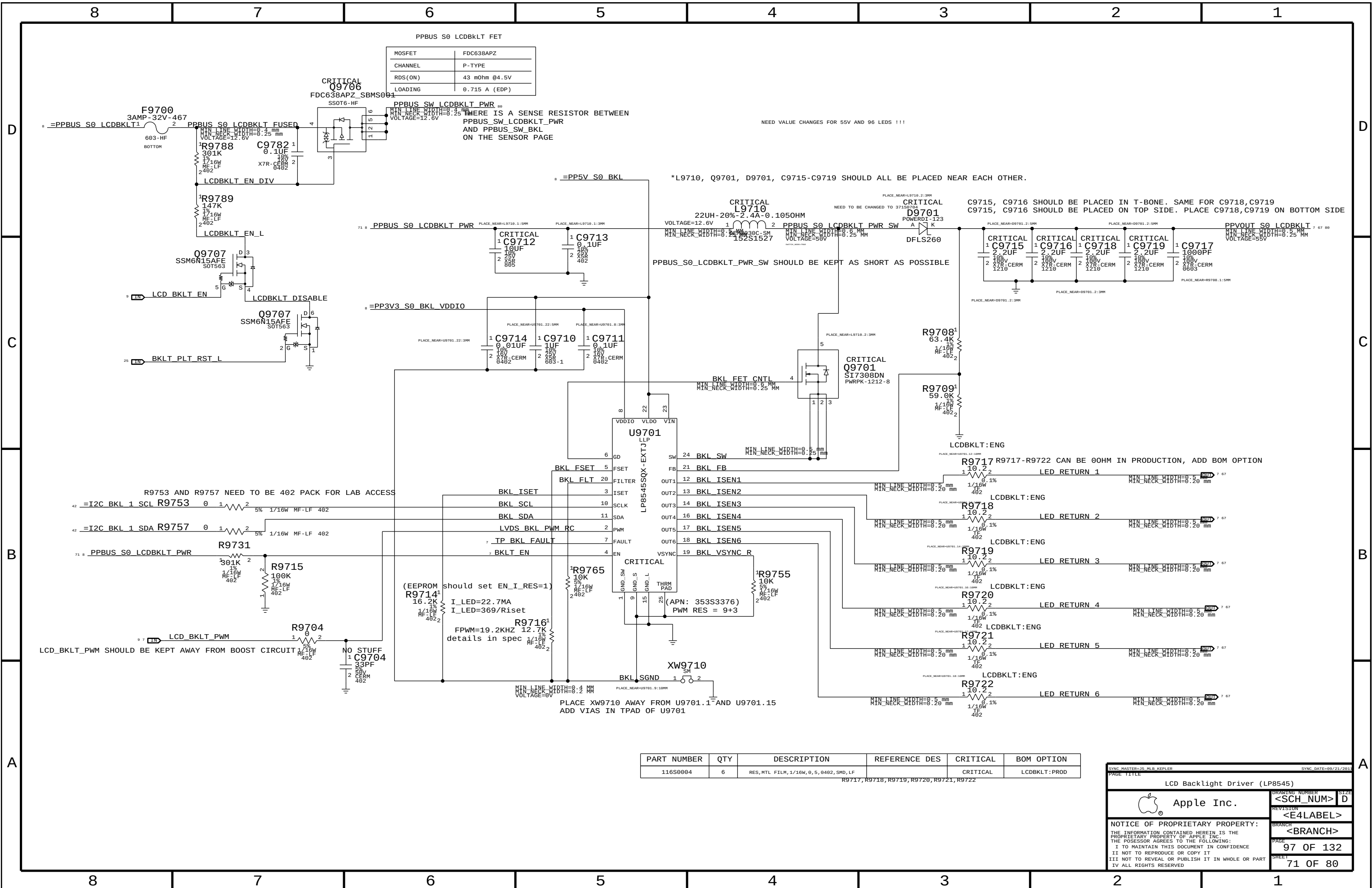
LCD Panel HPD & AUX strapping



SYNC_MASTER=D1_SENSORS		SYNC_DATE=07/11/2012	
PAGE TITLE			
eDP Display Connector			
	DRAWING NUMBER		SIZE
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	6	RES, MTL FILM, 1/16W, 0, 5, 0402, SMD, LF	R9717, R9718, R9719, R9720, R9721, R9722	CRITICAL	LCDBKLT: PROD

SYNC MASTER=J5, M.B. KEPLER

SYNC DATE=09/21/2011

LCD Backlight Driver (LP8545)

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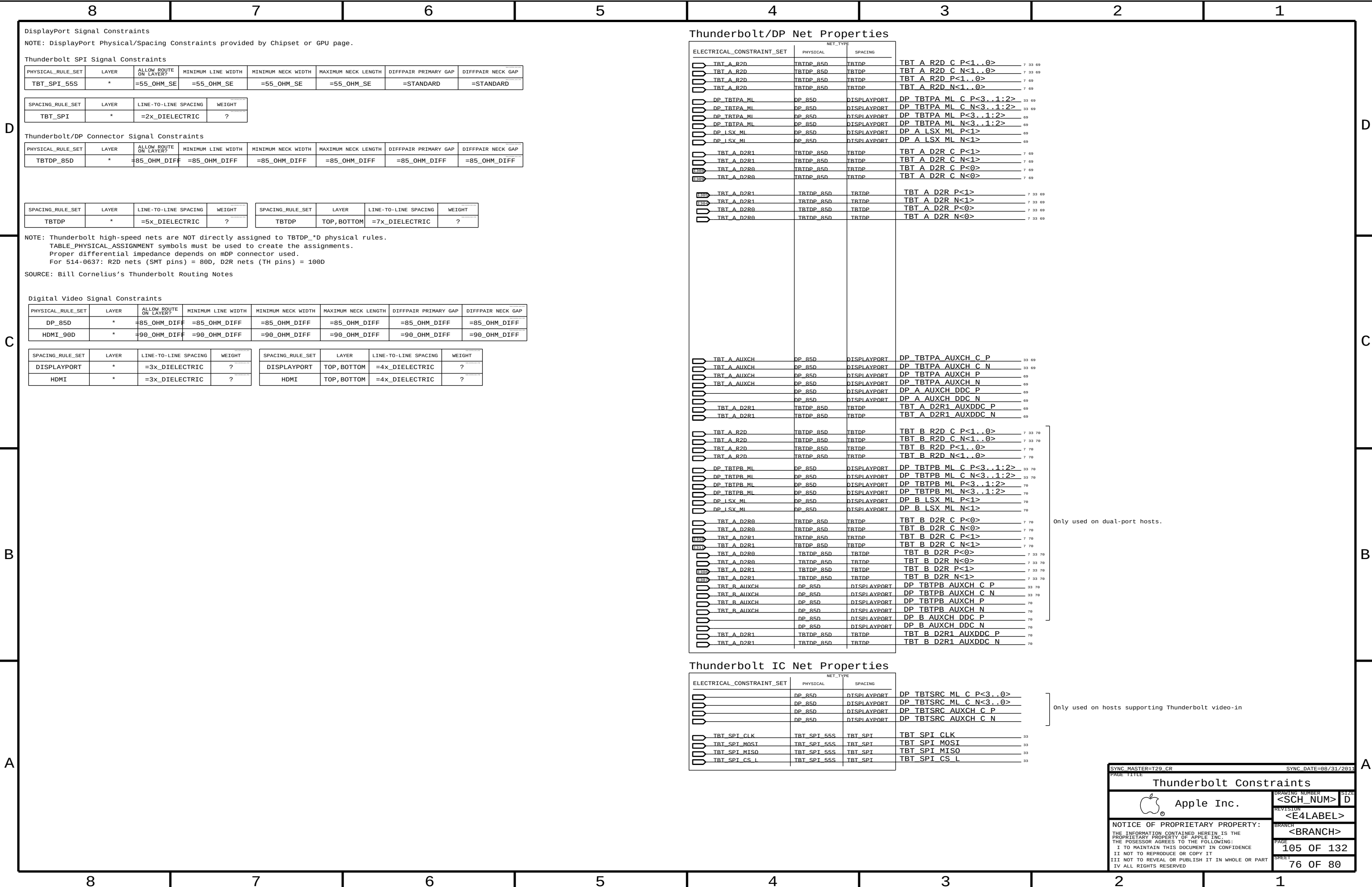
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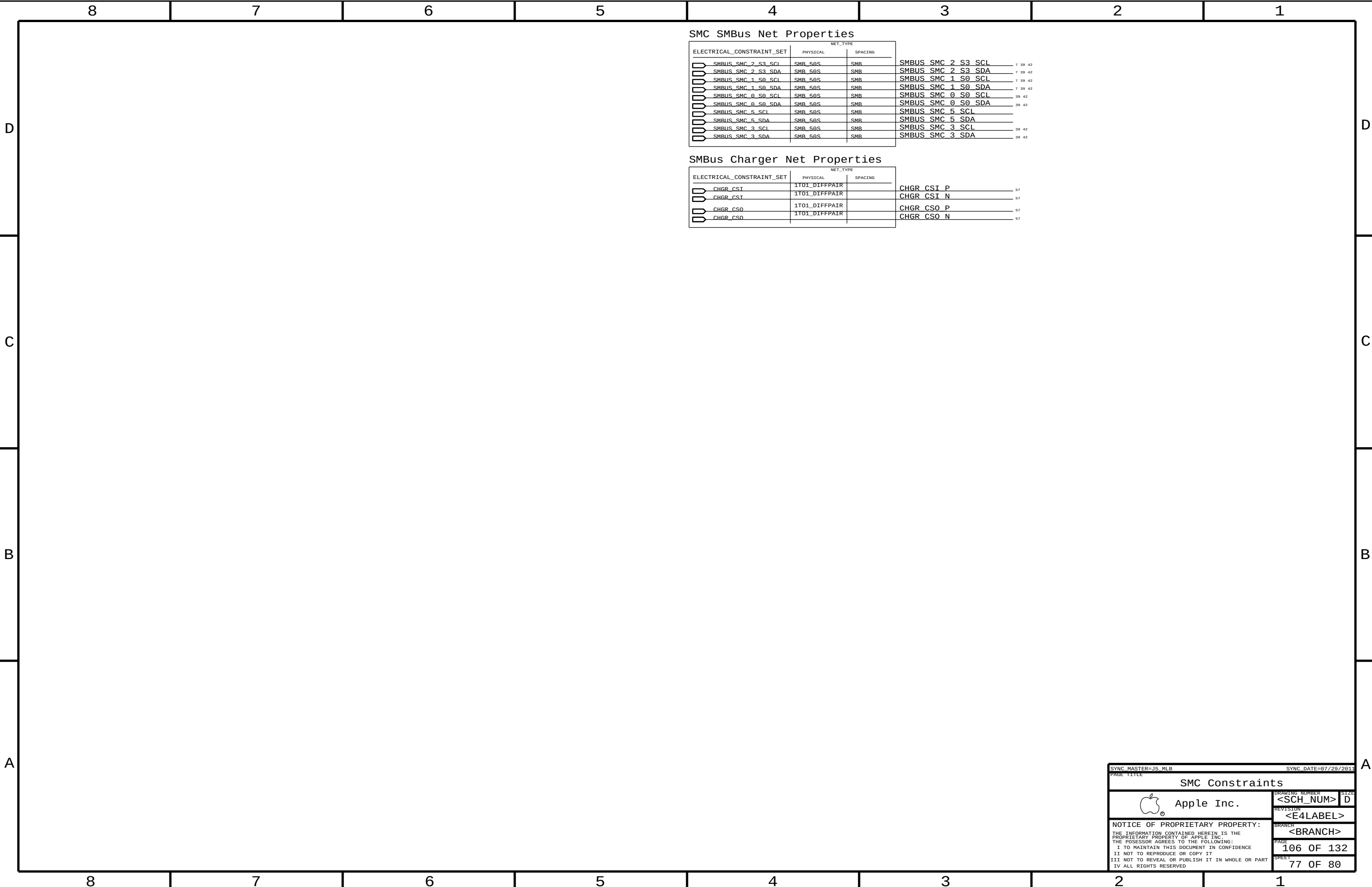
D

C

B

A





SMC SMBus Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
 SMBUS_SMC_2_S3_SCL	SMB_50S	SMB	SMBUS_SMC_2_S3_SCL	7 39 42
 SMBUS_SMC_2_S3_SDA	SMB_50S	SMB	SMBUS_SMC_2_S3_SDA	7 39 42
 SMBUS_SMC_1_S0_SCL	SMB_50S	SMB	SMBUS_SMC_1_S0_SCL	7 39 42
 SMBUS_SMC_1_S0_SDA	SMB_50S	SMB	SMBUS_SMC_1_S0_SDA	7 39 42
 SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0_S0_SCL	39 42
 SMBUS_SMC_0_S0_SDA	SMB_50S	SMB	SMBUS_SMC_0_S0_SDA	39 42
 SMBUS_SMC_5_SCL	SMB_50S	SMB	SMBUS_SMC_5_SCL	
 SMBUS_SMC_5_SDA	SMB_50S	SMB	SMBUS_SMC_5_SDA	
 SMBUS_SMC_3_SCL	SMB_50S	SMB	SMBUS_SMC_3_SCL	39 42
 SMBUS_SMC_3_SDA	SMB_50S	SMB	SMBUS_SMC_3_SDA	39 42

SMBus Charger Net Properties

NET_TYPE				
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
 CHGR_CST	1T01_DIFFPAIR		CHGR_CST_P	57
 CHGR_CST	1T01_DIFFPAIR		CHGR_CST_N	57
 CHGR_CS0	1T01_DIFFPAIR		CHGR_CS0_P	57
 CHGR_CS0	1T01_DIFFPAIR		CHGR_CS0_N	57

SYNC_MASTER=J5_MLB

SYNC_DATE=07/29/2011

PAGE

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SMC Constraints

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1T01_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_CLK	*	GND_P2MM
GND	MEM_CMD	*	GND_P2MM
GND	MEM_CTRL	*	GND_P2MM
GND	MEM_*_DQ_BYTE*	*	GND_P2MM
GND	MEM_DQS	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB3	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB3	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 mm OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.1 MM	500 MIL		
CPU_27P4S	BOTTOM			0.23 MM	100 MIL		
USB3_85D	TOP			0.1 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_37S	BGA_MEM	MEM_50S
MEM_40S	BGA_MEM	MEM_50S
MEM_72D	BGA_MEM	MEM_85D

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
1T01_DIFFPAIR	*	1:1_DIFFPAIR
SENSE_1T01_55S	*	SENSE_1T01_55S
THERM_1T01_55S	*	THERM_1T01_55S
DIFFPAIR	*	DIFFPAIR

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DP_85D	BGA	100_DIFF_BGA
SATA_90D	BGA	100_DIFF_BGA
CLK_PCIE_90D	BGA	100_DIFF_BGA

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.1 MM	6.35 MM		

D1 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	CPU THMSNS D2 P 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	CPU THMSNS D2 N 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	CPU THERMD P 9 10
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	CPU THERMD N 9 10
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	GPU THMSNS D P 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	GPU THMSNS D N 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	GPU TDIODE P 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	GPU TDIODE N 45
IC80	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	TBT THERMD P 45
IC80	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	TBT THERMD N 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	DDR3THMSNS D1 P 45
	SENSE_DIEFFPAIR	THERM_1T01_55S	THERM	DDR3THMSNS D1 N 45
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUVCCIO50 CS P 43 63
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPUVCCIO50 CS N 43 63
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPU_VDDQ_SENSE_P 13
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	CPU_VDDQ_SENSE_N 13
IC80	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_LCD_PANEL_P 43
IC80	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_LCD_PANEL_N 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_1V35_S3_MEM_P 37 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_1V35_S3_MEM_N 37 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_SSD_P 37 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_SSD_N 37 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_3V3_S0_SSD_R_P 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_3V3_S0_SSD_R_N 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_WLAN_P
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_WLAN_N
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_P
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_LCDBKLT_N
IC80	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_TBT_P 80
IC80	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_TBT_N 80
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_1V35_S3_MEM_R_P 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	ISNS_1V35_S3_MEM_R_N 43
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	VCCSA50 CS P 58 80
	SENSE_DIEFFPAIR	SENSE_1T01_55S	SENSE	VCCSA50 CS N 58 80
IC85	HDMT_CLK	HDMT_90D	HDMT	HDMI_IG_CLK_C_P 7 9 36
IC85	HDMT_CLK	HDMT_90D	HDMT	HDMI_IG_CLK_C_N 7 9 36
IC85	HDMT_DATA	HDMT_90D	HDMT	HDMI_IG_DATA_C_P<2..0> 7 9 36
IC85	HDMT_DATA	HDMT_90D	HDMT	HDMI_IG_DATA_C_N<2..0> 7 9 36

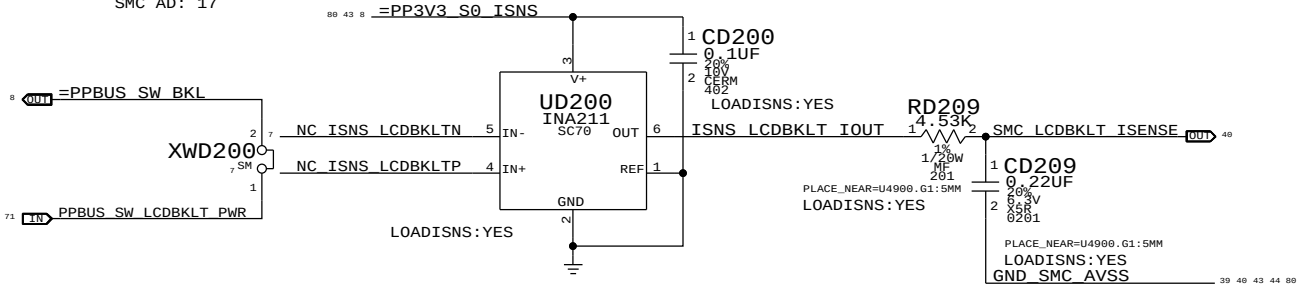
D1 Specific Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPECTRUM		
	PCIE_CLK100M_AP	CLK_PCIE_900	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	7 36
	PCIE_CLK100M_AP	CLK_PCIE_900	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	7 36
		1T01_DIFFEPAIR		CHGR_CSI_R_P	57
		1T01_DIFFEPAIR		CHGR_CSI_R_N	57
		1T01_DIFFEPAIR		CHGR_CSO_R_P	57
		1T01_DIFFEPAIR		CHGR_CSO_R_N	57
	USB_RT	USB_850	USB	USB_BT_CONN_P	36
	USB_BT	USB_850	USB	USB_BT_CONN_N	36
	USB_RT	USB_850	USB	USB_BT_WAKE_P	36
	USB_BT	USB_850	USB	USB_BT_WAKE_N	36
	AUDIO_DIFFEPAIR	DIFFEPAIR	AUDIO	SPKRCONN_SL_OUT_P	7 53 55
	AUDIO_DIFFEPAIR	DIFFEPAIR	AUDIO	SPKRCONN_SL_OUT_N	7 53 55
	AUDIO_DIFFEPAIR	DIFFEPAIR	AUDIO	SPKRCONN_SR_OUT_P	7 53 55
	AUDIO_DIFFEPAIR	DIFFEPAIR	AUDIO	SPKRCONN_SR_OUT_N	7 53 55
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_L_OUT_P	7 53 55 78
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_L_OUT_N	7 53 55 78
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_R_OUT_P	7 53 55 78
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_R_OUT_N	7 53 55 78
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNSG_P	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNSG_N	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS1G_P	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS1G_N	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS2G_P	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS2G_N	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISUMG_R_P	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISUMG_R_N	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		ISNS_HS_OTHER_P	44
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		ISNS_HS_OTHER_N	44
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		ISNS_HS_COMPUTING_P	44
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		ISNS_HS_COMPUTING_N	44
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS_P	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS_N	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS1_P	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS1_N	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS2_P	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISNS2_N	43 62
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISUM_R_P	43
	SENSE_DIFFEPAIR	SENSE_1T01_55S_SENSE		CPUI MVP_ISUM_R_N	43
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L01_L_P	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L01_L_N	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L01_R_P	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L01_R_N	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L02_L_P	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L02_L_N	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L02_R_P	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_L02_R_N	51 53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_P	51 54
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_MIC_INL_N	51 54
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_LIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_RIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_LSUBIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	AUD_SPKRAMP_RSUBIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	RSUBIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	RSUBIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	LSUBIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	LSUBIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRAMP_LIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRAMP_LIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRAMP_RIN_P	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRAMP_RIN_N	53
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	HS_MIC_HI_RC	54
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	HS_MIC_LO_RC	54
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	HS_MIC_HI	54
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	HS_MIC_LO	54
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_L_OUT_P	7 53 55 78
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_L_OUT_N	7 53 55 78
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_R_OUT_P	7 53 55 78
	AUDIO_DIFFEPAIR	AUDIODIFF	AUDIO	SPKRCONN_R_OUT_N	7 53 55 78
	USB_TP4D	USB_850	USB	USB_TP4D_R_P	26 47
	USB_TP4D	USB_850	USB	USB_TP4D_R_N	26 47
	USB_HUB	USB_850	USB	PU_USBHUB_DN4_P	9
	USB_HUB	USB_850	USB	PU_USBHUB_DN4_N	9
		SB_POWER		PP3V3_S5	7 8
		SB_POWER		PP3V3_S0	7 8
		SB_POWER		PP1V5_S3RS0_CPUDDR	8
		GND		GND	

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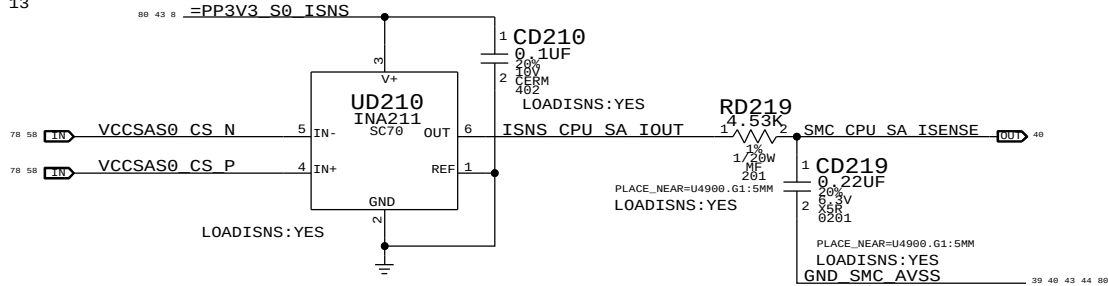
LCD Backlight Current Sense (IBLC)

Gain: 500x. EDP: 0.9 A
Rsense: 0.005 (RD200 / XWD200)
V across Rsense: 4.5 mV
SMC AD: 17



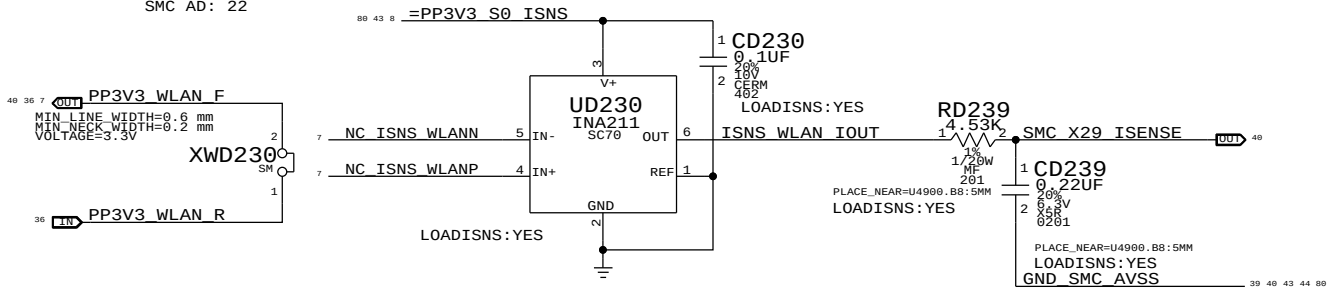
CPU SA Current Sense (IC2C)

Gain: 500x. EDP: 6 A
Rsense: 0.001 (R7140)
V across Rsense: 6 mV
SMC AD: 13



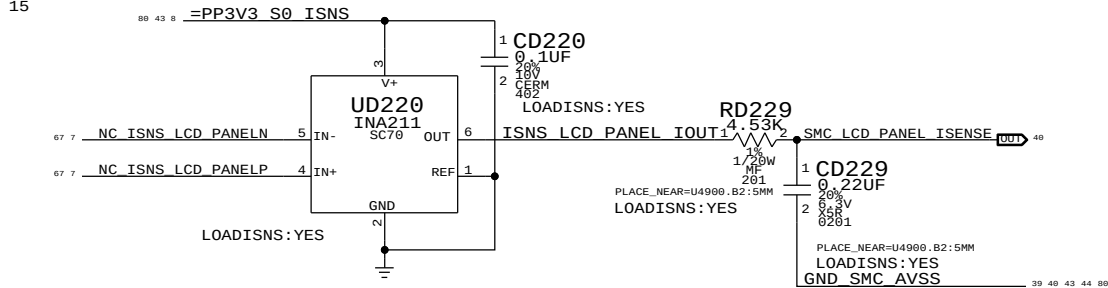
Airport X29 Current Sense (IAPC)

Gain: 500x. EDP: 1.06 A
Rsense: 0.005 (RD230 / XWD230)
V across Rsense: 5.3 mV
SMC AD: 22



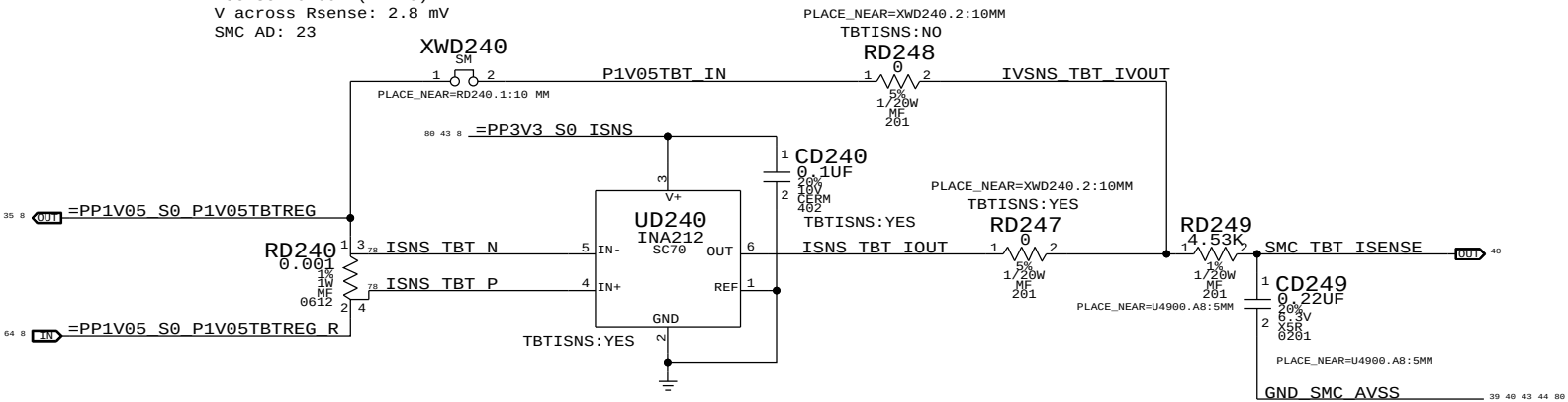
LCD Panel Current Sense (ILDC)

Gain: 500x. EDP: 1 A
Rsense: 0.005 (R9020, XW9020)
V across Rsense: 5 mV
SMC AD: 15



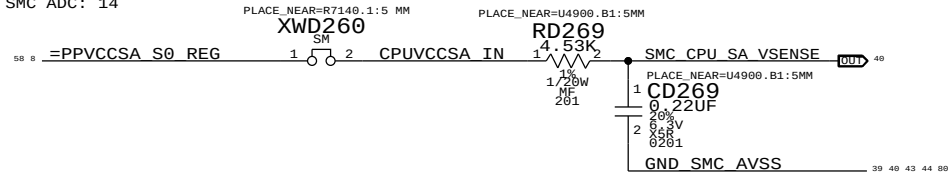
Thunderbolt TBT Current/Voltage Sense (IHSP/VHSP)

Gain: 1000x. EDP: 2.8 A
Rsense: 0.001 (RD240)
V across Rsense: 2.8 mV
SMC AD: 23



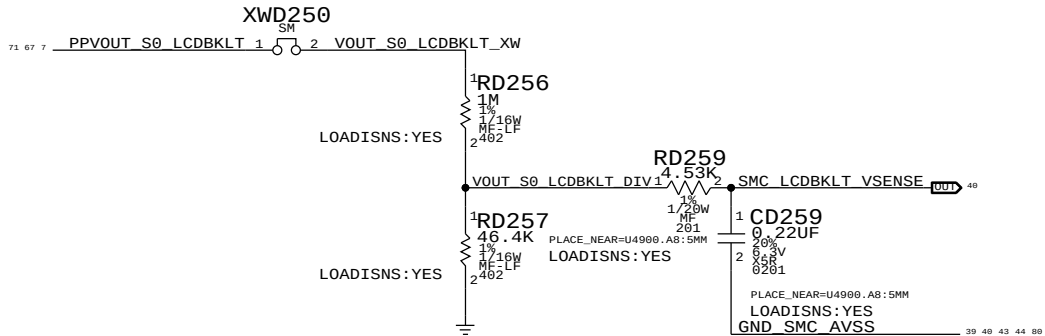
CPU SA Voltage Sense (VC2C)

Gain: 1x
SMC ADC: 14



LCD Backlight Voltage Sense (VBLC)

Gain: 0.04434



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	3	RES, MTL FILM, 100K, 1/16W, 0201, SMD, LF	CD209, CD219, CD229		LOADISNS:NO
117S0008	3	RES, MTL FILM, 100K, 1/16W, 0201, SMD, LF	CD239, CD259		LOADISNS:NO

SYNCH_MASTER=011_SENSORS		SYNCH_DATE=07/11/2013	
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